

Type0029 vendor unique capabilities

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1. Introduction

This document explains the vendor unique capabilities, which are used by Type0029 module (Type0029.md3, Type0029 module.bundle).

These definition values are defined in Maid3d1.h. Refer to the MAID 3.1 Specification for the details of capabilities.

NOTE) These unique capabilities may have different function at another module.

2.Supported camera

Type0029 module can control Z 7II, Z 6II, Z 6II_FU1(Z 6II firmup1) camera.

*Z 7II, Z 6II, Z 6II_FU1 camera doesn't have an exposure meter coupling lever.

3.Vendor Unique Capabilities

The vendor unique capabilities that are used by Type0029 module are described as follows.

The under line shows default value.

* "exposure mode" in the sentence refers to the exposure mode described below

- The exposure mode set in kNkMAIDCapability_ExposureMode
- When kNkMAIDCapability_ExposureMode is 15: U1 (User Mode 1)", "16: U 2 (User Mode 2)" or "18: U 3 (User Mode 3)", the exposure mode set in kNkMAIDCapability_UserMode1 ,2 ,or3.

The exposure mode is abbreviated as follows.

Program mode	: P
Aperture priority	: A
Speed priority	: S
Manual	: M

3.1.ImageSize

This will set the image size. (Photo shooting menu)

Capability kNkMAIDCapability_ImageSize

Object types Source

ulType kNkMAIDCapType_Enum

kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

During live view photography, this will set the image size of the photo shooting menu.

During movie live view, this capability specifies the image size of saving movie frame.

【Basic Image size (JPEG)】

[Z 7II]

Capability_CCDDDataMode	FX	DX	5:4	1:1	16:9
L	L(8256*5504)	L(5408*3600)	L(6880*5504)	L(5504*5504)	L(8256*4640)
M	M(6192*4128)	M(4048*2696)	M(5152*4120)	M(4128*4128)	M(6192*3480)
S	S(4128*2752)	S(2704*1800)	S(3440*2752)	S(2752*2752)	S(4128*2320)

[Z 6II, Z 6II_FU1]

Capability_CCDDDataMode	FX	DX	1:1	16:9
L	L(6048*4024)	L(3936*2624)	L(4016*4016)	L(6048*3400)
M	M(4528*3016)	M(2944*1968)	M(3008*3008)	M(4528*2544)
S	S(3024*2016)	S(1968*1312)	S(2000*2000)	S(3024*1696)

【During movie live view (JPEG)】

During movie live view, the value of this capability will be changed as fellows.

Capability_MovieScreenSize	4K	Full HD
size	3840*2160	1920*1080

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie live view.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
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NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.2. RawimageSize

This will set the raw size of image. (Photo shooting menu)

Capability	kNkMAIDCapability_RawImageSize
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	The data definitions are the same as Capability_ImageSize.

RawImage size may change according to the value of Capability_CCDDDataMode . The data definitions are the same as “S Size”, “M Size” or “L Size” definitions of Capability_ImageSize.

The following cases, this Capability is read only.

- During movie file recording .
- The value of Capability_SpotWBMode is “1(ON)”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.3. CompressionLevel

This will set the compression level of a picture. (Photo shooting menu)

Capability	kNkMAIDCapability_CompressionLevel
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	JPEG Basic, JPEG Basic*, <u>JPEG Normal</u> ,

JPEG Normal*,
JPEG Fine,
JPEG Fine*,
RAW,
RAW + JPEG Basic,
RAW + JPEG Basic*,
RAW + JPEG Normal,
RAW + JPEG Normal*,
RAW + JPEG Fine
RAW + JPEG Fine*

This capability value does not mean current setting value, but means current control value. If [+ NEF (RAW)] function is applied, this capability will returns [RAW+XXX].

The following cases, the value of this Capability is “JPEG Fine”.

- During movie live view

*.However, when the camera setting is changed to movie live view,the menu of “Image quality” of the camera maintains the setting at the time of still image live view.

The following cases, this Capability is read only.

- During movie live view.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.
CameraNotFound	If the camera is disconnected.

3.4. WBMode

This will set the white balance mode. (Photo shooting menu)

Capability	NkMAIDCapability_WBMode
Object type	Source
ulType	kMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	NkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	<u>Auto</u> , Incandescent, Fluorescent, Sunny, Flash, Shade, Cloudy, Preset1, Preset2, Peset3, Preset4, Preset5, Preset6, Color Temperature Natural

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.5. Sensitivity

This will set the sensitivity of camera (Photo shooting menu)

Capability kNkMAIDCapability_Sensitivity

Object types Source

ulType kMAIDCapType_Enum
kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data The value which can be set is changed by the setting of Capability_EVInterval

[Z 7II]

Capability_EVInterval	
1/3 step	1/2 step
Auto, LO-1, LO-0.7, LO-0.3, 64, 80, <u>100</u> , 125, 160, 200, 250, 320, 400, 500, 640, 800, 1000, 1250, 1600, 2000, 2500, 3200, 4000, 5000, 6400, 8000, 10000, 12800, 16000, 20000, 25600, Hi-0.3, Hi-0.7, Hi-1.0, Hi-2.0	Auto, LO-1, LO-0.5, 64, 72, <u>100</u> , 140, 200, 280, 400, 560, 800, 1100, 1600, 2200, 3200, 4500, 6400, 9000, 12800, 18000, 25600, Hi-0.5, Hi-1.0, Hi-2.0

[Z 6II, Z 6II_FU1]

Capability_EVInterval	
1/3 step	1/2 step
Auto, LO-1, LO-0.7, LO-0.3, <u>100</u> , 125, 160, 200, 250, 320, 400, 500, 640, 800, 1000, 1250, 1600, 2000, 2500,	Auto, LO-1, LO-0.5, <u>100</u> , 140, 200, 280, 400, 560, 800, 1100, 1600, 2200, 3200, 4500, 6400,

3200, 4000,5000, 6400, 8000, 10000, 12800, 16000, 20000, 25600, 32000, 40000, 51200, Hi-0.3, Hi-0.7, Hi-1.0, Hi-2.0,	9000, 12800, 18000, 25600, 36000, 51200, Hi-0.5,Hi-1.0, Hi-2.0,
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During movie live view, the ISO sensitivity is controlled by Capability_MovieSensitivity, not this capability.

The following cases, this Capability cannot set “Auto”

- When the exposure mode are setting except P, S, A, and M
- During movie file recording.

The following cases, the upper value of this Capability will be limited. If the range which can be set is changed, kNkMAIDEvent_CapChange will be issued.

- The value of Capability_LiveViewPhotoShootingMode is 1 (ON).

The following case, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.6. ResetMenuBank

This will reset the Photo shooting menu. (Photo shooting menu)

Capability kNkMAIDCapability_ResetMenuBank

Object types Source

ulType kMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

Data None

The following cases, the ulVisibility of this capability is set to invalid and ulOperations of this capability cannot perform.

- During movie file recording.
- The value of Capability_SpotWBMode is 1 (ON).

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.7. CompressRAWEx

This will set “NEF(RAW) recording – NEF(RAW) compression”. (Photo shooting menu)

Capability	kNkMAIDCapability_CompressRAWEx
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDCompressRAWEx 0 : Uncompressed 1 : Compressed <u>2 : Lossless compressed</u>

The following cases, this Capability is read only.

- During live view photography and then the value of Capability_RawImageSize is other than L.
- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.8. WB TuneAuto

This will set the white balance adjustment when the WBMode is “Auto”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneAuto

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224step=1 (Default: 612)

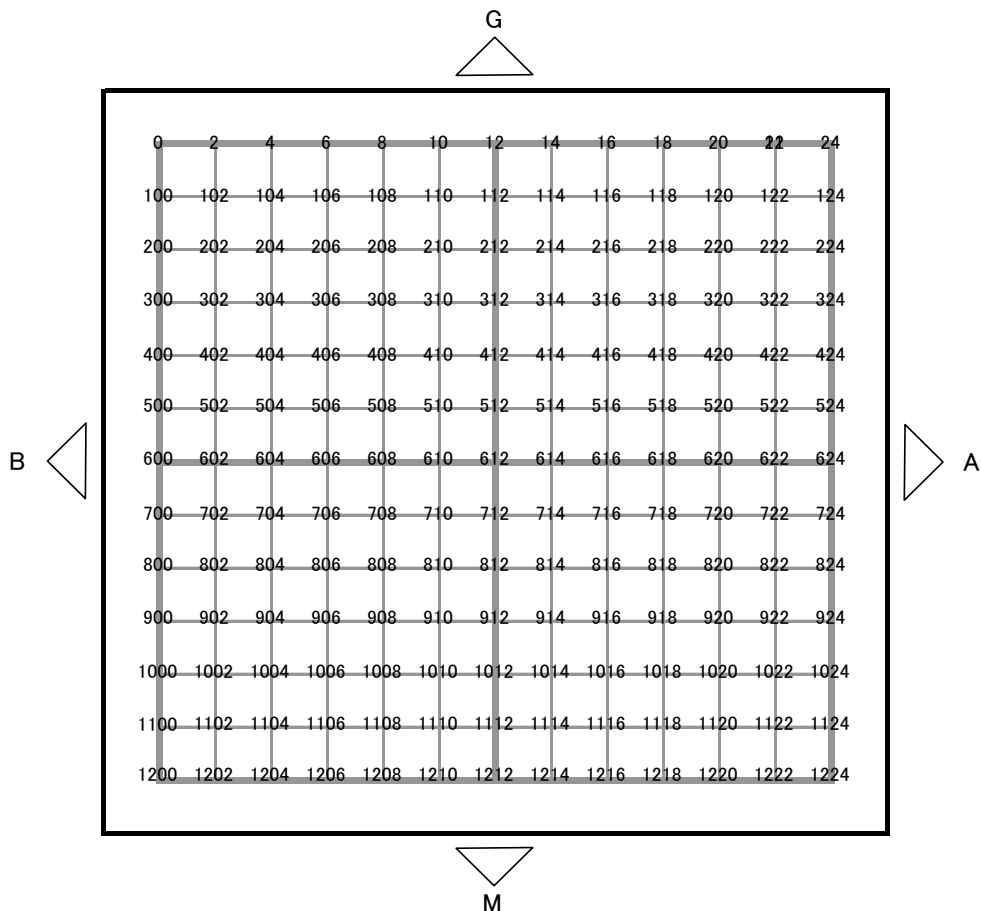
The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

The relationship between white balance adjustment value and the coordinates is shown in following figure.

A-B: 0.5step

G-M: 0.25step



Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.9. WBAutoType

This will set the option when the white balance auto. (Photo shooting menu)

Capability	kNkMAIDCapability_WBAutoType
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkWBAutoType 1 : Keep warm lighting colors 2 : Keep white(reduce warm colors) <u>3: Keep atmosphere</u>

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.10. WBTuneIncandescent

This will set the white balance adjustment when the WBMode is “Incandescent”. (Photo shooting menu)

Capability	kNkMAIDCapability_WBTuneIncandescent
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except

“Same as photo settings”.

- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.11. WBFluorescentType

This will set the fluorescent type when the WBMode is “Fluorescent”. (Photo shooting menu)

Capability	kNkMAIDCapability_WBFluorescentType
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkWBFluorescentType 0: Sodium-vapor lamps 1: Warm-white fluorescent 2: White fluorescent <u>3: Cool-white fluorescent</u> 4: Day white fluorescent 5: Daylight fluorescent 6: High temp. mercury-vapor

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.12. WBTuneFluorescent

This will set the white balance adjustment when the WBMode is “Fluorescent”. (Photo shooting menu)

Capability	kNkMAIDCapability_WBTuneFluorescent
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.13. WBTuneSunny

This will set the white balance adjustment when the WBMode is “Sunny”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneSunny

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.14. WBTuneFlash

This will set the white balance adjustment when the WBMode is “Flash”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneFlash

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
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NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.15. WBTuneShade

This will set the white balance adjustment when the WBMode is “Shade”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneShade

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is except the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.16. WB Tune Cloudy

This will set the white balance adjustment when the WBMode is “Cloudy”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneCloudy

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.17. WB Tune ColorTempEx

This will set the color temperature when the WBMode is “Color Temperature”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneColorTempEx

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 2500 to 10000 step=10 (Default: 5000)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
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NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.18. WB Tune Color Adjust

This will set the white balance adjustment when the WBMode is “Color Temperature”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTuneColorAdjust

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 48 step=1 (Default: 24)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.19. WB Tune Natural

This will set the white balance adjustment when the WBMode is “Natural” (Photo Shooting Menu).

Capability kNkMAIDCapability_MovieWB Tune Natural

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and the white balance correction value is the same as for WB Tune Auto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is except “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.20. WBTunePreset1

This will set the white balance adjustment when the WBMode is “Preset1”.

(Photo shooting menu)

Capability kNkMAIDCapability_WBTunePreset1

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- When the value of Capability_WBTunePresetProtect1 is “1(ON)”.
- During movie file recording and the value of Capability_MovieWBMode is “Same as photo settings”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- The value of Capability_WBMode is except “Preset1”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

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3.21. WB TunePreset2

This will set the white balance adjustment when the WBMode is “Preset2”. (Photo shooting menu)

Capability kNkMAIDCapability_WBTunePreset2

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- When the value of Capability_WBTunePresetProtect2 is “1(ON)”.
- During movie file recording and the value of Capability_MovieWBMode is “Same as photo settings”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- The value of Capability_WBMode is except “Preset2”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.22. WBTunePreset3

This will set the white balance adjustment when the WBMode is “Preset3”.

(Photo shooting menu)

Capability kNkMAIDCapability_WBTunePreset3

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- When the value of Capability_WBTunePresetProtect3 is “1(ON)”.
- During movie file recording and the value of Capability_MovieWBMode is “Same as photo settings”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- The value of Capability_WBMode is except “Preset3”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.23. WB TunePreset4

This will set the white balance adjustment when the WBMode is “Preset4”.

(Photo shooting menu)

Capability kNkMAIDCapability_WBTunePreset4

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- When the value of Capability_WBTunePresetProtect4 is “1(ON)”.
- During movie file recording and the value of Capability_MovieWBMode is “Same as photo settings”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- The value of Capability_WBMode is except “Preset4”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.24. WBTunePreset5

This will set the white balance adjustment when the WBMode is “Preset5”.

(Photo shooting menu)

Capability kNkMAIDCapability_WBTunePreset5

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- When the value of Capability_WBTunePresetProtect5 is “1(ON)”.
- During movie file recording and the value of Capability_MovieWBMode is “Same as photo settings”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- The value of Capability_WBMode is except “Preset5”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.25. WB TunePreset6

This will set the white balance adjustment when the WBMode is “Preset6”.

(Photo shooting menu)

Capability kNkMAIDCapability_WBTunePreset6

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability is read only.

- When the value of Capability_WBTunePresetProtect6 is “1(ON)”.
- During movie file recording and the value of Capability_MovieWBMode is “Same as photo settings”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- The value of Capability_WBMode is except “Preset6”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.26. WBPresetsProtect1

This will set the protection of white balance preset 1. (Photo shooting menu)

Capability kNkMAIDCapability_WBPresetsProtect1

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
kNkMAIDCapOperation_Set

Data one of eNkMAIDWBPresetsProtect1

0: Off

1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.27. WBPresetProtect2

This will set the protection of white balance preset 2. (Photo shooting menu)

Capability	kNkMAIDCapability_WBPresetProtect2
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDWBPresetProtect2 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.28. WBPresetProtect3

This will set the protection of white balance preset 3. (Photo shooting menu)

Capability	kNkMAIDCapability_WBPresetProtect3
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDWBPresetProtect3 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.29. WBPresetProtect4

This will set the protection of white balance preset 4. (Photo shooting menu)

Capability kNkMAIDCapability_WBPresetProtect4

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
kNkMAIDCapOperation_Set

Data one of eNkMAIDWBPresetProtect4

0: Off

1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.30. WBPresetsProtect5

This will set the protection of white balance preset 5. (Photo shooting menu)

Capability kNkMAIDCapability_WBPresetsProtect5

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
kNkMAIDCapOperation_Set

Data one of eNkMAIDWBPresetsProtect5

0: Off

1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.31. WBPresetProtect6

This will set the protection of white balance preset 6. (Photo shooting menu)

Capability	kNkMAIDCapability_WBPresetProtect6
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDWBPresetProtect6 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.32. WBPresetNumber

This will set the value of Photo shooting menu [White balance] – [Preset manual].

Capability kNkMAIDCapability_WBPresetNumber

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data Preset 1, Preset 2, Preset 3, Preset 4, Preset5, Preset6

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.33. WBPresetName

This will set the name of white balance preset data. (Photo shooting menu)

Capability kNkMAIDCapability_WBPresetName

Object types Source

ulType kNkMAIDCapType_String
kNkMAIDCapType_Array

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
kNkMAIDCapOperation_GetArray

Data NkMAIDArray

When the client sends to the module kNkMAIDCapOperation_GetArray, the module set string array of the name of white balance preset data to “NkMAIDArray.pData” in order of preset1-6. If the client set string longer than 36 bytes, the module uses 36 bytes from the head. The character, which can be included in the string, is only an ASCII 90 characters.(refer to the ASCII code table .) The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the client set to the white balance preset data with Capability_WBPresetProtectX setting is On(1).
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.34. CCDDataMode

This will set how to read data from CCD. (Photo shooting menu)

Capability	kNkMAIDCapability_CCDDataMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDCCDDataMode

[Z 7II]

3: FXformat

4: DXformat

5: 5:4

8: 1:1

9: 16:9

[Z 6II, Z 6II_FU1]

3: FXformat

4: DXformat

8: 1:1

9: 16:9

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- DX lens is attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.35. ImageColorSpace

This will set color space. (Photo shooting menu)

Capability kNkMAIDCapability_ImageColorSpace

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
kNkMAIDCapOperation_GetDefault

Data one of eNkMAIDImageColorSpace
0: sRGB,
1: AdobeRGB

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.36. IsoControl

This will set whether auto sensitivity control is used when you take a picture. (Photo shooting menu)

Capability kNkMAIDCapability_IsoControl
Object types Source
ulType kNkMAIDCapType_Boolean
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data True: used False: not used

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.37. NoiseReduction

This will set whether noise reduction is used or not used. (Photo shooting menu)

Capability kNkMAIDCapability_NoiseReduction
Object types Source
ulType kNkMAIDCapType_Boolean
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data True: used False: not used

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording
- The value of Capability_LiveViewPhotoShootingMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.38. NoiseReductionHighISO

This will set whether noise reduction is used or not used when high ISO.

(Photo shooting menu)

Capability	kNkMAIDCapability_NoiseReductionHighISO
Object type	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDNoiseReductionHighISO 0: OFF <u>1: ON (Normal)</u> 2: ON (High) 3: ON (Low)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1 (ON).
- During movie file recording

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.39. CompressRAWBitMode

This will set bit depth for RAW(NEF). (Photo shooting menu)

Capability	kNkMAIDCapability_CompressRAWBitMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDCompressRAWBitMode 0: 12-bit <u>1: 14-bit</u>

The following cases, this Capability is read only.

- During live view photography and then the value of Capability_RawImageSize is other than “L”.
- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.40. PictureControl

This will set Picture Control. (Photo shooting menu)

Capability	kNkMAIDCapability_PictureControl
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDPictureControl 1: Standard 2: Neutral 3: Vivid 4: Monochrome 5: Portrait 6: Landscape 7: Flat <u>8: Auto</u> 101: Dream 102: Morning 103: Pop 104: Sunday 105: Somber 106: Dramatic 107: Silence 108: Breached 109: Melancholic 110: Pure 111: Denim 112: Toy 113: Sepia 114: Blue 115: Red 116: Pink 117: Charcoal 118: Graphite 119: Binary 120: Carbon 201 - 209: Custom Picture Control 1 – 9

This capability shows the current selected Picture Control.

When the client sends `kNkMAIDCapOperation_GetArray`, the module returns the all Picture control enumeration value including unused Custom Picture Control.

The client can know whether the Picture Control is used or not by checking “CustomFlag” in Picture Control Data format.(see `Capability_PictureControldata`)

When the Picture Control selected currently is changed, `kNkMAIDEvent_CapChangeValueOnly` is issued about this capability.

The following cases, this Capability is read only.

- The value of `Capability_SpotWBMode` is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the client sends <code>kNkMAIDCapOperation_Set</code> with unused Picture Control.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.41. PictureControlDataEx2

This will set the assigned Picture Control data. And this will get Picture Control data. (Photo shooting menu)

Capability kNkMAIDCapability_PictureControlDataEx2

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Set, kNkMAIDCapOperation_Get
kNkMAIDCapOperation_GetDefault

Data pointer to NkMAIDPicCtrlData structure

typedef struct tagNkMAIDPicCtrlData

{

ULONG ulPicCtrlItem;----- The target Picture Control

ULONG ulSize;-----The size of Picture Control data (Max: 614 bytes)

bool bModifiedFlag; -----Modification flag

(false: initial registration, true: edit)

void* pData;----- The pointer of Picture Control data.

} NkMAIDPicCtrlData, FAR* LPNkMAIDPicCtrlData;

The range of value sets to “ulPicCtrlItem” is enumerated by Capability_PictureControl.

The following case, this Capability is read only.

- During movie file recording.

[In case of Set]

When the client sends kNkMAIDCapOperation_Set, the client must set the all the member of NkMAIDPicCtrlData.

If “bModifiedFlag” is false (initial registration), the module updates the current value and default value of Picture Control, by the content of “pData”. If “bModifiedFlag” is true (edit), the module updates the current value of Picture Control only, by the content of “pData”.

[In case of Get]

When the client sends kNkMAIDCapOperation_Get, the client must set the maximum Picture Control data size, 614, to "ulSize", and set the allocation space for 614 bytes to "pData".

The module sets the size of the picture control data actually set to “pData” to “ulSize” when succeeding in acquisition.

It is possible to get Picture Control data about unused Picture Control data.

The client can know whether the Picture Control data is used or not by referring “CustomFlag”. The format of the Picture Control data is shown below.

※The format of Picture Control data and the available values at set is shown at “[PictureControl DataSet](#)”

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	· If you run the Operation that it is not supported against Capability. · When the following all conditions will be met. - The ulPicCtrlItem is the CustomPictureControl - The bModifiedFlag is true. - There is the defferece ulPicCtrlItem and BasePictureControl of PictureControlData · When unusable value is set for the PictureControlData. · The invalid PictureControlData is specified.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.41.1. PictureControl DataSet

PictureControl DataSet is sent in Capability_PictureControlDataEx2 or Capability_PictureControlDataEx2.

PictureControl DataSet indicates information about B.

Field	Size	Data
BasePictureControl	2	BasePictureControl
RegistrationName	20	RegistrationName
ApplyLevel	1	ApplyLevel
QuickSharpFlag	1	QuickSharpFlag
QuickSharp	1	QuickSharp
Sharpening	1	Sharpening
MiddleRangeSharpening	1	MiddleRangeSharpening
Clarity	1	Clarity
Contrast	1	Contrast
Brightness	1	Brightness
Saturation	1	Saturation
Hue	1	Hue
FilterEffects	1	FilterEffects
Toning	1	Toning
ToningDensity	1	ToningDensity

CustomCurveFlag	1	CustomCurveFlag
CustomCurveData	0 or 578	CustomCurveData

The values used for Sharpening, MiddleRangeSharpening, Clarity, Contrast, Brightness, Saturation, Hue and ToningDensity are shown below.

Value	Description
-128	Auto
-N	-N x 0.25
...	
-1	-0.25
0	0
1	0.25
...	
N	N x 0.25

BasePictureControl

Indicates the type of PictureControl that is the base of the CustomPictureControl.

In the case of preinstalled PictureControl, it is indicated its PictureControl.

When the unused CustomPictureControl is got, it becomes 0

Value	Description
1	Standard
2	Neutral
3	Vivid
4	Monochrome
5	Portrait
6	Landscape
7	Flat
8	Auto
101	Dream
102	Morning
103	Pop
104	Sunday
105	Somber
106	Dramatic
107	Silence
108	Breached
109	Melancholic
110	Pure
111	Denim
112	Toy
113	Sepia
114	Blue
115	Red
116	Pink
117	Charcoal
118	Graphite
119	Binary
120	Carbon

RegistrationName

Indicates the RegistrationName of the PictureControl.

The size is fixed at 20 bytes (ASCII: 19 bytes + NULL).

ApplyLevel

Indicates the ApplyLevel of the PictureControl.

Value	Description
0	0
10	10
20	20
30	30
40	40
50	50
60	60
70	70
80	80

90	90
100	100

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
0			0 10 20 30 40 50 60 70 80 90 100			

QuickSharpFlag

Indicates whether the QuickSharp of the PictureControl is ON or OFF.

When the PictureControl is set and the QuickSharpFlag is ON, the value of Sharpening, MiddleRangeSharpening and Clarity are ignored,

Value	Description
0	OFF
1	ON

QuickSharp

Indicates the QuickSharp of the PictureControl.

Value	Description
-128	Auto
-2	-2
-1	-1
0	0
1	1
2	2

Values available for each PictureControl are shown below.

When the PictureControl is set and the QuickSharpFlag is ON, it cannot be set Auto

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
-2~2	-128 -2~2		-2~2			

Sharpening

Indicates the Sharpening of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
-8 -4 0	-128 (When the QuickSharp is Auto) -12~36					

4	
8	

MiddleRangeSharpening

Indicates the MiddleRangeSharpening of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
-8 -4 0 4 8	-128 (When the QuickSharp is Auto) -20~20					

Clarity

Indicates the Clarity of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
-8 -4 0 4 8	-128 (When the QuickSharp is Auto) -20~20					

Contrast

Indicates the Contrast of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
-8 -4 0 4 8	-128 -12~12		-12~12			

Brightness

Indicates the Brightness of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
0	-6~6		0			

Saturation

Indicates the Saturation of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
-8 -4 0 4 8	-128 -12~12	0	-12~12		0	

Hue

Indicates the Hue of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
0	-12~12	0	-12~12	0		

FilterEffects

Indicates the FilterEffects of the PictureControl.

Value	Description
0	OFF
1	Yellow
2	Orange
3	Red
4	Green

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
0	0~4		0	0~4		

Toning

Indicates the Toning of the PictureControl.

The value of ToningDensity will be ignored when the following all conditions will be met

- When the PictureControl is setting.
- The Toning can be set.
- The value of the Toning is B&W

Value	Description
0	B&W
1	Sepia
2	Cyanotype
3	Red
4	Yellow

5	Green
6	Blue Green
7	Blue
8	Purple Blue
9	Red Purple

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
0	0~9	0	0~9	0	0~9	0~9

ToningDensity

Indicates the ToningDensity of the PictureControl.

Values available for each PictureControl are shown below.

Auto	Standard Vivid Portrait Landscape Neutral Flat	Monochrome	Dream Morning Pop Sunday	Somber Dramatic Silence Breached Melancholic Pure Denim Toy	Sepia Blue Red Pink	Charcoal Graphite Binary Carbon
0	0~28	0	0~28	0	0~28	0~28

CustomCurveFlag

Indicates whether the CustomCurveFlag of the PictureControl is ON or OFF.

In case of the only CustomPictureControl,it can set ON

Value	Description
0	OFF
1	ON

CustomCurveData

Indicates the CustomCurveData of the PictureControl.

The CustomCurveData use LUT data(refer to LUT format).

When the CustomCurveFlag is OFF, this data is not used.

LUT format

For the LUT data, the 64-byte header to be used for the host is added to the 514-byte (2 byte x 257 point) actual data. The header format is specified by the host individually (storage position of the spline point of the LUT to be sent, etc., data to reproduce the LUT when reading is performed), and the camera does not care the contents. However, because the two bytes of the header are used for the camera to decide whether the header data is present or not, the data needs to be set in the header.

Byte	contents
0 ~ 63	Lut Header
64,65	Data0
66,67	Data1
...	...
576,577	Data256

[LUT header format]

As an example of Lut Header, the contents of the header set by the Nikon application are shown below.

Byte	contents	Range
1	AriaID (Byte1)	0x49
2	AriaID (Byte2)	0x30
3	Input Minimum (Black Point)	0-255
4	Input Maximum	0-255
5	Output Minimum	0-255
6	Output Maximum	0-255
7	Gamma (integer portion)	0-20
8	Gamma (fractional portion)	0-100
9	Number of Spline Points	2-20
10, 11	Splime Point1 (x, y)	0-255, 0-255
12, 13	Splime Point2 (x, y)	0-255, 0-255
. . .		
48, 49	Splime Point20 (x, y)	0-255, 0-255
50 to 64	Reserved	0

3.42. DeleteCustomPictureControl

This will delete Custom Picture Control. (Photo shooting menu)

Capability kNkMAIDCapability_DeleteCustomPictureControl

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Set

Data Custom Picture Control Item

When the client set the one of Custom Picture Control enumerated by

Capability_PictureControl and executes kNkMAIDCapOperation_Set, the specified Custom Picture Control will be deleted.

The following cases, this Capability can not set.

- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.43. Active_D_Lighting

This will set Active D-Lighting. (Photo shooting menu)

Capability kNkMAIDCapability_Active_D_Lighting

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
kNkMAIDCapOperation_GetDefault

Data one of meNkMAIDActive_D_Lighting

0: High
1: Normal
2: Low
<u>3: Off</u>
5: Extra high 1
6: Auto

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.44. ISOAutoShutterTime

This will set the shutter speed when ISO is controlled automatically. (Photo shooting menu)

Capability kNkMAIDCapability_ISOAutoShutterTime

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

Data one of eNkMAIDISOAutoShutterTime

Threshold	eNkMAIDISOAutoShutterTime	Threshold	eNkMAIDISOAutoShutterTime
1/4000	20	1/8	4
1/3200	21	1/6	37
1/2500	22	1/5	38
1/2000	23	1/4	5
1/1600	24	1/3	39
1/1250	25	1/2.5	40
1/1000	26	1/2	6
1/800	27	1/1.6	41
1/640	28	1/1.3	42
1/500	29	1	7
1/400	30	1.3	43
1/320	31	1.6	44
1/250	13	2	8
1/200	14	2.5	45
1/160	15	3	46
1/125	0	4	9
1/100	16	5	47
1/80	17	6	48
1/60	1	8	10
1/50	19	10	49
1/40	18	13	50
1/30	2	15	11
1/25	33	20	51
1/20	34	25	52
1/15	3	30	12
1/13	35	Auto	32
1/10	36		

The following cases, this Capability is read only.

- The value of Capability_IsoControl is OFF.
- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.45. ISOAutoShutterTimeAutoValue

This will set compensation value when the value of kNkMAIDCapability_ISOAutoShutterTime is 32(Auto). (Photo shooting menu)

Capability kNkMAIDCapability_ISOAutoShutterTimeAutoValue

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data -2~+2EV(Default value: 0)

This capability is valid when the value of Capability_IsoControl is ON and the value of Capability_ISOAutoShutterTime is 32(Auto).

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.46. ISOAutoHiLimit

This will set the max sensitivity when ISO is controlled automatically. (Photo shooting menu)

Capability kNkMAIDCapability_ISOAutoHiLimit

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

Data one of eNkMAIDISOAutoHiLimit3

The actually assignable value changes by setting Capability_EVInterval and the assignable values follows a value of Capability_Sensitivity.

[Z 7II]

eNkMAIDISOAutoHiLimit4	ISO	EVInterval	
		1/3	1/2
2	100	o	o
3	125	o	x
4	140	x	o
5	160	o	x
6	200	o	o
7	250	o	x
8	280	x	o
9	320	o	x
10	400	o	o
11	500	o	x
12	560	x	o
13	640	o	x
14	800	o	o
15	1000	o	x
16	1100	x	o
17	1250	o	x
18	1600	o	o
19	2000	o	x
20	2200	x	o
21	2500	o	x
22	3200	o	o
23	4000	o	x
24	4500	x	o
25	5000	o	x
26	6400	o	o
27	8000	o	x
28	9000	x	o
29	10000	o	x
30	12800	o	o
36	16000	o	x
37	18000	x	o
38	20000	o	x
39(Default)	25600	o	o

31	Hi-0.3	o	x
32	Hi-0.5	x	o
33	Hi-0.7	o	x
34	Hi-1.0	o	o
35	Hi-2.0	o	o

[Z 6II, Z 6II_FU1]

eNkMAIDISOAutoHiLimit3	ISO	EVInterval	
		1/3	1/2
0	200	o	o
1	250	o	x
2	280	x	o
3	320	o	x
4	400	o	o
5	500	o	x
6	560	x	o
7	640	o	x
8	800	o	o
9	1000	o	x
10	1100	x	o
11	1250	o	x
12	1600	o	o
13	2000	o	x
14	2200	x	o
15	2500	o	x
16	3200	o	o
17	4000	o	x
18	4500	x	o
19	5000	o	x
20	6400	o	o
26	8000	o	x
27	9000	x	o
28	10000	o	x
29	12800	o	o
32	16000	o	x
33	18000	x	o
34	20000	o	x
35	25600	o	o
36	32000	o	x
37	36000	x	o
38	40000	o	x
<u>39(Default)</u>	51200	o	o
21	Hi-0.3	o	x
22	Hi-0.5	x	o
23	Hi-0.7	o	x
24	Hi-1.0	o	o
25	Hi-2.0	o	o

The following cases, this Capability is read only.

- The value of Capability_IsoControl is OFF.
- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the value that cannot set is set to this capability according to the EVInterval setting.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.47. VignetteControl

This will set the Vignette control. (Photo shooting menu)

Capability	kNkMAIDCapability_VignetteControl
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDVignetteControl 0: High <u>1: Normal</u> 2: Low 3: Off

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.48. MovieScreenSize

This will set the value “Frame size/frame rate” of “Movie settings”. (Movie shooting menu)

Capability	kNkMAIDCapability_MovieScreenSize
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieScreenSize8

[Z 7II, Z 6II_FU1]

eNkMAIDMovieScreenSize8	size
22	3840x2160 60p
23	3840x2160 50p
0	3840x2160 30p
1	3840x2160 25p
2	3840x2160 24p
18	1920x1080 120p

19	1920x1080 100p
<u>3</u>	<u>1920x1080 60p</u>
4	1920x1080 50p
5	1920x1080 30p
6	1920x1080 25p
7	1920x1080 24p
15	1920x1080 30p x4(slow-mo)
16	1920x1080 25p x4(slow-mo)
17	1920x1080 24p x5(slow-mo)

[Z 6II]

eNkMAIDMovieScreenSize8	size
0	3840x2160 30p
1	3840x2160 25p
2	3840x2160 24p
18	1920x1080 120p
19	1920x1080 100p
<u>3</u>	<u>1920x1080 60p</u>
4	1920x1080 50p
5	1920x1080 30p
6	1920x1080 25p
7	1920x1080 24p
15	1920x1080 30p x4(slow-mo)
16	1920x1080 25p x4(slow-mo)
17	1920x1080 24p x5(slow-mo)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

[Z 7II]

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • During saving movie frame. • When the DX lens is attached to the camera, the value [22:3840x2160 60p] or [23: 3840x2160 50p] is set. • When the value of Capability_HDMIOutputDataDepth is 1(10bit) and an HDMI recorder is connected, the value [22:3840x2160 60p] or [23: 3840x2160 50p] is set.
DeviceBusy	If the camera is in the Busy state.

UnexpectedError	By the unexpected error, if it fails to execute the command.
-----------------	--

[Z 6II]

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • During saving movie frame. • When the value [15: 1920x1080 30p x4(slow-mo)], [16:1920x1080 25p x4(slow-mo)], [17:1920x1080 24p x5(slow-mo)], [18:1920x1080 120p] or [19:1920x1080 100p] is set, while a DX lens is attached to the camera.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

[Z 6II_FU1]

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • During saving movie frame. • When the value [15: 1920x1080 30p x4(slow-mo)], [16:1920x1080 25p x4(slow-mo)], [17:1920x1080 24p x5(slow-mo)], [18:1920x1080 120p] or [19:1920x1080 100p] is set, while a DX lens is attached to the camera. • When the value of Capability_HDMIOutputDataDepth is 1(10bit) and an HDMI recorder is connected, the value [22:3840x2160 60p] or [23: 3840x2160 50p] is set.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.49. MovieImageQuality

This will set the value “Movie quality” of “Movie settings”. (Movie shooting menu)

Capability	kNkMAIDCapability_MovieImageQuality
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieImageQuality 0: Normal <u>1: High quality</u>

The following cases, this Capability is read only.

[Z 7II, Z 6II_FU1]

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- The value of Capability_MovieScreenSize is “22: 3840 x 2160 60p”, “23: 3840 x 2160 50p”, “0: 3840 x 2160 30p”, “1: 3840 x 2160 25p”, “2: 3840 x 2160 24p”, “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”, “17 : 1920x1080 24p x5(slow-mo)”, “18 : 1920x1080 120p” or “19 : 1920x1080 100p”.
- The camera is performing output the raw image data to HDMI.

[Z 6II]

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- The value of Capability_MovieScreenSize is “0: 3840 x 2160 30p”, “1: 3840 x 2160 25p”, “2: 3840 x 2160 24p”, “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”, “17 : 1920x1080 24p x5(slow-mo)”, “18 : 1920x1080 120p” or “19 : 1920x1080 100p”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.50. MovieRecMicrophone

This will set the value “Microphone” of “Movie settings”. (Movie shooting menu)

Capability	kNkMAIDCapability_MovieRecMicrophone
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovRecMicrophone <u>0: Auto sensitivity (A)</u> 4: Microphone off 5: Manual sensitivity

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording and when the value of this capability is “4(Microphone off)”.
- The value of Capability_ MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)” or “17 : 1920x1080 24p x5(slow-mo)”,

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If this capability is set “4: Microphone off” when during movie file recording and the value of this capability is “0: Auto sensitivity” or “5: Manual sensitivity”.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.51. MovieRecMicrophoneValue

This will set the microphone sensitivity when Capability_MovieRecMicrophone is manual sensitivity. (Movie shooting menu)

Capability kNkMAIDCapability_MovieRecMicrophoneValue
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data 1 to 20 step=1 (Default: 15)

The following cases, this Capability is read only.

- When the value of Capability_MovieRecMicrophone is not 5(Manual sensitivity).
- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)” or “17 : 1920x1080 24p x5(slow-mo)”,

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.52. AutoDistortion

This will set the“Auto distortion control”. (Photo shooting menu)

Capability kNkMAIDCapability_AutoDistortion
Object types Source
ulType kNkMAIDCapType_Unsigned
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data one of eNkMAIDAutoDistortion
0: Off
1: On

The following cases, this Capability is read only.

- When Capability_LensType indicates 0x00000080(Always correct distortion)
- During movie file recording.
- When the value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.

DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.53. UserMode1

This will get the exposure mode set up for " U1(User Mode1)".

Capability	kNkMAIDCapability_UserMode1
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
Data	one of eNkMAIDUserMode <u>19 : P (Program mode)</u> 20 : S (Speed priority) 21 : A (Aperture priority) 22 : M (Manual) 23 : Auto

This Capability is used when the value of kNkMAIDCapability_ExposureMode is 15(U1).

When the value of kNkMAIDCapability_ExposureMode is not 15(U1),the value of this Capability is not guaranteed.

Registering and resetting U1 can do by only the camera operation

3.54. UserMode2

This will get the exposure mode set up for " U2(User Mode2)".

Capability	kNkMAIDCapability_UserMode2
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
Data	one of eNkMAIDUserMode <u>19 : P (Program mode)</u> 20 : S (Speed priority) 21 : A (Aperture priority) 22 : M (Manual) 23 : Auto

This Capability is used when the value of kNkMAIDCapability_ExposureMode is 16(U2).

When the value of kNkMAIDCapability_ExposureMode is not 16(U2),the value of this Capability is not guaranteed.

Registering and resetting U2 can do by only the camera operation

3.55. UserMode3

This will get the exposure mode set up for " U3(User Mode3)".

Capability	kNkMAIDCapability_UserMode3
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
Data	one of eNkMAIDUserMode <u>19 : P (Program mode)</u> 20 : S (Speed priority) 21 : A (Aperture priority) 22 : M (Manual) 23 : Auto

This Capability is used when the value of kNkMAIDCapability_ExposureMode is 18(U3).

When the value of kNkMAIDCapability_ExposureMode is not 18(U3),the value of this Capability is not guaranteed.

Registering and resetting U3 can do by only the camera operation

3.56. HDRMode

This will set the Photo shooting menu, [HDR(high dynamic range) - HDR mode]. (Photo shooting menu)

Capability	kNkMAIDCapability_HDRMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDHDRMode <u>0: Off</u> 1: On(single photo) 2: On(series)

When the release mode is continuous shooting, the HDR photo will be only the first photo.

The following cases, this Capability is read only.

- Capability_EnableBracketing is ON.
- When the exposure mode are setting except P, S, A, and M.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.57. HDRExposure

This will set the Photo shooting menu, [HDR(high dynamic range) – Exposure differential]. (Photo shooting menu)

Capability kNkMAIDCapability_HDRExposure

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault,
kNkMAIDCapOperation_Set

Data one of eNkMAIDHDRExposure

0: Auto

1: 1EV

2: 2EV

3: 3EV

The following cases, this Capability is read only.

- The value of Capability_EnableBracketing is ON.
- When the exposure mode are setting except P, S, A, and M.
- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.58. HDRSmoothing

This will set “HDR(high dynamic range) – Smoothing”. (Photo shooting menu)

Capability	kNkMAIDCapability_HDRSmoothing
Object types	Source ulType kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDHDRSmoothing 0: High <u>1: Normal</u> 2: Low

The following cases, this Capability is read only.

- The value of Capability_EnableBracketing is ON.
- When the exposure mode are setting except P, S, A, and M.
- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.59. HDRSaveIndividualImages

This will set “HDR(high dynamic range) - Save individual images(NEF)” of Photo Shooting Menu.

Capability	kNkMAIDCapability_HDRSaveIndividualImages
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDHDRSaveIndividualImages <u>0: OFF</u> 1: ON

The following cases, this Capability is read only.

- When Capability_EnableBracketing is ON

- When the exposure mode are setting except P, S, A, and M.
- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.60. FlashISOAutoHighLimit

This will set the Photo shooting menu, “ISO sensitivity settings – Auto ISO sensitivity control - Maximum sensitivity with flash”. (Photo shooting menu)

Capability kNkMAIDCapability_FlashISOAutoHighLimit

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
kNkMAIDCapOperation_GetDefault

Data one of eNkMAIDFlashISOAutoHighLimit

The value which can be set up by setup of Capability_EVInterval changes.

[Z 7II]

eNkMAIDFlashISOAutoHighLimit	Maximum sensitivity	EVInterval	
		1/3	1/2
48	100	o	o
49	125	o	x
50	140	x	o
51	160	o	x
0	200	o	o
1	250	o	x
2	280	x	o
3	320	o	x
4	400	o	o
5	500	o	x
6	560	x	o
7	640	o	x
8	800	o	o
9	1000	o	x
10	1100	x	o
11	1250	o	x
12	1600	o	o
13	2000	o	x
14	2200	x	o
15	2500	o	x
16	3200	o	o
17	4000	o	x

18	4500	x	o
19	5000	o	x
20	6400	o	o
21	8000	o	x
22	9000	x	o
23	10000	o	x
24	12800	o	o
25	16000	o	x
26	18000	x	o
27	20000	o	x
28	25600	o	o
29	32000	o	x
30	36000	x	o
31	40000	o	x
32	51200	o	o
37	Hi-0.3	o	x
38	Hi-0.5	x	o
39	Hi-0.7	o	x
40	Hi-1.0	o	o
41	Hi-2.0	o	o
<u>45(Default)</u>	Same as without flash		

[Z 6II, Z 6II_FU1]

eNkMAIDFlashISOAutoHighLimit	Maximum sensitivity	EVInterval	
		1/3	1/2
0	200	o	o
1	250	o	x
2	280	x	o
3	320	o	x
4	400	o	o
5	500	o	x
6	560	x	o
7	640	o	x
8	800	o	o
9	1000	o	x
10	1100	x	o
11	1250	o	x
12	1600	o	o
13	2000	o	x
14	2200	x	o
15	2500	o	x
16	3200	o	o
17	4000	o	x
18	4500	x	o
19	5000	o	x
20	6400	o	o
21	8000	o	x
22	9000	x	o
23	10000	o	x
24	12800	o	o

25	16000	o	x
26	18000	x	o
27	20000	o	x
28	25600	o	o
29	32000	o	x
30	36000	x	o
31	40000	o	x
32	51200	o	o
37	Hi-0.3	o	x
38	Hi-0.5	x	o
39	Hi-0.7	o	x
40	Hi-1.0	o	o
41	Hi-2.0	o	o
<u>45(Default)</u>	Same as without flash		

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- The value of Capability_IsoControl is OFF.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the value that cannot set is set to this capability according to the EVInterval setting.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.61. **ResetCustomSetting**

 This will reset the custom settings, which is selected by Capability_CustomSettings. (Custom setting menu)

- Capability** kNkMAIDCapability_ResetCustomSetting
- Object types** Source
- ulType** kNkMAIDCapType_Process
- ulOperations** kNkMAIDCapOperation_Start
- Data** None

 The following cases, the ulVisibility of this capability is set to invalid and ulOperations of this capability cannot perform.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.62. AFcPriority

This will set the continuous AF area priority. (Custom setting menu a1)

Capability kNkMAIDCapability_AFcPriority
Object types Source
ulType kNkMAIDCapType_Enum
 kNkMAIDArrayType_PackedString
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set

Data

Focus	“AF-C Focus”
<u>Release</u>	“AF-C Shutter”

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.63. AFsPriority

This will set the single AF area priority. (Custom setting menu a2)

Capability kNkMAIDCapability_AFsPriority
Object types Source
ulType kNkMAIDCapType_Enum
 kNkMAIDArrayType_PackedString
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set

Data

<u>Focus</u>	“AF-S Focus”
Release	“AF-S Shutter”

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.64. AFLockOnAcross

This will set “Focus tracking with lock-on – Blocked shot AF response”. (Custom setting menu a3)

Capability	kNkMAIDCapability_AFLockOnAcross
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDAFLockOnAcross 1(Quick)~5(Delayed) Default:3

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.65. AFAreaPoint

This will set “Focus points used”. (Custom setting menu a5)

Capability	kNkMAIDCapability_AFAreaPoint
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDAFAreaPoint <u>5: All points</u> 6: Every other point

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.66. EVInterval

This will set the EV interval of the Capability_ShutterSpeed, Capability_Aperture, Capability_FlexibleProgram, and Capability_AEBracketingStep. (Custom setting menu b1)

Capability	kNkMAIDCapability_EVInterval
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	

<u>1/3step</u>	“1/3 Step”
1/2step	“1/2 Step”

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.67. CWMeteringDiameter

This will set the center weighted metering diameter. (Custom setting menu b3)

- Capability** kNkMAIDCapability_CWMeteringDiameter
- Object types** Source
- ulType** kNkMAIDCapType_Enum
 kNkMAIDArrayType_PackedString
- ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set

Data

Average	“Average”
<u>φ12mm</u>	“12 mm”

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.68. ExpBaseMatrix

This will set the exposure base when the metering mode is matrix. (Custom setting menu b4)

Capability kNkMAIDCapability_ExpBaseMatrix
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data -1 - +1EV (1/6EV step) (Default : 0)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.69. ExpBaseCenter

This will set the exposure base when the metering mode is center weighted. (Custom setting menu b4)

Capability kNkMAIDCapability_ExpBaseCenter
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data -1 - +1EV (1/6EV step) (Default : 0)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.70. ExpBaseSpot

This will set the exposure base when the metering mode is spot. (Custom setting menu b4)

Capability kNkMAIDCapability_ExpBaseSpot

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data -1 - +1EV (1/6EV step) (Default : 0)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.71. ExpBaseHighlight

This will set the exposure base when the metering mode is highlight-weighted metering. (Custom setting menu b4)

Capability kNkMAIDCapability_ExpBaseHighlight
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data -1 - +1EV (1/6EV step) (Default : 0)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.72. ShootingSpeedHigh

This will set frame rate for continuous highlow-speed(CH). (Custom setting menu d1)

Capability	kNkMAIDCapability_ShootingSpeedHigh
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get._kNkMAIDCapOperation_GetArray. kNkMAIDCapOperation_Set

Data

<u>5 fps</u>	“5 frames / second”
4 fps	“4 frames / second”
3 fps	“3 frames / second”
2 fps	“2 frames / second”
1 fps	“1 frame / second”

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.73. ShootingLimit

This will set shooting limit number in continuous shooting. (Custom setting menu d2)

Capability kNkMAIDCapability_ShootingLimit

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data 1 - 200 (Default:200)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.74. ExposureDelayEx

This will set exposure delay mode. (Custom setting menu d4)

Capability	kNkMAIDCapability_ExposureDelayEx
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDExposureDelayEx 0:3 sec 1:2 sec 2:1 sec <u>3: Off</u> 4:0.5 sec 5:0.2 sec

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.75. **NumberingMode**

 This will set a Numbering Mode. (Custom setting menu d7)

Capability kNkMAIDCapability_NumberingMode

Object types Source

ulType kNkMAIDCapType_Enum
 kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set

Data

Off	“Normal filename assignment”
<u>On</u>	“Sequential filename assignment ”

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the value of CapabilityCapability_HDRMode is except “Off”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.76. **ResetFileNumber**

 This resets the number of the file, which will be stored in card. (Custom setting menu d7)

Capability	kNkMAIDCapability_ResetFileNumber
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start
Data	None

 The following cases, the ulVisibility of this capability is set to invalid and ulOperations of this capability cannot perform.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the value of CapabilityCapability_HDRMode is except “Off”.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.77. FlashSyncTime

This will set flash sync speed on shooting with speedlight. (Custom setting menu e1)

Capability kNkMAIDCapability_FlashSyncTime

Object types Source

ulType kNkMAIDCapType_Enum
 kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set

Data

1/200 sec(FP Auto)	“1/200 sec (FP Auto)”
1/200 sec	“1/200 sec”
1/160 sec	“1/160 sec”
1/125 sec	“1/125 sec”
1/100 sec	“1/100 sec”
1/80 sec	“1/80 sec”
1/60 sec	“1/60 sec”

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.78. FlashSlowLimit

This will set the lowest shutter speed on shooting with speedlight. (Custom setting menu e2)

Capability kNkMAIDCapability_FlashSlowLimit
Object types Source
ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

<u>1/60 sec</u>	"1/60 sec"
1/30 sec	"1/30 sec"
1/15 sec	"1/15 sec"
1/8 sec	"1/8 sec"
1/4 sec	"1/4 sec"
1/2 sec	"1/2 sec"
1 sec	"1 sec"
2 sec	"2 sec"
4 sec	"4 sec"
8 sec	"8 sec"
15 sec	"15 sec"
30 sec	"30 sec"

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.79. ExposureCompFlashUsed

This will set “CUSTUM SETTING MENU – e. Bracketing/flash – e3.Exposure comp. for flash”.
(Custom setting menu e3)

Capability	kNkMAIDCapability_ExposureCompFlashUsed
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDExposureCompFlashUsed

0 : Entire frame
1 : Background only

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.80. BracketingVary

This will set "Auto bracketing - Auto bracketing set" (Photo Shooting Menu).

Capability kNkMAIDCapability_BracketingVary

Object types Source

ulType kNkMAIDCapType_Enum

kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

AE bracketing	"AE Only"
Flash only	"Flash Only"
<u>AE & flash</u>	"AE & Flash"
WB bracketing	"White Balance"
ADLbracketing	"ADL bracketing"

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.81. BracketingFactor

This will set the bracketing variation on manual mode. (Custom setting menu e6)

Capability kNkMAIDCapability_BracketingFactor

Object types Source

ulType kNkMAIDCapType_Enum

kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

<u>Flash/speed</u>	“Exposure time + TTL control”
Flash/speed/aperture	“Exposure time + Aperture + TTL control”
Flash/aperture	“Aperture + TTL control”
Flash only	“TTL control”

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.82. BracketingOrder

This will set the bracketing order. (Custom setting menu e7)

Capability kNkMAIDCapability_BracketingOrder

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

[0] -> [-] -> [+]	“Same as Auto Bracketing”
[-] -> [0] -> [+]	“Negative to Positive”

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the exposure mode are setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.83. ShutterSpeedLockSetting

This will set the function of ShutterSpeed lock. (Custom setting menu f4)

Capability	kNkMAIDCapability_ShutterSpeedLockSetting
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDShutterSpeedLockSetting <u>0 : Off</u> 1 : On

The following cases, this Capability is read only.

- The exposure mode are except “Speed priority” and “Manual”.
- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the lens that is failed to firm up is attached.
- Under warning of the shutter speed or the aperture value display

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • On the card unformatted error.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.84. ApertureLockSetting

This will set the function of Aperture lock. (Custom setting menu f4)

Capability kNkMAIDCapability_ApertureLockSetting

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
 kNkMAIDCapOperation_GetDefault

Data one of eNkMAIDApertureLockSetting

0 : Off

 1 : On

The following cases, this Capability is read only.

- The exposure mode are except S, A, and M.
- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the lens that is failed to firm up is attached.
- Under warning of the shutter speed or the aperture value display.
- When the mount adapter is attached.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • On the card access error. • On the card unformatted error.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.85. **MovieReleaseButton**

 This will set the function assigned for shutter button during movie live view. (Custom setting menu g2)

- Capability** kNkMAIDCapability_MovieReleaseButton
- Object types** Source
- ulType** kNkMAIDCapType_Enum
 kNkMAIDArrayType_Unsigned
- ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set
- Data** one of eNkMAIDMovieReleaseButton
 0: Take photos
 1: Record movies

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.86. SBWirelessMode

This will set wireless flash options.

Capability	kNkMAIDCapability_SBWirelessMode
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDSBWirelessMode <u>0 : Off</u> 1 : Radio AWL 2 : Optical AWL 3 : Optical + Radio AWL

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie live view.
- When the not supported external flash is attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.87. SBWirelessMultipleFlashMode

This will set remote flash control.

Capability	kNkMAIDCapability_SBWirelessMultipleFlashMode
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDSBWirelessMultipleFlashMode <u>0 : Group flash</u> 1 : Quick wireless control 2 : Remote repeating

When the value of Capability_SBWirelessMode is [3:Optical + Radio AWL], the value of this capability is set to [0:Group flash].

By the flash type attached to camera or the existence of wireless communication accessory, when there is no supported value, the current value of the Capability is set to default. And in order to show the indefinite state, Visibility is set to Invalid and operation is set to read only.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie live view.
- The value of Capability_SBWirelessMode is [0: Off].
- When the not supported external flash is attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.88. SBUsableGroup

This will get the available group of the external flash.

Capability	kNkMAIDCapability_SBUstableGroup
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDSBUstableGroup <u>0 : Use prohibited</u> or the logical disjunction of <u>eNkMAIDSBGroupID</u> .

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.89. WirelessCLSEntryMode

This will set “Remote (WR) setting – Link mode” (SETUP).

Capability	kNkMAIDCapability_WirelessCLSEntryMode
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDWirelessCLSEntryMode <u>0 : Pairing</u> 1 : PIN

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- The attached wireless remote controller doesn't support the radio-controlled AWL.
- The wireless remote controller is not attached.
- When the not supported external flash is attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.

UnexpectedError	By the unexpected error, if it fails to execute the command.
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3.90. SBPINCode

This will set “Wireless remote (WR) options – Link mode - PIN”. (SETUP)

Capability	kNkMAIDCapability_SBPINCode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDault, kNkMAIDCapOperation_Set
Data	0 ~ 9999(Defalut:0)

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- The attached wireless remote controller doesn't support the radio-controlled AWL.
- The wireless remote controller is not attached.
- When the not supported external flash is attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.91. RadioMultipleFlashChannel

This will get the communication channel of the wireless remote controller.

Capability	kNkMAIDCapability_RadioMultipleFlashChannel
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDault,
Data	one of eNkMAIDRadioMultipleFlashChannel <u>0 : Unknown</u> 5 : 5CH 10 : 10CH 15 : 15CH

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.92. OpticalMultipleFlashChannel

This will get the channel of the optical communication to the external flash.

Capability	kNkMAIDCapability_OpticalMultipleFlashChannel
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDOpticalMultipleFlashChannel <u>1 : 1CH</u> 2 : 2CH 3 : 3CH 4 : 4CH

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_SBWirelessMode is “0:Off” or “1 : Radio AWL”.
- When the not supported external flash is attached to the camera.

The following cases, the ulVisibility of this capability is set to invalid and ulOperations of this capability is set to read-only.

- The value of this capability is undefined.
- During movie live view

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the control possible flash is not attached
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.93. FlashRangeDisplay

This will get the unit of distance for the setting of the external flash.

Capability	kNkMAIDCapability_FlashRangeDisplay
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
Data	one of eNkMAIDFlashRangeDisplay <u>0 : Meter(m)</u> 1 : feet(ft)

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.94. AllTestFiringDisable

It represents a prohibition state of external flash all test firing.

Capability	kNkMAIDCapability_AllTestFiringDisable
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDault,
Data	one of eNkMAIDAllTestFiringDisable <u>0 : Flash permit</u> 1 : Flash off

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.95. SBSettingMemberLock

This will get the locked status of the external flash setting member.

Capability	kNkMAIDCapability_SBSettingMemberLock
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDSBSSettingMemberLock <u>0 : unlock</u> 1 : lock

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_ExternalFlashStatus is [2: Not Exist].
- When the not supported external flash is attached to the camera.
- During movie live view.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.96. ShootNoCard

This will set “SETUP MENU – Slot empty release lock”. (SETUP)

Capability kNkMAIDCapability_ShootNoCard

Object types Source

ulType kNkMAIDCapType_Boolean

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
 kNkMAIDCapOperation_GetDefault

Data True: OK False: Lock

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.97. UserComment

This will set a description of an image. (SETUP)

Capability kNkMAIDCapability_UserComment
Object types Source
ulType kNkMAIDCapType_String
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data String shorter than 36 bytes (not including termination ‘\0’)

If the client set string longer than 36 bytes, the module uses 36 bytes from the head. The character, which can be included in the string, is only an ASCII characters. (refer to the ASCII code table.)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an invalid character has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.98. EnableComment

This will enable to add UserComment to an image file. (SETUP)

Capability kNkMAIDCapability_EnableComment
Object types Source
ulType kNkMAIDCapType_Boolean
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
 kNkMAIDCapOperation_GetDefault

Data True: Enable False: Disable

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.99. ClockDateTime

This will set the built-in clock of camera. (SETUP)

Capability	kNkMAIDCapability_ClockDateTime
Object types	Source
ulType	kNkMAIDCapType_DateTime
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	kNkMAIDDataType_DateTimePtr

The following case, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.100. ManualSetLensNo

This will set the number of the lens referred to by Capability_FmmManual and Capability_F0Manual. (SETUP)

Capability	kNkMAIDCapability_ManualSetLensNo
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	0 - 19 (Default : 0)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

When the current value changes, kNkMAIDEvent_CapChange and kNkMAIDEvent_CapChangeValueOnly of this Capability are not issued. kNkMAIDEvent_ManualSettingLensDataChanged is issued instead.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.

UnexpectedError	By the unexpected error, if it fails to execute the command.
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3.101. FmmManual

This will set the focal length[mm] of the lens specified by kNkMAIDCapability_ManualSetLensNo.
(SETUP)

Capability kNkMAIDCapability_FmmManual

Object types Source

ulType kNkMAIDCapType_Enum
 kNkMAIDArrayType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
 kNkMAIDCapOperation_Set

Data 0(N/A), 6, 8, 13, 15, 16, 18, 20, 24, 25, 28, 35, 43, 45, 50, 55, 58, 70, 80, 85,
 86, 100, 105, 135, 180, 200, 300, 360, 400, 500, 600, 800, 1000, 1200,1400, 1600,
 20002900, 2800, 3200, 4000 (Default: 0)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

When the current value changes, kNkMAIDEvent_CapChange and kNkMAIDEvent_CapChangeValueOnly of this Capability are not issued. kNkMAIDEvent_ManualSettingLensDataChanged is issued instead.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.102. F0Manual

This will set the maximum aperture of the lens specified by kNkMAIDCapability_ManualSetLensNo. (SETUP)

Capability kNkMAIDCapability_F0Manual
Object types Source
ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_Unsigned
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set
Data 0(N/A), 12, 14, 18, 20, 25, 28, 33, 35, 40, 45, 50, 56, 63, 71, 80,
95, 110, 130 150,160 190, 220 (Default: 0)

This capability returns the aperture value multiplied by 10.
(e.g.: If aperture is F1.2 , the module returns 12, if aperture is F19, returns 190)
The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

When the current value changes, kNkMAIDEvent_CapChange and kNkMAIDEvent_CapChangeValueOnly of this Capability are not issued. kNkMAIDEvent_ManualSettingLensDataChanged is issued instead.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.103. EnableCopyright

This will set whether attach copyright information. (SETUP)

Capability kNkMAIDCapability_EnableCopyright

Object types Source

ulType kNkMAIDCapType_Boolean

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
 kNkMAIDCapOperation_GetDefault

Data True: attach False: none

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.104. **ArtistName**

This will set the artist information. (SETUP)

- Capability** kNkMAIDCapability_ArtistName
- Object types** Source
- ulType** kNkMAIDCapType_String
- ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
- Data** String shorter than 36 bytes (not including termination ‘\0’)

If the client set string longer than 36 bytes, the module uses 36 bytes from the head, and the space (0x20) before termination ‘\0’ of string will be ignored.

for example.)

ABCD EFG H I J K L M N O P Q R S T U V W X Y Z

□ shows a space(0x20) , so 8 space is ignored.

The character, which can be included in the string, is only an ASCII 95 characters. (refer to the ASCII code table.)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an invalid character has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.105. CopyrightInfo

This will set the copyright information. (SETUP)

Capability	kNkMAIDCapability_CopyrightInfo
Object types	Source
ulType	kNkMAIDCapType_String
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	String shorter than 54 bytes (not including termination ‘0’)

If the client set string longer than 54 bytes, the module uses 54 bytes from the head, and the space(0x20) before termination ‘¥0’ of string will be ignored.

for example.)

ABCD EFG ‘¥0’

shows a space(0x20) , so 8 space is ignored.

The character, which can be included in the string, is only an ASCII 95 characters. (refer to the ASCII code table.)

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an invalid character has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.106. ShutterSpeed

This will set the shutter speed.

Capability	kNkMAIDCapability_ShutterSpeed
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	Strings of shutter time in second. (e.g.) "30", ... "1", "1/1.3", "1/1.6", ... "1/4000" "x 1//200", "x 1/160", "x 1/125",... "x 1/60" (Default : Minimum value of setting range)

Capability_MovieShutterSpeed is used for during movie live view, during movie file recording.
Capability_ShutterSpeed is used for taking photos, during live view photography.

When the Capability_ExternalFlashStatus is not "Not Exist", the maximum shutter speed value is limited to the Capability_FlashSyncTime setting and the array data is changed.

In addition, both of the following exposure modes and flash modes are satisfied, the minimum shutter speed value is limited to the Capability_FlashSlowLimit setting.

Flash mode	Exposure mode
Red-eye reduction	P
Fill flash	A

The following cases, this capability cannot set Bulb and Time.

- The exposure mode is except "Manual".
- The value of Capability_HDRMode is except "Off".

When the current value is "bulb", the above condition is met, and when "bulb" is not included in the enumeration, the current value is changed to the slowest shutter speed.

After that, when the above condition is canceled and the "bulb" is included in the enumeration, the current value holds the changed value.

The following case, this minimum value is limited to 1/2000.

- The value of Capability_ElectronicFrontCurtainShutterEx is "1:Electronic front-curtain shutter".

The following case, the values greater than 30s are added.

- When the exposure mode is M and the value of Capability_ExtendedShutterSpeedsManual is "1:On".

The following cases, this capability is read only.

- The exposure mode is except “Manual”, or “Speed priority”
- The value of Capability_ShutterSpeedLockSetting is “1(On)”.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode is “Manual”, the value of Capability_Capability_EnableBracketing is “ON”, the value of Capability_BracketingVary is “AE only”, or “AE & Flash”, and “TTL control” is contained in the value of Capability_BracketingFactor.
*However, even when Capability_EnableBracketing is OFF, it may be read only. See EnableBracketing for details.
- The value of Capability_RetractableLensWarningStatus is “True”.
- The value of Capability_ExposureMode is other than Manual and the CPU lens does not attached.
- During movie file recording.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When sequence error has occurred.
- Aperture ring is not minimum.
- When the exposure mode is other than A or M and the value of Capability_RetractableLensWarningStatus is “True”.
- When the exposure mode is other than A or M and a CPU lens is not attached.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.107. FlexibleProgram

This will set the Flexible program value.

Capability kNkMAIDCapability_FlexibleProgram

Object types Source

ulType kNkMAIDCapType_Range

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Data -5~+5EV(Default value: 0)

The module set the step values same as Capability_EVInterval. When the Capability_EVInterval, the exposure mode is changed, the capability is set to default(0), and the

module sends to the client kMAIDEvent_CapChange or kMAIDEvent_CapChangeValueOnly.

The following cases, this Capability is read only.

- The exposure mode is except “Program”.
- During an error which becomes program shift setting prohibited, or a warning is dispiled.
- During movie live view.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.108. Aperture

This will set the aperture.

Capability kNkMAIDCapability_Aperture

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data String of F value (e.g.) “1.4”, “1.6”, “1.8”, ”2”... (Default : Minimum value of setting range)

Capability_MovieAperture is used for during movie live view, during movie file recording.
Capability_Aperture is used for taking photos, during live view photography.

When the aperture value is an integer, the decimal point is not included in the string. When aperture is not set to minimum(FEE), the string of “EE” is set.

When CPU lens is not attached, this capability returns the following value, depending on the value of the Capability_F0Manual and the status of exposure meter coupling lever.

status		The value of this Capability
an exposure meter coupling lever supported camera	The value of Capability_F0Manual is other than ” 0(N/A)”.	It changes depending on the value of Capability_F0Manual and the status of exposure meter coupling lever.
	The value of Capability_F0Manual is ” 0(N/A)”.	"--"
an exposure meter coupling lever unsupported camera	-	

*Regarding the exposure meter coupling lever supporting, refer to “[2.Supported camera](#)”.

The following cases, this Capability is read only.

- Lens is not set to minimum aperture(FEE, Aperture value error)
- When the CPU lens is not attached.
- When the exposure mode are except “A” and “M”.
- When the value of Capability_ApertureLockMode is set to “1(On)”.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_SpotWBMode is “1(On)”.
- When the exposure mode is “Manual”, the value of Capability_EnableBracketing is

“ON”, the value of Capability_BracketingVary is “AE only”, or “AE & Flash”, and Aperture is contained in the value of Capability_BracketingFactor.

*However, even when Capability_EnableBracketing is OFF, it may be read only. See EnableBracketing for details.

The following cases, this Capability is read only and this Visibility is Invalid.

- When sequence error has occurred.
- When the exposure mode is other than A or M and the value of Capability_RetractableLensWarningStatus is “True”. When the exposure mode is other than A or M and a CPU lens is not attached.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.109. MeteringMode

This will get the metering mode.

Capability kNkMAIDCapability_MeteringMode

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
kNkMAIDCapOperation_GetDefault

Data one of eNkMAIDMeteringMode

0: Matrix

1: Center weighted

2: Spot

4: Highlight

For the metering mode during movie live view, it is necessary to use Capability_MovieMeteringMode.

The table below shows this Capability’s control value when the following conditions are met.

The condition	Control value
When the following all conditions are matched. • The “SETUP MENU” – “Non-CPU lens data” is not set. • When CPU lens is not attached. • The value of this Capability is ” 0: <u>Matrix</u>”.	Center weighted
When the following all conditions are matched.	Center weighted

• When CPU lens is not attached. • The value of this Capability is "4: Highlight".	
---	--

The following cases, this Capability is read only

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- The value of Capability_LockExposure is True: Lock.

The following case, this Capability is read only.

- When the exposure mode is setting except P, S, A, and M

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.110. ExposureMode

This will select the exposure mode.

Capability	kNkMAIDCapability_ExposureMode
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDExposureMode <u>0: Program mode</u> 1: Aperture priority 2: Speed priority 3: Manual 5: Auto 15: U1(UserMode1) 16: U2(UserMode2) 18: U3(UserMode3)

If the exposure mode is setting “15: U1(User Mode1)”, “16: U2(User Mode2)” or “17: U3(User Mode3)”, the exposure mode set in Capability_UserMode1, Capability_UserMode2 or Capability_UserMode3 will be used.

The following case, the array data includes only P, S, A, and M.

- When the value of Capability_SpotWBMode is 1(ON).

The following cases, this Capability is read only.

- During movie file recording.
- When the Capability_LockCamera is “False:Unlock(PC-camera mode).”

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.111. ExposureComp

This will set the exposure compensation value.

- Capability kNkMAIDCapability_ExposureComp
- Object types Source
- ulType kNkMAIDCapType_Range
- ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
- Data -5~+5EV(Default value: 0)

The module sets the step value same as Capability_EVInterval.

Capability_MovieExposureComp is used for exposure compensation setting during movie live view.

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode is “Auto”

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.112. ShootingMode

This will set “release mode” (Photo Shooting Menu). **Capability**

kNkMAIDCapability_ShootingMode

Object types Source

ulType kNkMAIDCapType_Enum

kNkMAIDArrayType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,

kNkMAIDCapOperation_Set

Data one of eNkMAIDShootingMode

0: <u>Single</u>
1: Continuous L
2: Continuous H
3: Self-Timer
11: Continuous H(extended)

CapabilityCapabilityCapability

During continuous shooting, the maximum number of shots that can be taken in one burst is equal to the lowest of the following values:

- Capability_ContinuousShootingNum
- Capability_RemainContinuousShooting

When the value of Capability_LiveViewSelector is “1: movie live view”, the Capability_StillCaptureModeSaveFrame is referred as the release mode.

Further, depending on the release mode and the camera state, the number of possible shots can vary:

Release mode	Capability_ContinuousShootingNum	Number of shots
Single frame	Disabled	Only 1.
Continuous L Continuous H Continuous H(extended)	Enabled	The capture of new objects is possible until the number of shots reaches the lowest of the following values: • The value of Capability_ContinuousShootingNum. • The maximum number of objects that can be transferred to the SDRAM (calculated by the camera). • The number of shots remaining during bracket shooting. If the internal flash is in a state in which it can be fired, or if HDR mode is active, only 1 shot is possible.
Self-timer	Disabled	Only 1. On release, the value of this Capability is temporarily set to Single frame.

The following case, this Capability is read only.

- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.113. ContinuousShootingNum

This will set the number of shots in continuous shooting by host.

Capability kNkMAIDCapability_ContinuousShootingNum
Object types Source
ulType kNkMAIDCapType_Unsigned
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
 kNkMAIDCapOperation_GetDefault
Data 1 – 65535 (Default 65535)

The following case, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.114. EnableBracketing

This will set whether bracketing is applied or not.

Capability kNkMAIDCapability_EnableBracketing

Object types Source

ulType kNkMAIDCapType_Boolean

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
kNkMAIDCapOperation_GetDefault

Data True: ON False: OFF

In the case of below table condition of other Capability, the value of this Capability is set “False: OFF”.

Condition		Capability_BracketingVary		
		"AE & Flash"/ "AE Only"/ "Flash Only"	“White Balance”	“ADL bracketing”
Changing Capability_BracketingVary	To “WB bracketing “ from the others. To the other value from “WB bracketing “.	False: OFF	False: OFF	False: OFF
	To “ADL bracketing “ from the others. To the other value from “ADL bracketing “.	False: OFF	False: OFF	False: OFF
Changing Capability_EVInterval		False: OFF	N.A.	N.A.

CapabilityThe following cases, this Capability is read only

- When the value of Capability_HDRMode is except “OFF”.
- When the value of Capability_ LiveViewPhotoShootingMode is 1(ON) and the value of Capability_BracketingVary is “Flash Only”
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.
- During movie file recording.

The following cases, this Capability is read only & invalid.

- When the value of Capability_BracketingVary is “White Balance” and the value of Capability_CompressionLevel is “RAW”, ”RAW+JPEG”,.

This Capability has restrictions as follows.

- When the value of Capability_LiveViewSelector is 1(movie live view),the setting of bracketing of the camera is ON. However, the camera can not perform bracketing. Therefore, the Module sets the value of this Capability to "False: OFF" and sets it to Read Only.(There are differences between the value and the setting prohibited state of Module and camera)
- When there are differences between Module and camera, and bracketing is included in

setting prohibition condition of other Capability.

The setting prohibited state of other Capability will be judged by the setting of bracketing of not Capability_EnableBracketing but the camera.(There is no difference between the setting prohibited state of Module and camera in other Capability)

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.115. AEBracketingStep

This will set the exposure increment for AE, SB, AE/SB bracketing.

Capability kNkMAIDCapability_AEBracketingStep

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data one of eNkMAIDAEBracketingStep
0: 1/3EV
1: 1/2EV
2: 2/3EV
3: 1EV
7: 2EV
8: 3EV

The array data is affected by the Capability_EVInterval setting as following table.

EVInterval	AEBracketingStep
1/3EV	1/3EV, 2/3EV, 1EV, 2EV, 3EV
1/2 EV	1/2EV, 1EV, 2EV, 3EV

When the value of Capability_EVInterval is changed, this capability is changed to 3(1EV).

The following cases, this Capability is read only

- When the value of Capability_BracketingVary is “White Balance” or “ADL blacketing”.
- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- When the value of Capability_HDRMode is except “Off”.
- During movie file recording.
- The value of Capability_LiveViewPhotoShootingMode is 1(ON) and the value of Capability_BracketingVary is “Flash Only”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.116. WBBracketingStep

This will set the white balance increment for WB bracketing.

Capability	kNkMAIDCapability_WBBracketingStep
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDWBBracketingStep <u>0: 1Step</u> 1: 2Step 2: 3Step

The following cases, this Capability is read only

- When the value of Capability_BracketingVary is except “White Balance”
- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- When the value of Capability_HDRMode is except “Off”.
- During movie file recording.
- When the value of Capability_CompressionLevel includes “RAW”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.117. BracketingType

This will set the combination bracketing shots and direction when AE, Flash, AE and Flash, White balance bracketing.

Capability	kNkMAIDCapability_BracketingType
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDBracketingType 0: Minus_2 1: Plus_2 2: Minus_3 3: Plus_3 4: Both_3 5: Both_5 6: Both_7 7: Both_9 8: None_0

When Capability_BracketingVary is “AE only” or “AE & flash” or “Flash only”, and Capability_AEBracketingStep is set to 3EV or 2EV, "6: Both_7", "7: Both_9" are not enumerated.

When Capability_EVInterval is changed, the value of this Capability becomes 8 (None - 0).

The following cases, this Capability is read only

- When the value of Capability_BracketingVary is “ADL blacketing”.
- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- When the value of Capability_HDRMode is except “Off”.
- During movie file recording.
- The value of Capability_LiveViewPhotoShootingMode is 1(ON) and the value of Capability_BracketingVary is “Flash Only”.
- When the value of Capability_BracketingVary is “White Balance” and the value of Capability_CompressionLevel includes “RAW”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.118. ADLBracketingType

This will set the bracketing shots when ADL bracketing.

Capability	kNkMAIDCapability_ADLBracketingType
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
Data	one of eNkMAIDADLBracketingType <u>0 : 2 shots(Off – UserSetting)</u> 1 : 3 shots(Off – Low – Normal) 2 : 4 shots(Off – Low – Normal – High – Extra High) 3 : 5 shots(Off – Low – Normal – High – Extra High) 5 : 0 shots ※User settings use the setting of Capability_ADLBracketingStep.

The following cases, this Capability is read only.

- When the value of Capability_BracketingVary is not “ADL blacketing”.
- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- When the value of Capability_HDRMode is except “Off”.
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.119. ADLBracketingStep

This will set ADL bracketing level when two shots ADL bracketing is selected.

Capability	kNkMAIDCapability_ADLBracketingStep
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDADLBracketingStep <u>0: Auto</u> 1: Low 2: Normal 3: High 4: Extra high1

The following cases, this Capability is read only.

- When the value of Capability_BracketingVary is not “ADL blacketing”.
- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M
- When the value of Capability_HDRMode is except “Off”.
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.120. LiveViewStatus

This will start or stop Live view and show status of Live view.

Capability	kNkMAIDCapability_LiveViewStatus
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDLiveViewStatus <u>0: OFF</u> 3: ON(Remote live view) 4:ON(Camera live view) *It can use only kNkMAIDCapOperation_Get

When the client start Remote live view, the client must set the value of this capability to 3(ON(Remote live view)). And when the client stop Live view, the client set the value of this capability to 0(OFF). If it is stopped Remote live view, the camera changes to Camera live view. In case of kNkMAIDCapOperation_Get, the value of this Capability will show the current status of Live view.

When the value of this Capability is 3(ON(Remote live view)), it can be gotten Live view image by Capability_GetLiveViewImage

The client should check this value before closing Source object, and if the value of this capability is 3(ON(Remote live view)), should set to 0(OFF).

When the Live view is started, the status of camera will be changed to Lock camera internally, but the value of Capability_LockCamera kept the current value.

The client has to check the value of Capability_LiveViewProhibit, and when the value of Capability_LiveViewProhibit is not 0, Live view will not be started.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.121. LiveViewProhibit

This will get the status of Live view prohibition.

Capability kNkMAIDCapability_LiveViewProhibit

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data one of eNkMAIDLIVEViewProhibit

The live view prohibition is shown by logical disjunction value of the following definition value.

value	Forbidden condition
0x10000000	Lens factor. • If the mount adapter is attached, it warns version update.
0x00020000	High temperature and can not start live view.
0x00008000	During photo shooting. *Until finish photo shooting
0x00000100	battery shortage
0x00000004	Sequence error

When the value of this capability is not 0, it shows the status of Live view prohibition.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.122. CameraInclination

This will get inclination of camera.

Capability kNkMAIDCapability_CameraInclination

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault

Data one of eNkMAIDCameraInclination

0: Horizontal or Unsettled

1: Vertical (Grip is top)

2: Vertical (Grip is bottom)

3: Horizontal (Up Down)

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.123. **RemainCotinuuousShooting**

This will get the number of shot that can be recorded on SDRAM or the card in the continuous shooting mode by the command.

Capability	kNkMAIDCapability_RemainContinuousShooting
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault
Data	0 – 360 (Default: 360)

The value of this capability will be changed by the following setting.

- Capability_CompressionLevel
- Capability_ImageSize
- Capability_CompressRAWEx
- Capability_CompressRAWBitMode□
- Capability_NoiseReduction
- Capability_NoiseReductionHighISO
- Capability_ShootingLimit
- Capability_CCDDataMode
- Capability_HDRMode
- Capability_LiveViewPhotoShootingMode
- Capability_LiveViewSelector

If the current value is changed because of the above capability setting, the module sends to the client kMAIDEvent_CapChangeValueOnly.

Further, when in continuous shooting mode, the number of possible shots can vary depending on the value of this Capability. For details, see Capability_ShootingMode.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.124. RemainCountInMedia

This will get the number of shot that can be saved in Card in current image quality.

Capability	kNkMAIDCapability_RemainCountInMedia
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault
Data	0 - 4294967295 (Default:0)

When a card is not inserted, the value of this capability is 0.

The value of this capability is changed by the setting of camera.

If the total number is over 4294967295, the upper limit is 4294967295.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.125. LockExposure

This will get lock status of auto exposure.

Capability	kNkMAIDCapability_LockExposure
Object types	Source
ulType	kNkMAIDCapType_Boolean
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	True: Lock <u>False: Unlock</u>

When Set operation is executed on this Capability and kNkMAIDResult_NoError is returned, it indicates that the request of the switching AE lock is accepted but the completion of the switching AE lock is not guaranteed.

The completion of switching can be confirmed by issuing the CapChangeValueOnly event from this Capability.

When Get operation is executed on this Capability, AE lock status at that time can be acquired.

Therefore, the set value (request content) is not held in this capability.

The following cases, this Capability is read only and the Visibility of this capability is invalid.

- When sequence error has occurred.
- Aperture ring is not minimum.
- The value of Capability_RetractableLensWarningStatus is "True".
- When the value of Capability_SpotWBMode is 1 (ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. In the aperture preview.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.126. LockFocus

This will get lock status of auto focus.

Capability kNkMAIDCapability_LockFocus

Object types Source

ulType kNkMAIDCapType_Boolean

ulOperations kNkMAIDCapOperation_Get

Data True: Lock False: Unlock

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.127. LockFV

This will get the status of FV lock.

Capability kNkMAIDCapability_LockFV

Object types Source

ulType kNkMAIDCapType_Boolean

ulOperations kNkMAIDCapOperation_Get

Data True: Lock False: Unlock

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.128. ExposureStatus

This will get the exposure indicator status of Camera.

Capability kNkMAIDCapability_ExposureStatus

Object types Source

ulType kNkMAIDCapType_Float

ulOperations kNkMAIDCapOperation_Get

Data -10EV~ +10EV step by 1/6 (Default : 0.0EV)

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.129. InfoDisplayErrStatus

This will show error display status on the information panel.

Capability kNkMAIDCapability_InfoDisplayErrStatus

Object types Source

ulType kNkMAIDCapType_Boolean

ulOperations kNkMAIDCapOperation_Get

Data True: ON(Error display) False: OFF

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.130. FocalLength

This will get the focal length of the lens.

Capability kNkMAIDCapability_FocalLength

Object types Source

ulType kNkMAIDCapType_Float

ulOperations kNkMAIDCapOperation_Get

Data lfValue (mm)

When a CPU lens is not attached, the value of this capability is set to zero.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.131. FocusMode

This will get the focus mode.

Capability kNkMAIDCapability_FocusMode

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data one of eNkMAIDFocusMode

0: MF

1: AF-S

2: AF-C

4: AF-F

When no lens is attached, this Capability is always MF.

The value of 4(AF-F) can be got only during movie live view.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.132. BracketingCount

This will get the number of shots on AE bracketing or ADL bracketing.

Capability kNkMAIDCapability_BracketingCount

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data [AE Bracketing] 1– 9
[ADL Bracketing] 1 – 5

When the Capability_EnableBracketing is true and the Capability_BracketingVary is “AE Only” or “Flash Only” or “AE & Flash” or “ADL bracketing”, this capability is valid. If this capability is invalid, returns zero.

When the value of Capability_LiveViewSelector is 1(movie live view), the value of Capability_EnableBracketing becomes OFF. Therefore, the value of this Capability will become 0(Disable).

However, there is a possibility that the camera's bracketing shot is ON.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.133. ExternalFlashStatus

This will shows the status of External flash.

Capability kNkMAIDCapability_ExternalFlashStatus

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data one of eNkMAIDExternalFlashStatus
0: Ready
1:Not Ready
2: Not Exist

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.134. ExternalFlashComp

This will set the flash compensation of the external speedlight.

- Capability kNkMAIDCapability_ExternalFlashComp
- Object types Source
- ulType kNkMAIDCapType_Range
- ulOperations kNkMAIDCapOperation_Get
- Data -3~+3EV (1/6EV step) (Default : 0.0EV)

This capability is valid when Capability_ExternalNewTypeFlashMode is either “iTTL-BL(1)”, “iTTL(2)”, “AA(3)” or “GN(5)”.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.135. ExternalFlashSort

This will get the sort of external speedlight.

Capability kNkMAIDCapability_ExternalFlashSort

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data one of eNkMAIDExternalFlashSort

0: non- communication.
2: new communication (with setting display)
4: new communication (without setting display)
3: Not exist.

The camera cannot detect “1: old communication.”, so this capability returns always “0: noncommunication.” when an old communication speedlight is attached.

The relationship of external speedlight type and the speedlight made by Nikon is shown in the following table.

New communication (with setting display)	New communication (without setting display)	Old communication	Non- communication	Not detected
SB-5000 SB-910 SB-900 SB-800 SB-700 SB-600 SU-800	SB-500 SB-400 SB-300	SB-80DX SB-50DX SB-28DX SB-28D SB-28 SB-27 SB-26 SB-25 SB-24	SB-30 SB-29 SB-29S SB-23 SB-22 SB-22S SB-21A SB-21B SB-20 SB-19 SB-18 SB-17 SB-16A SB-16B SB-15 SB-14 SB-12 SB-11 SB-10 SB-E	SB-9 SB-8 SB-7 SB-6 SB-5 SB-4 SB-3 SB-2 SB-1

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.136. ExternalNewTypeFlashMode

This will get flash mode when the Capability_ExternalFlashSort is “2 :new communication (with setting display)” or ”4:new communication (without setting display)”.

- Capability kNkMAIDCapability_ExternalNewTypeFlashMode
- Object types Source
- ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_Unsigned
- ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
- Data one of eNkMAIDExternalNewTypeFlashMode
 - 0: OFF
 - 1: iTTL-BL
 - 2: iTTL
 - 3: AA(Auto aperture)
 - 4: A(Non-TTL auto)
 - 5: GN(Range-priority manual)
 - 6: M(manual)
 - 7: Repeating flash
 - 8: The external speed light, new communication does not exist.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.137. LensInfo

This will get the focal length and minimum F number.

- Capability kNkMAIDCapability_LensInfo
- Object types Source
- ulType kNkMAIDCapType_String
- ulOperations kNkMAIDCapOperation_Get
- Data (e.g.)”35-70/F3.3-4.5D”

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.138. AFCapture

This will take a picture after auto focus and save an image to specified media.

- Capability kNkMAIDCapability_AFCapture
- Object types Source
- ulType kNkMAIDCapType_Process
- ulOperations kNkMAIDCapOperation_Start

This will take a picture after auto focus. If the Capability_FocusMode is MF (0) or lens is not attached, the camera does shooting immediately without auto focus.

When auto focus failed, whether taking a picture or returning out of focus error, that is depend on the setting of Capability_FocusMode, Capability_AFsPriolity, Capability_LiveViewSelector and Capability_AFcPriolity.

When Capability_LiveViewSelector is "1: movie live view", it always shifts to shooting operation irrespective of the state after AF driving. Since the camera is fixed “priority release”.

When the release mode is set to continuous shooting, refer to [Capability ShootingMode](#) for the maximum number of shots possible.

When the module prepared to get a main image, the module issues kNkMAIDEvent_AddChild to source object.

The taken image will be saved to the destination specified by Capability_SaveMedia.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

[Z 7II, Z 6II_FU1]

- When the value of Capability_ShutterSpeed is “Time”, or greater than 30s.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_MovieLogOutput is “1(Outputting)”
- When the value of Capability_SpotWBMode is 1(ON).
- When the value of Capability_MovieScreenSize is “22 : 3840x2160 60p”, “23 : 3840x2160 50p”, during movie live view.

[Z 6II]

- When the value of Capability_ShutterSpeed is “Time”, or greater than 30s.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_MovieLogOutput is “1(Outputting)”
- When the value of Capability_SpotWBMode is 1(ON).

Result Codes

OutOfFocus	When the following all conditions are met. •The AF operation mode is the focus priority. •The AF operation is out-of-focus, •The value of Capability_LiveViewSelector is “0: live view photography”.
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NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media
NoMedia	•If the card is under being formatted •If no card is inserted and Capability_SaveMedia is "card", or "card and SDRAM". •When battery is low.
InvalidMedia	If the card error has occurred on camera and Capability_SaveMedia is "card", or "card and SDRAM".
UnformattedMedia	If the card has unformatted and Capability_SaveMedia is "card", or "card and SDRAM".
MediaReadOnly	If the card is protected.
ApertureFEE	Aperture ring is not minimum.
BatteryDontWork	When battery is low.
ShutterBulb	When shutter speed is bulb.
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.139. ContrastAF

This will control contrast AF during live view.

Capability	kNkMAIDCapability_ContrastAF
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set one of eNkMAIDContrastAF 0x00: start AF (effective only as the Set value) 0x01: stop AF (effective only as the Set value) 0x10: AF finishes in focus (effective only as the Get value) 0x11: AF finishes out of focus (effective only as the Get value) 0x12: It is operating AF (effective only as the Get value)

Contrast AF will start when the client set 0x00 (start AF). And the module will return the response without wait for AF finish.

The client can confirm whether contrast AF finish correctly by getting value(0x10, 0x11, 0x12) of this capability, or referring “focus drive state” of “display information” in Live view image. (Please refer NkMAIDCapability_GetLiveViewImage)

When the client wants to stop contrast AF, the client will set 0x01 (stop AF). After contrast AF finish, the module returns response.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capability_FocusMode is “MF”.
- When CPUlens is not attached
- When the value of Capability_LiveViewStatus is “OFF”.
- When the value of Capability_RetractableLensWarningStatus “True”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
OutOfFocus	AF mode of operation is the focus priority, if the AF operation of out-of-focus.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.140. MFDriveStep

This will set the driving step of lens for adjusting focus position during live view.

Capability	kNkMAIDCapability_MFDriveStep
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	driving step (Number of pulses) 1 to 32767

This capability will save the driving step internally, does not send request for adjusting focus position to camera. Capability_MFDrive will send request for adjusting focus position to camera with this capability value actually.

The following cases, the ulVisibility of this capability is invalid and read only.

- When the value of Capability_FocusMode is “MF”.
- When CPUlens is not attached.
- When the value of Capability_LiveViewStatus is “OFF”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.141. MFDrive

This will adjust focus position during live view.

Capability	kNkMAIDCapability_MFDrive
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Set
Data	one of eNkMAIDMFDrive 0: infinity -> close 1: close -> infinity

This will send request to adjust focus position with the setting of this capability and the step of Capability_MFDriveStep. The module will return response as soon as the camera starts adjusting manual focus position, the module doesn't wait to finish manual focus driving.

After this capability is executed correctly, the client can confirm whether manual focus driving finish correctly by performing kNkMAIDCapability_DeviceReady. In addition, when the value of Capability_LiveViewStatus is 3(ON(Remote live view)), it can be also confirmed by "focus drive state" of "Display information".

The following cases, the ulVisibility of this capability is invalid and read only.

- When the value of Capability_FocusMode is "MF" or "AF-F"
- When CPUlens is not attached.
- When the value of Capability_LiveViewStatus is "OFF".

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.142. ContrastAFArea

This will change focus point of contrast AF during remote live view.

Capability kNkMAIDCapability_ContrastAFArea

Object types Source

ulType kNkMAIDCapType_Point

ulOperations kNkMAIDCapOperation_Set

Data typedef struct tagNkMAIDPoint
 {
 SLONG x; -----Coordinates of X axis
 SLONG y; -----Coordinates of Y axis
 } NkMAIDPoint, FAR* LPNkMAIDPoint;

This capability set the focus point by using x and y of NkMAIDPoint structure.

The value range of x and y is defined by “total size” of “Display information” in Live view image.
(please refer NkMAIDCapability_GetLiveViewImage)

But the range that can be actually set becomes an area where "size of the AF frame" length and breadth size half was subtracted from the length and breadth size of "total size" respectively. When the value that exceeds the range that can be set to x and y is set, the maximum or minimum value will be used as this value.

Even though Capability_FocusMode is 0(MF), or CPU lens is not attached, this capability can be set.

The following cases, the ulVisibility of this capability is invalid and read only.

- When the value of Capability_LiveViewStatus is except “3:ON(remote live view)”.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
NotLiveView	If remote live view does not be started.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.143. CaptureDustImage

This will take a dust off ref photo and saved to specified media.

Capability kNkMAIDCapability_CaptureDustImage

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

The format type of dust off ref photo is kNkMAIDFileType_NDF.

When the client deletes a dust off ref photo by Capability_DeleteDramImage, the client must use Item ID notified by data parameter of kNkMAIDEvent_AddChild as

Capability_CurrentItemID. The taken image will be saved to the destination specified by Capability_SaveMedia.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When CPUlens is not attached.
- When the value of Capability_ShutterSpeed is "Time".
- When the value of Capability_LiveViewSelector is "1: movie live view".
- Capability_RetractableLensWarningStatus is "True".
- When the value of Capability_MovieLogOutput is 1(Outputting)
- When the value of Capability_SpotWBMode is 1(ON)
- When the value of Capability_HDRMode is other than 0(OFF).
- DX lens is attached to the camera.

Result Codes

CaptureDustFailure	If the camera failed in taking a dust off ref photo.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media
NoMedia	If the card is under being formatted or no card is inserted and Capability_SaveMedia is "card", or "card and SDRAM". When battery is low.
InvalidMedia	If the card error has occurred on camera and Capability_SaveMedia is "card", or "card and SDRAM".
UnformattedMedia	If the card has unformatted and Capability_SaveMedia is "card", or "card and SDRAM".
MediaReadOnly	If the card is protected.
ApertureFEE	Aperture ring is not minimum.

BatteryDontWork	When battery is low.
ShutterBulb	Under bulb warning
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.144. DeleteDramImage

This will delete DRAM image specified by Capability_CurrentItemID.

Capability kNkMAIDCapability_DeleteDramImage

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

The DRAM image to be deleted is specified by Capability_CurrentItemID.

This capability execution timing is limited to the following case.

- After issuing kNkMAIDCapability_Acquire for Image Object, and before issuing kNkMAIDCommand_Close

The client will issue Capability_Acquire for Image object and cancel Capability_Acquire by kNkMAIDCommand_Abort, and set Capability_CurrentItemID and execute this capability, so, the deletion will be completed.

In case of deletion of RAW+JPEG, if the client executes this capability for JPEG, the both of RAW and Jpeg files will be deleted at the same time.

When the client deletes DRAM image after receiving kNkMAIDEvent_AddChild, the client must close Item object. The module does not close Item object.

This capability is not supported when an image is saved on Card.

To Delete SDRAM image by this capability is prohibited, when it is applied for either of condition below.

- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.145. RawJpegImageStatus

This will get whether the image is taken on RAW+JPEG mode.

Capability kNkMAIDCapability_RawJpegImageStatus

Object types Image

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data one of eNkMAIDRawJpegImageStatus

0: Single 1: Raw+JPEG

Result Codes

NoMedia	If the card is being initialized. If no card is inserted. When battery is low.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.146. CurrentItemID

This will specify the DRAM image operated now.

Capability kNkMAIDCapability_CurrentItemID

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

Item ID is used as an identifier that specifies the image data in SDRAM.

Item ID is notified by data parameter of kNkMAIDEvent_AddChild.

The value of this capability is referred by Capability_DeleteDramImage.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.147. GetLiveViewImage

This will get Live view image.

Capability kNkMAIDCapability_GetLiveViewImage

Object types Source

ulType kNkMAIDCapType_Array
kNkMAIDArrayType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray

The client will get the size of Live view image by kNkMAIDCapOperation_Get, and get an actual Live view data by kNkMAIDCapOperation_GetArray.

Every time, the size of Live view image need not be confirmed with kNkMAIDCapOperation_Get in this capability before execution of kNkMAIDCapOperation_GetArray because the specification of Live view image is always maximum size.

The quality of the live view image that can be got by this Module is determined by the connection method with the camera.

When the client wants to get Live view image with kNkMAIDCapOperation_GetArray, the client must allocate the buffer for the maximum size, and set buffer to kNkMAIDArray.pData, and set allocate size to kNkMAIDArray.ulElements.

After reading Item image, the Live view image will be set to kNkMAIDArray.pData.

The following cases, this Capability is read only.

- When the value of Capability_LiveViewStatus is except 3(ON(Remote live view))
- When the value of Capability_LiveViewImageStatus is 0(Live view image can't be acquired)

Live view image is consisted of “Display information” and “Live view image(JPEG).”

The pixel size of Live view image is different in each Live view data, each detailed information is set to “Display information” area.

The format of the live view data is shown [“LiveView data format”](#).

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.147.1. LiveView data format

	Field	Size	Description
Major version	Version	2Byte	Major version(0x01)

			2Byte	Minor version(0x00)
Reserved			4Byte	
Display information area size			4Byte	
Live view image area size			4Byte	
Whole size		Horizontal size	2Byte	Standard of the coordinates
		Vertical size	2Byte	
Display area size		Horizontal size	2Byte	The whole size is equal to the display area size when the image is not enlarged.
		Vertical size	2Byte	
Display center coordinates		Horizontal size	2Byte	
		Vertical size	2Byte	
Live view image size		Horizontal size	2Byte	This size is equal to the size setting by LiveViewImageSize.
		Vertical size	2Byte	
Live view image quality				The image quality will be the same as the value of LiveViewImageCompression 0:BASIC(Size priority) 1:BASIC(Image quality priority) 2:NORMAL(Size priority) 3:NORMAL(Image quality priority) 4:FINE(Size priority) 5:FINE(Image quality priority).
Reserve			7Byte	
AF driving enabled status			1Byte	0: AF driving disabled, 1: AF driving enabled
Focus driving status			1Byte	0: Not driving, 1: Driving
Focusing judgement result			1Byte	0: No information, 1: Not focused, 2: Focused
AF mode state			1Byte	0: Other 1: Face/eye/Animal-priority AF 2:Auto area AF
Selected focus area			1Byte	From 0 to 42
Selected face index			1Byte	From 0 to 34
Tracking status			1Byte	0:Waiting 1:Selecting 2:Tracking
Reserve			1Byte	
0 to 41	AF frame size	Horizontal size	2Byte	Area of the AF frame size and the AF frame center coordinates for 42 byte
		Vertical size	2Byte	
	AF frame center coordinates	Horizontal position	2Byte	
		Vertical position	2Byte	
Remaining time of movie recording			4Byte	From 0 to 1200000 [msec] * It is valid during the movie recording state.
Sound indicator (peak value)	L	1Byte	From 0 to 14	
	R	1Byte	From 0 to 14	
Sound indicator (current value)	L	1Byte	From 0 to 14	
	R	1Byte	From 0 to 14	
Movie recording information			1Byte	0: During LV execution 1: During movie recording
Synchronous recording state of external recording device			1Byte	Reserved(fixed to 0)
Reserve			1Byte	

	Timecode status		1Byte	0:OFF 1:ON
	Timecode count		1Byte	Hour
			1Byte	Minute
			1Byte	Second
			1Byte	Frame
	Countdown time		2Byte	Countdown every one second starting from 3600 (one hour) ; countdown starting from thirty seconds with a rise in temperature
	Spot WB condition		1Byte	0: Off 1: On standby (not acquired) 2: During acquisition operation 3: On standby (acquisition success) 4: On standby (acquisition failure)
	Rotation direction		1Byte	0: No rotation, 1: Rotate counterclockwise, 2: Rotate clockwise, 3: Upside down
	Level angle information	Rolling	4Byte	
		Pitching	4Byte	
		Yawing	4Byte	
Live view image	Use of white balance for live view photography		1Byte	Reserved(fixed to 0)
	Reserve		95Byte	
	Image data			

The data specification of LiveViewObject (with version) is shown below.

No.	Description
1	The display information size is 512Byte.
2	When the ElectronicVR is ON, the through image has a field angle of 90%.

The AF frame size and the AF frame center coordinates are shown below.

No.	Description
1	The AF frame size and the AF frame center coordinates are valid only for the number of areas.
2	If the priority is set to Face/eye/Animal while the AF area mode is set to "Wide(L-people/L-animals)", the first area will be the data for the AF frame size and the AF frame center coordinates of Wide-area AF(L). The second and the subsequent area will be the data for the AF frame size and the AF frame center coordinates of Face/eye/Animal-priority AF.

Spirit level angle status

■ Rolling

See [AngleLevel](#).

■ Pitching

The data is in an signed fixed-point 32bit format, with the upper 16 bits representing the integer part, and the lower 16 bits representing the decimal part.

When the camera is horizontal, the angle is either 0.0° or 180.0°. The angle range is ±15° from 0.0° or 180.0°.

Camera position	Lens orientation	Angle range	Horizon
Horizontal	Up	0.0 ~ 60.0	0.0
	Down	0.0 , 359.9999847412109375 ~ 300.0	
Horizontal (upside down)	Up	180.0 ~ 120.0	180.0
	Down	180.0 ~ 240.0	

In the level display of the setup menu, the display becomes green if the angle value is horizontal.

If the angle value after having added $+0.5^\circ$ and truncated the decimal part is equal to 0.0° or 180.0° , it is considered horizontal.

■ Yawing

The data is in a signed fixed-point 32bit format, with the upper 16 bits representing the integer part, and the lower 16 bits representing the decimal part.

When the camera is horizontal, the angle is either 0.0° or 180.0° . The angle range is $\pm 15^\circ$ from 0.0° or 180.0° .

Camera position	Lens orientation	Angle range	Horizon
Vertical (grip up)	Up	0.0 ~ 60.0	0.0
	Down	0.0 , 359.9999847412109375 ~ 300.0	
Vertical (grip down)	Up	180.0 ~ 120.0	180.0
	Down	180.0 ~ 240.0	

In the level display of the setup menu, the display becomes green if the angle value is horizontal. If the angle value after having added $+0.5^\circ$ and truncated the decimal part is equal to 0.0° or 180.0° , it is considered horizontal.

The details of AF mode state are shown below.

No.	Description
1	AF mode state will be “1:Face/eye/Animal-priority AF”, when the following conditions will be met. •If the priority is set to Face/eye/Animal while the AF area mode is set to “Auto(people/animals)”. •If the priority is set to Face/eye/Animal while the AF area mode is set to “Wide(L-people/L-animals)”.
2	AF mode state will be “2:Auto area AF”, when the following conditions will be met. •If the AF area mode is set to “Auto area AF”. •If the priority is not set to Face/eye/Animal while the AF area mode is set to “Auto area AF”.
3	AF mode state will be “0: Other”, when the following conditions will be met. •When the conditions for No.1 or No.2 doesn't match. ※This includes the condition when the priority is not set to Face/eye/Animal while the AF area mode is set to “Wide(L-people/L-animals)”.

3.148. GetVideoImageEx

This will get Movie image.

Capability kNkMAIDCapability_GetVideoImageEx

Object types Video

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray

Data pointer to NkMAIDGetVideoImageEx structure

```
typedef struct tagNkMAIDGetVideoImageEx
{
```

```
    NK_UINT_64    ullOffset;----- Offset address of file data before transfer
```

```
    NK_UINT_64    ullReadSize;-----Size of acquired data
```

```
    NK_UINT_64    ullDataSize;-----Size of buffer set to“pData”
```

```
    LPVOID    pData;-----Pointer to buffer
```

} NkMAIDGetVideoImageEx, FAR* LPNkMAIDGetVideoImageEx;

The client will get the size of Movie image by kNkMAIDCapOperation_Get, and get an actual Movie data by kNkMAIDCapOperation_GetArray.

[In case of Get]

The unacquired data of Movie image is set to NkMAIDGetVideoImageEx.ullDataSize.

[In case of GetArray]

When the client want to get Movie image with kNkMAIDCapOperation_GetArray, the client must allocate the buffer for size to be acquired, and set buffer to

NkMAIDGetVideoImageEx.pData, and set allocate size to NkMAIDGetVideoImageEx.ullDataSize.

After reading, the Movie image will be set to NkMAIDGetVideoImageEx.pData, the size of data actually read will be set to

NkMAIDGetVideoImageEx.ullReadSize and the offset address of file data before transfer will be set to NkMAIDGetVideoImageEx.ullOffset. (The Movie image of the

NkMAIDGetVideoImageEx.ullReadSize size is stored in NkMAIDGetVideoImageEx.pData.

The data stored in NkMAIDGetVideoImageEx.pData begin from the position indicated in NkMAIDGetVideoImageEx.ullOffset.)

If the size more than unacquired data of Movie image is set to NkMAIDGetVideoImageEx.ullDataSize, module will read only the actual unacquired data size.

Please set 0 to NkMAIDGetVideoImageEx. ullDataSizewhen the client want to stop the acquisition of Movie image on the way.

During the division acquisition of the Movie image data, if the following operation is executed, the camera resets the own division acquisition information and then the value of NkMAIDGetVideoImageEx.ullOffset becomes zero(0).

- Removal of a card
- Delete of a file.
- Still image shooting.
- Starting movie recordingStarting movie playback.
- Issuing interval of kNkMAIDCapability_GetVideoImageEx is 60 seconds or more.
- Execution of kNkMAIDCapability_SaveCameraSetting.
- Execution of kNkMAIDCapability_InitiatePixelMapping.

If the camera resets the own division acquisition information, it is necessary to stop data acquisition halfway or to reacquire from the beginning.

Result Codes

NoMedia	• If the card is being initialized. • If no card is inserted. • When battery is low.
NotSupported	• If you run the Operation that it is not supported against Capability.

	• During movie file recording. • When the movie is playing.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.149. LockCamera

This will lock camera. When the camera is locked, user can't operate it directly.

Capability kNkMAIDCapability_LockCamera
Object types Source
ulType kNkMAIDCapType_Boolean
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
 kNkMAIDCapOperation_GetDefault
Data True: Lock(Remote mode) False: Unlock(PC-camera mode)

The following cases, this Capability is read only.

- During movie file recording.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state. If this Capability can not be changed by the following operation. - Release operation - AF operation - Image sensor cleaning - Pixel mapping
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.150. CameraType

This will get the camera type.

Capability kNkMAIDCapability_CameraType
Object types Source
ulType kNkMAIDCapType_Unsigned
ulOperations kNkMAIDCapOperation_Get
Data one of eNkMAIDCameraType
 0x53: Z 6II
 0x56: Z 7II
 0x58: Z 6II_FU1

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.151. LensType

This will get the lens type about CPU lens.

Capability	kNkMAIDCapability_LensType
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get
Data	one of eNkMAIDLensType
	0x00000001: D type
	0x00000010: G type
	0x00000100: VR
	0x00001000: DX
	0x00010000:AF-S
	0x00100000: Auto distortion control
	0x01000000: Retractable
	0x00000020: E type
	0x00000040:STM
	0x00000080:Always correct distortion

The value of this capability is expressed by logical disjunction value.

When CPU lens is not attached, the module returns 0

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.152. MovRecInCardStatus

This will start or stop movie recording to the card. In addition this will show status of movie recording.

Capability	kNkMAIDCapability_MovRecInCardStatus
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDMovRecInCardStatus <u>0: OFF</u> 1: ON

When the client starts movie recording to the card, the client must set the value of this capability to 1(ON). And when the client stops movie recording to the external recording device or the card, the client must set the value of this capability to 0(OFF).However, the current value is switched to 0 (OFF) when the camera completes movie file recording. And the Module issues CapChangeValueOnly

If movie recording has been normally started, the Record destination is returned as Result of this Capability. (Refer to Result Codes.)

In case of kNkMAIDCapOperation_Get, the value of this capability will show the current status of movie recording.

If movie recording has been started form camera side, kNkMAIDEvent_StartMovieRecord is issued and the value of this Capability is switched to 1(ON).

It is recommended to check the value of Capability_MovRecInCardProhibit before issuing this capability before starting movie recording. If the Capability_MovRecInCardProhibit is a value other than 0, the client can not start movie recording.

During movie recording, even if it is stopped the remote live view by the setteing Capability_LiveViewStatus to 0 (OFF) , the camera switches to camera live view and keeps movie recording.

Taking a picture of the still picture is possible while movie recording.

The following cases, this capability is read only.

- The value of Capability_LiveViewStatus is "OFF".
- The value of Capability_LiveViewSelector is "0: live view photography".
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

[When starting the movie recording is successful]

RecInCard	Recording by card
-----------	-------------------

[When starting the movie recording has failed]

NotLiveView	If the remote live view does not be started.
ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.153. MovRecInCardProhibit

This will get the status of movie recording prohibition.

Capability kNkMAIDCapability_MovRecInCardProhibit

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data one of eNkMAIDMovRecInCardProhibit

The movie recording prohibition is shown by logical disjunction value of the following definition value. When 0 returns, the status is not movie recording prohibition.

Value	Prohibition condition
0x00080000	Warning or error by the lens factor.
0x00040000	Sequence error
0x00020000	When the following all conditions will be met. [Z 7II, Z 6II_FU1] • The value of Capability_HDMIOutputDataDepth is “1: 10bit” and HDMI output is being executed. • The value of Capability_MovieScreenSize is “22: 3840x2160 60p” “23: 3840x2160 50p” “0: 3840x2160 30p” “1: 3840x2160 25p” “2: 3840x2160 24p” “15: 1920x1080 30p x4(slo-mo)” “16: 1920x1080 25p x4(slo-mo)” “17: 1920x1080 24p x5(slo-mo)” “18: 1920x1080 120p” “19: 1920x1080 100p” • Card is inserted [Z 6II] • The value of Capability_HDMIOutputDataDepth is “1: 10bit” and HDMI output is being executed. • The value of Capability_MovieScreenSize is “0: 3840x2160 30p” “1: 3840x2160 25p” “2: 3840x2160 24p” “15: 1920x1080 30p x4(slo-mo)” “16: 1920x1080 25p x4(slo-mo)” “17: 1920x1080 24p x5(slo-mo)” “18: 1920x1080 120p” “19: 1920x1080 100p” • Card is inserted
0x00008000	When the following conditions will be met. • The value of Capability_MovieLogOutput is 1(Outputting) and the value of Capability_ExternalRecordingControl is 0(OFF).

	• The value of Capability_LiveViewStatus is 3(Remote live view)
0x00002000	Capability_LiveViewSelector is 0 (live view photography)
0x00001000	During enlarged display of Live view
0x00000800	Card is write protected.
0x00000400	During movie file recording
0x00000200	There is movie data in the buffer.
0x00000008	No free area in the card
0x00000004	Card not formatted
0x00000002	Card access error
0x00000001	No card inserted.

This Capability becomes effective only during live view.

When the Capability_LiveViewStatus is 0(OFF), the ulVisibility of this capability is invalid and the current value is invalid.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.154. AngleLevel

This will get the virtual horizon angle information.

Capability kNkMAIDCapability_AngleLevel
Object types Source
ulType kNkMAIDCapType_float
ulOperations kNkMAIDCapOperation_Get
Data normal : 0.0°- 359.9999847412109375°

When angle information cannot be acquired : -1

The angle becomes 0.0 degree when the camera is horizontal, and increases when the camera is turned to an anti-clock surroundings in view of the photographer.

The range of the angle is from 0.0 degree to 359.9999847412109375 degrees.

It becomes 0.0 degree or more if it turns anti-clockwise from the state of 359.9999847412109375 degrees.

It becomes 359.9999847412109375 degrees or less if it turns clockwise from the state of 0.0 degree.

The event is not issued even if there is a change in angle information on the camera.

The module returns -1 when angle information cannot be acquired or there is no reliability in the angle.

When the numerical value that adds 0.5 to the real number value and cuts down the fraction part is a multiple of 0 or 90, it is judged it is the horizontal and vertical.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.155. SaveMedia

This will set the recording media by shooting, shutter-release button or Capability_Capture or Capability_AFCapture, Capability_CaptureDustImage, Capability_CaptureAsync, Capability_AFCaptureAsync.

- Capability kNkMAIDCapability_SaveMedia
- Object types Source
- ulType kNkMAIDCapType_Unsigned
- ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
- Data one of eNkMAIDSaveMedia
 - 0: Card
 - 1: SDRAM
 - 2: Card + SDRAM

The following cases, this Capability is read only.

- During movie file recording.
- When the value of Capability_SpotWBMode is 1(ON).

Setting change of this capability is taken as prohibition until it receives kNkMAIDEvent_CaptureComplete about SDRAM from the execution start of Capability_Capture, Capability_AFCapture, or Capability_CaptureDustImage, Capability_CaptureAsync, Capability_AFCaptureAsync.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.156. TerminateCapture

This capability will stop the shooting operation run from the host(module).

Capability kNkMAIDCapability_TerminateCapture

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Start

Data pointer to NkMAIDTerminateCapture structure

```
typedef struct tagNkMAIDTerminateCapture
```

```
{
```

```
    ULONG    ulKeyCode;----- Specify 0.
```

```
    ULONG    ulReleaseTime;----- Specify 0.
```

(This will be specified by 100msec.)

```
    } NkMAIDTerminateCapture, FAR* LPNkMAIDTerminateCapture;
```

This capability stops the following shooting operation run from the host(module)

shooting operation	the operation after stop shooting
Bulb shooting	The camera records the image until stop shooting, to card or to SDRAM.
continuous low/high speed shooting	The camera stops shooting.

The following cases, the ulVisibility of this capability is set to invalid and the ulOperations of this capability cannot be performed.

- During movie file recording.
- When all the conditions below are not met.
 - The exposure mode is Manual and Capability_ShutterSpeed is “bulb” .
 - The value of Capability_LiveViewSelector is ”0: live view photography” and the value of Capability_ShootingMode is “1: Low-speed continuous shooting”, “2: High-speed continuous shooting”, “11: High-speed continuous shooting (extended) ”.
 - The value of Capability_LiveViewSelector is ”1: movie live view” and the value of Capability_ShootingMode is “2: High-speed continuous shooting”.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.157. BlinkingStatus

This will get the status of the display a shutter speed and an aperture of the camera.

Capability	kNkMAIDCapability_BlinkingStatus
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get
Data	one of eNkMAIDBlinkStatus

- 0: Both a shutter speed and an aperture are displayed with normal status.
- 1: Only a shutter speed is displayed with blinking status.
- 2: Only an aperture is displayed with blinking status.
- 3: Both a shutter speed and an aperture are displayed with blinking status.

When in the bilb or time warning, the value of this Capability is “1: Only a shutter speed is displayed with blinking status.”

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.158. LiveViewSelector

This will set live view photography or movie live view.

Capability	kNkMAIDCapability_LiveViewSelector
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDLiveViewSelector <u>0: live view photography</u> 1: movie live view

The following cases, this Capability is read only.

- When the Capability_LockCamera is “False:Unlock(PC-camera mode).”.
- During movie file recording.
- When the value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.159. ResetWBMode

This will reset the white balance adjustment of Capability_WBMode.

Capability	kNkMAIDCapability_ResetWBMode
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start
Data	none

The following cases, the ulOperations of this capability cannot be performed.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.160. MovieShutterSpeed

This will set the shutter speed during movie live view

Capability	kNkMAIDCapability_MovieShutterSpeed
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	Strings of shutter time in second. (e.g.) "1/30", "1/40", "1/50" (Default : Minimum value of setting range)

It is used for the exposure time of each frame of the movie.

Capability_MovieShutterSpeed is used for during movie live view, during movie file recording, during "Live frame grab". Capability_ShutterSpeed is used for taking photos, during live view photography.

The value range of this capability is changed by the setting of Capability_EVInterval, Capability_ExternalFlashStatus, Capability_ExternalFlashSort, Capability_MovieScreenSize.

The lowest shutter speed is changed by the setting of movie frame rate as follows.

movie frame rate	Lowest shutter speed
120fps	1/125
100fps	1/100
60fps	1/60
50fps	1/50
30fps	1/30
25fps	1/25
24fps	1/25

The following cases, this Capability is read only.

- When the value of Capability_LiveViewStatus is "OFF".
- When the value of Capability_LiveViewSelector is "0: live view photography".
- When the value of Capability_ExposureMode is except."Manual"
- When the value of Capability_SpotWBMode is 1(ON).
- When the value of Capability_RetractableLensWarningStatus is "True: Warning about retraction"

The following cases, this Capability is read only and the Visibility of this capability is invalid.

- When sequence error has occurred.
- Aperture ring is not minimum.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.161. MovieAperture

This will set the aperture during movie live view

Capability kNkMAIDCapability_MovieAperture

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data String of F value. (e.g.) "1.4", "1.6", "1.8", "2" ...
(Default : Minimum value of setting range)

It is used for the exposure time of each frame of the movie.

Capability_MovieAperture is used for during movie live view, during movie file recording, during "Live frame grab". Capability_Aperture is used for taking photos, during live view photography.

When the aperture value is an integer, the decimal point is not included in the string.

When CPU lens is not attached, this capability returns the following value, depending on the value of Capability_F0Manual and the status of exposure meter coupling lever.

status		The value of this Capability
an exposure meter coupling lever supported camera.	The value of Capability_F0Manual is other than "0(N/A)".	It changes depend on the value of Capability_F0Manual and the status of exposure meter coupling lever.
	The value of Capability_F0Manual is "0(N/A)".	"--"
an exposure meter coupling lever unsupported camera	-	

*Regarding the exposure meter coupling lever supporting, refer to "[2.Supported camera](#)".

When the movie live view is not executed, the Visibility of this capability is invalid.

The following cases, this Capability is read only.

- When aperture is not set to “minimum” (FEE, Aperture value error).
- When the exposure mode is setting except “Auto” and “Manual”.
- When CPUlens is not attached.
- When sequence error has occurred.
- When the value of Capability_LiveViewStatus is "OFF".
- When the value of Capability_LiveViewSelector is "0: live view photography ".
- When the value of Capability_SpotWBMode is 1(ON).
- When the value of Capability_RetractableLensWarningStatus is “True: Warning about retraction”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.162. MovieSensitivity

This will set “ISO sensitivity(mode M)” for movie live view. (Movie Shooting Menu)

Capability kNkMAIDCapability_MovieSensitivity

Object types Source

ulType kNkMAIDCapType_Enum

kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data The value which can be set is changed by the setting of Capability_EVInterval.

[Z 7II]

Capability_EVInterval	
1/3 step	1/2 step
64, 80, <u>100</u> ,	64, 72, <u>100</u> ,
125, 160, 200,	140, 200, 280,
250, 320, 400,	400, 560, 800,
500, 640, 800,	1100, 1600, 2200,
1000, 1250, 1600,	3200, 4500, 6400,
2000, 2500, 3200,	9000, 12800,

4000, 5000, 6400, 8000, 10000, 12800, 16000, 20000, 25600, Hi 0.3, Hi 0.7, Hi 1.0, Hi 2.0	18000, 25600, Hi 0.5, Hi 1.0, Hi 2.0
--	---

[Z 6II, Z 6II_FU1]

Capability_EVInterval	
1/3 step	1/2 step
<u>100</u> , 125, 160, 200, 250, 320, 400, 500, 640, 800, 1000, 1250, 1600, 2000, 2500, 3200, 4000, 5000, 6400, 8000, 10000, 12800, 16000, 20000, 25600, 32000, 40000, 51200, Hi-0.3, Hi-0.7, Hi-1.0, Hi-2.0	<u>100</u> , 140, 200, 280, 400, 560, 800, 1100, 1600, 2200, 3200, 4500, 6400, 9000, 12800, 18000, 25600, 36000, 51200, Hi-0.5, Hi-1.0, Hi-2.0

In the following case, the range of value of this Capability can be set is limited from 800 to 25600 in Z 7II and 800 to 51200 in Z 6II or Z 6II_FU1. If the range which can be set is changed, kNkMAIDEvent_CapChange will be issued.

- When the value of Capability_MovieLogOutput is 1(Outputting).

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.
- During movie file recording, and the exposure mode is setting except Manual.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.

UnexpectedError	By the unexpected error, if it fails to execute the command.
-----------------	--

3.163. MovieExposureComp

This will set the exposure compensation for movie live view.

Capability	kNkMAIDCapability_MovieExposureComp
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	-3 - +3EV(Default value: 0)

The module sets the step value same as Capability_EVInterval. When the Capability_ EVInterval is changed, the module sends to the client kMAIDEvent_CapChange.

The following cases, this Capability is read only.

- When the value of Capability_LiveViewStatus is "OFF".
- When the value of Capability_LiveViewSelector is "0: live view photography ".
- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode is setting “Auto”

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.164. MovieRecFrameCount

This will show the number of frames that can take the picture during movie recording.

Capability	kNkMAIDCapability_MovieRecFrameCount
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault
Data	

[Z 7II]

0 – 50 (Default: 50)

[Z 6II, Z 6II_FU1]

0 – 40 (Default: 40)

When the Capability_MovieScreenSize is “22: 3840x2160 60p” or “23: 3840x2160 50p” during movie mode, the value of this Capability will be 0 in Z 7II and Z 6II_FU1.

3.165. RetractableLensWarningStatus

This will get the status of RetractableLensWarning.

Capability	kNkMAIDCapability_RetractableLensWarningStatus
Object types	Source
ulType	kNkMAIDCapType_Boolean
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault
Data	True: Warning about retraction <u>False: Not warning about retraction</u>

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.166. MovieWindNoiseReduction

This will set “Wind noise reduction”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWindNoiseReduction
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWindNoiseReduction <u>0: OFF</u> 1: ON

The following cases, this Capability is read only.

- During movie file recording and the value of Capability_MovieRecMicrophone is “4: Microphone off”.
- When having connected the external microphone.
- When the value of Capability_SpotWBMode is 1(ON).
- The value of Capability_ MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)” or “17 : 1920x1080 24p x5(slow-mo)”,

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.167. MovieRecordingZone

This will set “ Frequency response”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieRecordingZone
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieRecordingZone <u>0: Wide range</u> 1: Vocal range

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording and the value of Capability_MovieRecMicrophone is “4: Microphone off”.
- The value of Capability_ MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”or “17 : 1920x1080 24p x5(slow-mo)”,

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.168. MovieISOControl

This will set “Movie ISO sensitivity setting - Auto ISO control (mode M)”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieISOControl
Object types	Source
ulType	kNkMAIDCapType_Boolean
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	<u>True: On</u> False: Off

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording, and the exposure mode is setting except Manual.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.169. MovieISOAutoHiLimit

This will set “Movie ISO sensitivity setting - Maximum sensitivity”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieISOAutoHiLimit
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieISOAutoHiLimit3

[Z 7II]

eNkMAIDMovieISOAutoHiLimit3	Maximum sensitivity	EVInterval	
		1/3	1/2
0	200	o	o
1	250	o	x
2	280	x	o
3	320	o	x
4	400	o	o
5	500	o	x
6	560	x	o
7	640	o	x
8	800	o	o
9	1000	o	x
10	1100	x	o
11	1250	o	x
12	1600	o	o
13	2000	o	x
14	2200	x	o
15	2500	o	x
16	3200	o	o
17	4000	o	x
18	4500	x	o
19	5000	o	x
20	6400	o	o
21	8000	o	x
22	9000	x	o
23	10000	o	x
24	12800	o	o
30	16000	o	x
31	18000	x	o
32	20000	o	x
<u>33</u>	<u>25600</u>	o	o
25	Hi 0.3	o	x
26	Hi 0.5	x	o
27	Hi 0.7	o	x
28	Hi 1.0	o	o
29	Hi 2.0	o	o

[Z 6II, Z 6II_FU1]

eNkMAIDMovieISOAutoHiLimit3	Maximum sensitivity	EVInterval	
		1/3	1/2
0	200	o	o
1	250	o	x
2	280	x	o
3	320	o	x
4	400	o	o
5	500	o	x
6	560	x	o
7	640	o	x
8	800	o	o
9	1000	o	x
10	1100	x	o
11	1250	o	x
12	1600	o	o
13	2000	o	x
14	2200	x	o
15	2500	o	x
16	3200	o	o
17	4000	o	x
18	4500	x	o
19	5000	o	x
20	6400	o	o
21	8000	o	x
22	9000	x	o
23	10000	o	x
24	12800	o	o
30	16000	o	x
31	18000	x	o
32	20000	o	x
33	25600	o	o
34	32000	o	x
35	36000	x	o
36	40000	o	x
37	51200	o	o
25	Hi-0.3	o	x
26	Hi-0.5	x	o
27	Hi-0.7	o	x
28	Hi-1.0	o	o
29	Hi-2.0	o	o

The value which can be set up by setup of Capability_EVInterval changes.

In the following case, the range of ISO sensitivity for movie can be set is limited from 1600 to 25600 in Z 7II and 1600 to 51200 in Z 6II or Z 6II_FU1.

If the range which can be set is changed, kNkMAIDEvent_CapChange will be issued.

- When the value of Capability_MovieLogOutput is 1(Outputting).

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the value that cannot set is set to this capability according to the EVInterval setting.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.170. MoviePictureControlDataEx2

This will set and get the Movie Picture Control data. (Movie Shooting Menu)

Capability kNkMAIDCapability_MoviePictureControlDataEx2

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Set, kNkMAIDCapOperation_Get
kNkMAIDCapOperation_GetDefault

Data pointer to NkMAIDPicCtrlData structure

```
typedef struct tagNkMAIDPicCtrlData
{
    ULONG ulPicCtrlItem;----- The target Picture Control
    ULONG ulSize;-----The size of Picture Control data (Max: 614 bytes)
    bool bModifiedFlag; -----Modification flag
                                (false: initial registration, true: edit)
    void* pData;----- The pointer of Picture Control data.
} NkMAIDPicCtrlData, FAR* LPNkMAIDPicCtrlData;
```

The Picture control items to be specified in the ulPicCtrlItem, value ranges listed in Capability_MoviePictureControl: to (except "100: Same as photo settings").

The following cases, this Capability is Read Only.

- During movie file recording.

About how to use of this capability and the format of picture control, refer to [Capability PictureContorolDataEx2](#).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • When the following all conditions will be met. - The ulPicCtrlItem is the Custom Picture Control. - The bModifiedFlag is true. - The ulPicCtrlItem and the BasePictureControl of the Picture Control Data do not match. • If the unsupported value has been set to the Picture Control data. • If the invalid Picture Control data is specified.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.171. DeleteMovieCustomPictureControl

This will delete Custom Picture Control. (Movie Shooting Menu)

Capability	kNkMAIDCapability_DeleteMovieCustomPictureControl
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Set
Data	Custom Picture control item

Delete a custom picture control specified by the specified listed in Capability_MoviePictureControl, one of the 1-9 custom picture control area, you run the Set.

During movie file recording, the Capability Set is not allowed.

The following case, this Capability is read only.

- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.172. MovieCCDDataMode

This will set “Image area – Choose image area”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieCCDDataMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDMovieCCDDataMode <u>0: FX Format</u> 1: DX Format

The following cases, this Capability is read only.

[Z 7II, Z 6II_FU1]

- During movie file recording.
- When Capability_SpotWBMode is 1(ON).
- The value of Capability_ MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”, “17 : 1920x1080 24p x5(slow-mo)”, “22: 3840x2160 60p”, “23: 3840x2160 50p”, “18 : 1920x1080 120p” or “19 : 1920x1080 100p”.
- DX lens is attached to the camera.

[Z 6II]

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_ MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”, “17 : 1920x1080 24p x5(slow-mo)”, “18 : 1920x1080 120p” or “19 : 1920x1080 100p”.
- DX lens is attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.173. MovieWBMode

This will set the white balance mode. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBMode
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_PackedString
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	Auto, Incandescent, Fluorescent, Sunny, Shade, Cloudy, Preset1, Preset2, Preset3, Preset4, Preset5, Preset6 Color Temperature Natural <u>Same as photo settings</u>

If the camera is recording a movie, “Same as photo settings” is not included in the enumeration. However, if it is set to the current value, it is included in the enumeration.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.

Result Codes

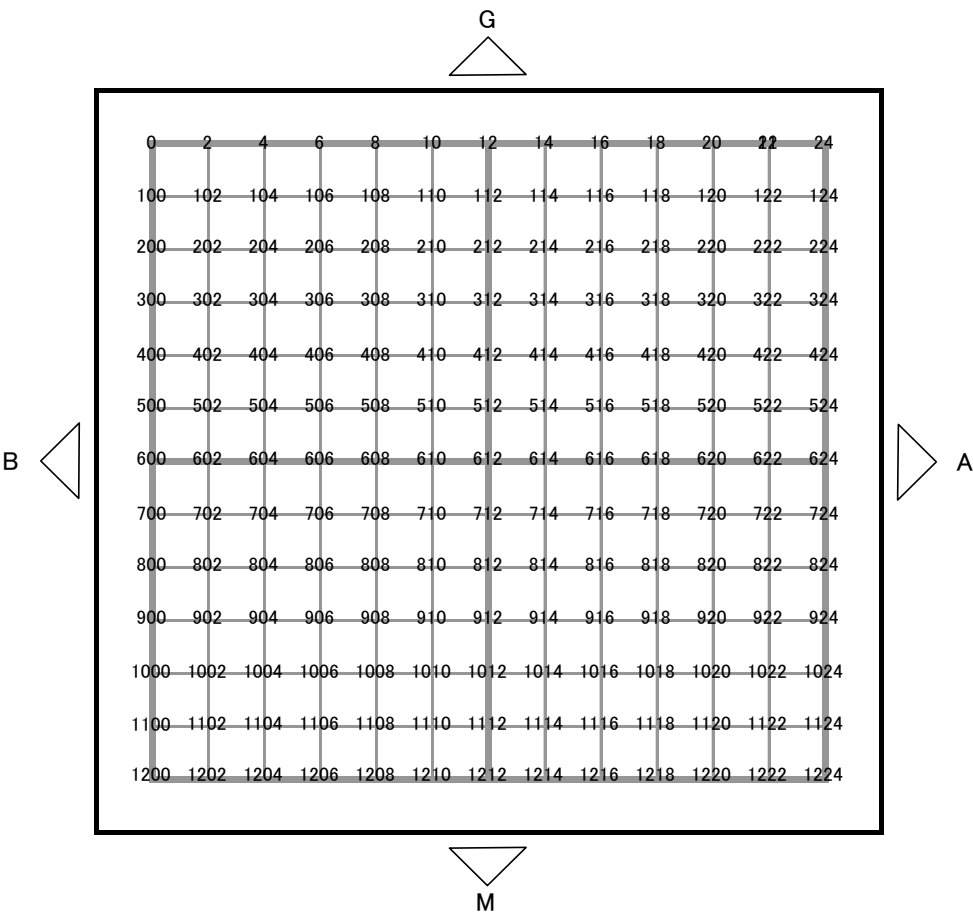
ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.174. **MovieWB TuneAuto**

This will set the white balance adjustment when the WBMode is “Auto”. (Movie Shooting Menu)

Capability kNkMAIDCapability_MovieWB TuneAuto
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data 0 to 1224 step=1 (Default: 612)

Relation to the set of coordinates and camera white balance correction value as in the figure below.
AB direction is 0.5 step increments, and GM direction is 0.25 step increments.



- The following cases, this Capability is read only.
- When Capability_SpotWBMode is 1(ON).
 - During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
 - When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.

DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.175. MovieWBAutoType

This will set the option when the white balance auto. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBAutoType
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMovieWBAutoType 1 : Keep warm lighting colors 2 : Keep white(reduce warm colors) <u>3: Keep atmosphere</u>

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.176. **MovieWB TuneIncandescent**

This will set the white balance adjustment when the WBMode is “Incandescent”.
(Movie Shooting Menu)

Capability kNkMAIDCapability_MovieWB TuneIncandescent
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data 0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.177. **MovieWBFluorescentType**

This will set the fluorescent type when the WBMode is “Fluorescent”.

(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBFluorescentType
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDMovieWBFluorescentType 0: Sodium-vapor lamps 1: Warm-white fluorescent 2: White fluorescent <u>3: Cool-white fluorescent</u> 4: Day white fluorescent 5: Daylight fluorescent 6: High temp. mercury-vapor

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.178. **MovieWB TuneFluorescent**

This will set the white balance adjustment when the WBMode is “Fluorescent”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TuneFluorescent
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.179. MovieWB TuneSunny

This will set the white balance adjustment when the WBMode is “Sunny”.

(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TuneSunny
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.180. **MovieWB TuneShade**

This will set the white balance adjustment when the WBMode is “Shade”.
(Movie Shooting Menu)

Capability kNkMAIDCapability_MovieWB TuneShade
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data 0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

- The following cases, this Capability is read only.
- When Capability_SpotWBMode is 1(ON).
 - During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
 - When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.181. **MovieWB TuneCloudy**

This will set the white balance adjustment when the WBMode is “Cloudy”.
(Movie Shooting Menu)

Capability kNkMAIDCapability_MovieWB TuneCloudy
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data 0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

- The following cases, this Capability is read only.
- When Capability_SpotWBMode is 1(ON).
 - During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
 - When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.182. **MovieWBtuneColorTempEx**

This will set the color temperature when the WBMode is “Color Temperature”.
(Movie Shooting Menu)

Capability kNkMAIDCapability_MovieWBtuneColorTempEx
Object types Source
ulType kNkMAIDCapType_Range
ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data 2500 to 10000 step=10 (Default: 5000)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWBtuneAuto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.183. MovieWB TuneColorAdjust

This will set the white balance adjustment when the WBMode is “Color Temperature”.

(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TuneColorAdjust
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 48 step=1 (Default: 24)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.184. MovieWB TuneNatural

This will set the white balance adjustment when the WBMode is “Natural” (Movie Shooting Menu).

Capability	kNkMAIDCapability_MovieWB TuneNatural
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and the white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- When Capability_SpotWBMode is 1(ON).

- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.185. **MovieWB TunePreset1**

This will set the white balance adjustment when the WBMode is “Preset1”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TunePreset1
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieWb TunePresetProtect1 is set to “ON”.
- The value of Capability_ MovieWBMode is except “Preset1”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.186. **MovieWB TunePreset2**

This will set the white balance adjustment when the WBMode is “Preset2”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TunePreset2
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

- The following cases, this Capability is read only.
- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
 - When Capability_SpotWBMode is 1(ON).
 - The value of Capability_MovieWb TunePresetProtect2 is set to “ON”.
 - The value of Capability_ MovieWBMode is except “Preset2”.
 - When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.187. **MovieWB TunePreset3**

This will set the white balance adjustment when the WBMode is “Preset3”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TunePreset3
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieWb TunePresetProtect3 is set to “ON”.
- The value of Capability_ MovieWBMode is except “Preset3”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.188. **MovieWB TunePreset4**

This will set the white balance adjustment when the WBMode is “Preset4”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TunePreset2
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- During movie file recording, and the value of Capability_ MovieWB Mode is “Same as photo settings”.
- When Capability_SpotWB Mode is 1(ON).
- The value of Capability_MovieWb TunePresetProtect4 is set to “ON”.
- The value of Capability_ MovieWB Mode is except “Preset4”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.189. **MovieWB TunePreset5**

This will set the white balance adjustment when the WBMode is “Preset5”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TunePreset5
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- During movie file recording, and the value of Capability_ MovieWB Mode is “Same as photo settings”.
- When Capability_SpotWB Mode is 1(ON).
- The value of Capability_MovieWb TunePresetProtect5 is set to “ON”.
- The value of Capability_ MovieWB Mode is except “Preset5”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.190. **MovieWB TunePreset6**

This will set the white balance adjustment when the WBMode is “Preset6”.
(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWB TunePreset6
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	0 to 1224 step=1 (Default: 612)

Relationship of setting the coordinates of the camera and white balance correction value is the same as for MovieWB TuneAuto.

The following cases, this Capability is read only.

- During movie file recording, and the value of Capability_ MovieWBMode is “Same as photo settings”.
- When Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieWb TunePresetProtect6 is set to “ON”.
- The value of Capability_ MovieWBMode is except “Preset6”.
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.191. MovieWBPresetProtect1

This will set the protection of white balance preset 1. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBPresetProtect1
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWBPresetProtect1 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.192. MovieWBPresetProtect2

This will set the protection of white balance preset 2. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBPresetProtect2
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWBPresetProtect2 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.193. MovieWBPresetProtect3

This will set the protection of white balance preset 3. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBPresetProtect3
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWBPresetProtect3 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.194. MovieWBPresetsProtect4

This will set the protection of white balance preset 4. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBPresetsProtect4
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWBPresetsProtect4 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.195. MovieWBPresetProtect5

This will set the protection of white balance preset 5. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBPresetProtect5
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWBPresetProtect5 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.196. MovieWBPresetProtect6

This will set the protection of white balance preset 6. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieWBPresetProtect6
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieWBPresetProtect6 <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.197. MovieWBPresetName

This will set the name of white balance preset data. (Movie Shooting Menu)

Capability kNkMAIDCapability_MovieWBPresetName

Object types Source

ulType kNkMAIDCapType_String

 kNkMAIDCapType_Array

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
 kNkMAIDCapOperation_GetArray

Data NkMAIDArray

kNkMAIDCapOperation_GetArray runtime, an array of NkMAIDString, Module to set the name of preset white balance data in the order of Preset1 ~ 6 to NkMAIDArray.pData.

If the string of more than 36 bytes in the name of each preset white balance data is set, it is ignored and subsequent 37 bytes.

Valid characters are only ASCII code of 90 characters.(refer to the ASCII code table).

The (kNkMAIDResult_NotSupported) error when performing Set for preset data is set to: "1" is Capability_MovieWBPresetProtect1 ~ 6.

The following cases, this Capability is read only.

- During movie file recording.
- When Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the client set to the white balance preset data with Capability_MovieWBPresetProtectX setting is On(1).
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.198. MovieResetWBMode

This will reset the white balance adjustment of Capability_MovieWBMode.

Capability kNkMAIDCapability_MovieResetWBMode

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

Data None

By the execution of this Capability, Capability of the following is reset.

- Capability_MovieWBMode
- Capability_MovieWBTuneAuto
- Capability_MovieWBAutoType
- Capability_MovieWBTuneIncandescent
- Capability_MovieWBTuneFluorescent
- Capability_MovieWBFluorescentType
- Capability_MovieWBTuneSunny
- Capability_MovieWBTuneCloudy
- Capability_MovieWBTuneShade
- Capability_MovieWBTuneNatural
- Capability_MovieWBTuneColorTemp
- Capability_MovieWBColorAdjust

The following case, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- During movie file recording.
- When Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.199. **MovieNoiseReductionHighISO**

This will set whether noise reduction is used or not used when high ISO. (Movie Shooting Menu)

Capability kNkMAIDCapability_MovieNoiseReductionHighISO

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,
 kNkMAIDCapOperation_GetDefault

Data one of eNkMAIDMovieNoiseReductionHighISO

 0: OFF

1: ON (Default)

 2: ON (High)

 3: ON (Low)

The following cases, this Capability is read only.

- During movie file recording.
- When Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieLogOutput is 1(Outputting).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.200. MoviePictureControl

This will set Picture Control. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MoviePictureControl
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMoviePictureControl 1: Standard 2: Neutral 3: Vivid 4: Monochrome 5: Portrait 6: Landscape 7: Flat 8: Auto <u>100: Same as photo settings</u> 101: Dream 102: Morning 103: Pop 104: Sunday 105: Somber 106: Dramatic 107: Silence 108: Breached 109: Melancholic 110: Pure 111: Denim 112: Toy 113: Sepia 114: Blue 115: Red 116: Pink 117: Charcoal 118: Graphite 119: Binary 120: Carbon

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Represents the picture control item that is set is currently active.

Custom picture control area are listed all the ones that are not currently registered. Whether the data to the picture control area is registered, it is determined in the "Custom flag" of the picture control data.

If the picture control item that used as the current configuration has been changed, CapChangeValueOnly events for this Capability is issued.

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.
- The value of Capability_MovieLogOutput is 1(Outputting).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the client sends kNkMAIDCapOperation_Set with unused Picture Control.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.201. ISOControlSensitivity

This will get the ISO sensitivity which the camera is controlling.

Capability kNkMAIDCapability_ISOControlSensitivity

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data

[Z 7II]

eNkMAIDISOControlSensitivity7	ISO	eNkMAIDISOControlSensitivity7	ISO
32	Lo1.0	3200	ISO3200
40	Lo0.7	3600	ISO3600
45	Lo0.5	4000	ISO4000
50	Lo0.3	4500	ISO4500
56	ISO56	5000	ISO5000
64	ISO64	5600	ISO5600
72	ISO72	6400	ISO6400
80	ISO80	7200	ISO7200
90	ISO90	8000	ISO8000
100	ISO100	9000	ISO9000
110	ISO110	10000	ISO10000
125	ISO125	11000	ISO11000
140	ISO140	12800	ISO12800
160	ISO160	14400	ISO14400
180	ISO180	16000	ISO16000
200	ISO200	18000	ISO18000
220	ISO220	20000	ISO20000
250	ISO250	22000	ISO22000
280	ISO280	25600	ISO25600
320	ISO320	28800	Hi0.2
360	ISO360	32000	Hi0.3
400	ISO400	36000	Hi0.5
450	ISO450	40000	Hi0.7
500	ISO500	45600	Hi0.8
560	ISO560	51200	Hi1.0
640	ISO640	57600	Hi1.2
720	ISO720	64000	Hi1.3
800	ISO800	72000	Hi1.5
900	ISO900	81200	Hi1.7
1000	ISO1000	91200	Hi1.8
1100	ISO1100	102400	Hi2.0
1250	ISO1250		
1400	ISO1400		
1600	ISO1600		
1800	ISO1800		
2000	ISO2000		
2200	ISO2200		
2500	ISO2500		
2800	ISO2800		

[Z 6II, Z 6II_FU1]

NkMAIDISOControlSensitivity6	ISO	NkMAIDISOControlSensitivity6	ISO
50	Lo1.0	10000	ISO10000
56	Lo0.8	11000	ISO11000
64	Lo0.7	12800	ISO12800
72	Lo0.5	14400	ISO14400

80	Lo0.3	16000	ISO16000
90	Lo0.2	18000	ISO18000
100	ISO100	20000	ISO20000
110	ISO110	22000	ISO22000
125	ISO125	25600	ISO25600
140	ISO140	28800	ISO28800
160	ISO160	32000	ISO32000
180	ISO180	36000	ISO36000
200	ISO200	40000	ISO40000
220	ISO220	45600	ISO45600
250	ISO250	51200	ISO51200
280	ISO280	57600	Hi0.2
320	ISO320	64000	Hi0.3
360	ISO360	72000	Hi0.5
400	ISO400	81200	Hi0.7
450	ISO450	91200	Hi0.8
500	ISO500	102400	Hi1.0
560	ISO560	115000	Hi1.2
640	ISO640	128000	Hi1.3
720	ISO720	144000	Hi1.5
800	ISO800	162000	Hi1.7
900	ISO900	182000	Hi1.8
1000	ISO1000	204800	Hi2.0
1100	ISO1100		
1250	ISO1250		
1400	ISO1400		
1600	ISO1600		
1800	ISO1800		
2000	ISO2000		
2200	ISO2200		
2500	ISO2500		
2800	ISO2800		
3200	ISO3200		
3600	ISO3600		
4000	ISO4000		
4500	ISO4500		
5000	ISO5000		
5600	ISO5600		
6400	ISO6400		
7200	ISO7200		
8000	ISO8000		
9000	ISO9000		

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.202. MovieMeteringMode

This will set “Metering”.(Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieMeteringMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDMovieMeteringMode <u>0: Matrix</u> 1: Center weighted 2: Highlight

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- The exposure mode is setting ”Auto”.
- The value of Capability_LockExposure is True: Lock

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.203. **LiveViewImageSize**

This will set the size for Live View image

Capability	kNkMAIDCapability_LiveViewImageSize
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDLiveViewImageSize 1: QVGA <u>2: VGA</u> 3: XGA

The following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.204. SpotWBMode

This will set the ON / OFF state of waiting to acquire SpotWB.

Capability	kNkMAIDCapability_SpotWBMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDSpotWBMode <u>0: OFF</u> 1: ON

When set to 1 (ON) to start the state of waiting to acquire SpotWB, the value of this Capability set to 0 (OFF), then stop.

This capability can not get dynamically “ON / OFF”, state of waiting to acquire SpotWB.

If acquired by Capability_GetLiveViewImage “SpotWB status” was “3: OFF (acquisition success)”, “State of waiting to acquire SpotWB” will be released automatically by the camera, so this capability must be set to “0: OFF”.

The destination specified in the Capability_MovieWBMode is used for this capability. When the value of Capability_MovieWBMode is "Same as photo settings", the destination that is specified in the Capability_WBMode is used for this capability.

The following cases, this Capability is read only.

- The value of Capability_LockCamera is False:Unlock(PC-camera mode).
- The value of Capability_LiveViewSelector is “0: live view photography”.
- The value of Capability_LiveViewStatus is except “3:ON(Remote live view)”.
- During movie file recording.
- CapabilityThe value of Capability_MovieWBMode are except “Preset1~6” and “same as photo settings”
- When state of “Preset1~6” that set to Capability_MovieWBMode is protected .
- When the value of Capability_MovieWBMode is “same as photo settings”, and the value of Capability_WBMode is except “Preset1~6”.
- When the value of Capability_MovieWBMode is “same as photo settings”, and the state of “Preset1~6” is protected that set to Capability_WBMode.
- CapabilityCapabilityWhen the value of Capability_ExposureMode is one of the following values.
 - 5: Auto
 - 15: U1(UserMode1)
 - 16: U2(UserMode2)

- 18: U3(UserMode3)

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotLiveView	If the remote live view does not be started.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.205. SpotWBMeasure

This capability obtain SpotWB information

Capability kNkMAIDCapability_SpotWBMeasure

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

Data None

This capability acquires SpotWB information for coordinates specified by Capability_ChangeSpotWBArea.

“SpotWB information” result of the acquisition will be notified in “SpotWB status” to get in the Capability_GetLiveViewImage.

Obtaining results of the spot WB information is notified by the display information of the live view data acquired in Capability_GetLiveViewImage.

It is possible to perform the operation of obtaining the WB information again by executing the Capability for release after the Spot WB acquisition result screen, the spot WB state is waiting.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capability_SpotWBMode is “0(OFF)”.
- During movie file recording.

Result Codes

NotLiveView	If the live view does not be started.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.206. SpotWBChangeArea

This Capability is used to change the SpotWB area while waiting for SpotWB acquisition.

Capability kNkMAIDCapability_SpotWBChangeArea

Object types Source

ulType kNkMAIDCapType_Point

ulOperations kNkMAIDCapOperation_Set

Data typedef struct tagNkMAIDPoint
 {
 SLONG x; Coordinates of the X-axis
 SLONG y; Coordinates of the Y-axis
 } NkMAIDPoint, FAR* LPNkMAIDPoint;

This capability set the focus point by using X and Y of NkMAIDPoint structure.

The value range of X and Y is defined by “total size” of “Display information” in Live View image.
(Please refer NkMAIDCapability_GetLiveViewImage)

But the range that can be actually set becomes an area where subtracted the half of "size of the AF frame" length and width from "total size", respectively.

When the value that exceeds the range of x and y is set, the maximum or minimum value will be used as this value.

The following cases, this Capability cannot be set into.

- When the value of Capability_SpotWBMode is “0(OFF)”.
- During movie file recording.

Result Codes

NotLiveView	If live view does not be started.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.207. SpotWBResultDispEnd

This capability releases the display of SpotWB result.

Capability	kNkMAIDCapability_SpotWBResultDispEnd
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start
Data	None

By executing this capability, the value of the "SpotWB status", retrieved by GetLiveViewImage, is modified as follows.

- 3: OFF(Acquisition success) → 0: OFF, or
- 4: Waiting(Acquisition failure) → 1: Waiting(Not acquired)

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capability_LiveViewStatus is except “3(ON(Remote live view))”.
- During movie file recording.

Result Codes

NotLiveView	If live view does not be started.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.208. RawJpegTransferStatus

This capability will set the type of SDRAM image which be notified and transmitted to a client, when the value of Capability_SaveMedia is “2: Card + SDRAM” and the value of Capability_CompressionLevel is “RAW + JPEG”.

Capability	kNkMAIDCapability_RawJpegTransferStatus
Object types	Source
ulType	kNkMAIDCapType_ Boolean
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	True : JPEG only <u>False : RAW & JPEG</u>

The following cases, the ulVisibility of this capability is invalid. (The ulOperations is can set.)

- When the value of Capability_CompressionLevel is except “RAW + JPEG”.
- When the value of Capability_SaveMedia is except “2: Card + SDRAM”.

This capability is always can be set into but setting change of this capability is taken as prohibition until it receives kNkMAIDEvent_CaptureComplete about SDRAM from the execution start of Capability_Capture, Capability_AFCapture, Capability_CaptureDustImage, Capability_CaptureAsync or Capability_AFCaptureAsync.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.209. GetSBHandles

This capability obtains the handle of the flash attached with the camera.

Capability kNkMAIDCapability_GetSBHandles

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Get

Data pointer to NkMAIDGetSBHandles structure

typedef struct tag NkMAIDGetSBHandles

{

ULONG ulSBGroupID;----0 : all flashes, N : one of SBGroupID
(one of [eNkMAIDSBGroupID](#))

ULONG ulNumber;-----The number of SBHandle in the list

ULONG ulSize;----- The size of pData.

void* pData;----- The pointer of [SBHandle](#) list※

} NkMAIDGetSBHandles, FAR* LPNkMAIDGetSBHandles;

※The format of SBHandle list is shown below.

N : The number of SBHandle in the list

Field	Size	DataType
SBHandle 1	4	UINT32
SBHandle 2	4	UINT32
...	4	UINT32
SBHandle N-1	4	UINT32
SBHandle N	4	UINT32

The client gets SBHandle in the following procedure.

Step.1. Get the number of SBHandle in the list.

Step.2. Get the list of SBHandle.

■Step.1. Get the number of SBHandle in the list.

The client must set the following member of NkMAIDGetSBHandles.

- ulSBGroupID : Specify the SBGroupID to be acquisition target.
- ulNumber : Specify 0.

After execution, the following members of NkMAIDGetSBHandles are set.

- ulNumber : The total number of SBHandle to be acquisition target.
- ulSize : The data size for acquisition of SBHandle list.

■Step.2. Get the list of SBHandle.

The client must set the following member of NkMAIDGetSBHandles.

- ulSBGroupID : Specify the SBGroupID to be acquisition target.

- ulNumber : Specify the number of SBHandle to be acquisition target.
- ulSize : Specify the size of pData (The buffer size for the acquisition to SBHandle list.)
- pData : Specify the pointer of SBHandle list.

After execution, the following members of NkMAIDGetSBHandles are set.

- ulSize : The buffer size of SBHandle list, which is set to pData.
- pData : SBHandle list.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.210. GetSBAttrDesc

This capability obtains the descriptor of the attribute for the flash.

Capability	kNkMAIDCapability_GetSBAttrDesc
Object types	Source
ulType	kNkMAIDCapType_Generic
ulOperations	kNkMAIDCapOperation_Get
Data	pointer to NkMAIDGetSBAttrDesc structure <pre>typedef struct tag NkMAIDGetSBAttrDesc { ULONG ulSBHandle;-----0 : all flashes, N : one of SBHandle (eNkMAIDSBHandle, or one of SBHandle) ULONG ulSBAttrID; -----0 : all SBAttrID, N : one of SBAttrID (one of eNkMAIDSBAttribute) ULONG ulSize;----- The size of pData. void* pData;----- The pointer of SBHandleAttributeDescList※ } NkMAIDGetSBAttrDesc, FAR* LPNkMAIDGetSBAttrDesc;</pre>

※The format of SBHandleAttributeDescList is shown at “[SBHandleAttributeDescList](#)”

The client gets the descriptor of the attribute for the flash in the following procedure.

Step.1. Get the size of the descriptor of the attribute for the flash.

Step.2. Get the descriptor of the attribute for the flash.

■Step.1. Get the size of the descriptor of the attribute for the flash.

The client must set the following member of `NkMAIDGetSBAttrDesc`.

- **ulSBHandle** : Specify the SBHandle to be acquisition target.
- **ulSBAttrID** : Specify the SBAttrID to be acquisition target.
- **ulSize** : Specify 0.

After getting value, the following members of `NkMAIDGetSBAttrDesc` are set.

- ulSize : The buffer size of descriptor of the attribute for the flash.

■Step.2. Get the descriptor of the attribute for the flash.

The client must set the following member of `NkMAIDGetSBAttrDesc`.

- **ulSBHandle** : Specify the SBHandle to be acquisition target.
- **ulSBAttrID** : Specify the SBAttrID to be acquisition target.
- **ulSize** : Specify the size of pData (The buffer size for the acquisition to SBHandleAttributeDescList.)
- **pData** : Specify the pointer of SBHandleAttributeDescList.

After getting value, the following members of NkMAIDGetSBAttrDesc are set.

- ulSize : The buffer size of SBHandleAttributeDescList, which is set to pData.
- pData : SBHandleAttributeDescList.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command. When valid SB information cannot be acquired.

3.210.1. SBHandleAttributeDescList

Field	DataSize	DataType or Data Set	Description
NumberOfElements	4	UINT32	The number of SBHandle in the list.
Element1 SBHandle	4	SBHandle	SBHandle for element1.
Element1 SBAttributeDescList	DTS	Refer to SBAttributeDescList	Attribute descriptor of the element1 flash.
Element2 SBHandle	4	SBHandle	SBHandle for element2.
Element2 SBAttributeDescList	DTS	Refer to SBAttributeDescList	Attribute descriptor of the element2 flash.
...			
ElementN SBHandle	4	SBHandle	SBHandle for element N.
ElementN SBAttributeDescList	DTS	Refer to SBAttributeDescList	Attribute descriptor of the element N flash.

3.210.2. SBAttributeDescList

Field	Size (Byte)	DataType or Data Set	Description
NumberOfElements	4	UINT32	The number of the attribute in the list.
Element1 AttributeDesc	DTS	Refer to SBAttributeDesc	SBAttributeDesc for element 1
Element2 AttributeDesc	DTS	Refer to SBAttributeDesc	SBAttributeDesc for element 2
...			
ElementN AttributeDesc	DTS	Refer to SBAttributeDesc	SBAttributeDesc for element N

3.210.3. SBAttributeDesc

Field	Size (Byte)	DataType	Description
AttributeID	2	UINT16	SBAttrID or SBGroupAttrID
DataType	2	UINT16	Data type of the attribute The value varies depending on the attribute. Refer to DataTypeCode
GetSet	1	UINT8	Availability of Get and Set operation. 0x00: Only Get operation is available (Get) 0x01: Get and Set operation are available (Get / Set)
DefaultValue	DTS	DTS	The default value in the camera The value varies depending on the attribute.
CurrentValue	DTS	DTS	The current value in the camera The value varies depending on the attribute.
FormFlag	1	UINT8	Type of the descriptor 0x00: None 0x01: Range 0x02: Enumeration
FORM	DTS	DTS	The value in the field is depending on FormFlagfield. If FormFlag is 0, this field doesn't exist.

3.211. SBAttrValue

This capability obtains or updates the current attribute value for the specified flash.

Capability	kNkMAIDCapability_SBAttrValue
Object types	Source
ulType	kNkMAIDCapType_Generic
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	pointer to NkMAIDSBAttrValue structure <pre>typedef struct tag NkMAIDSBAttrValue { ULONG ulSBHandle;-----0 : all flashes, N : one of SBHandle (eNkMAIDSBHandle, or one of SBHandle) ※Use only get ULONG ulSBAttrID; -----0 : the value for all SBAttribute, N : one of SBAttrID (one of eNkMAIDSBAttribute) ※Use only get ULONG ulSize;----- The size of pData. void* pData;----- The pointer of SBHandleAttributeList※ } NkMAIDSBAttrValue, FAR* LPNkMAIDSBAttrValue;</pre>

※The format of SBHandleAttributeList is shown at “SBHandleAttributeList”

[In case of Get]

The client gets the attribute for the flash in the following procedure.

Step.1. Get the size of the current attribute value for the flash.

Step.2. Get the current attribute value for the flash.

If you want to run the Get after running the Set, it is necessary to always executed from Step 1.

■Step.1. Get the size of the current attribute value for the flash.

The client must set the following member of NkMAIDSBAttrValue.

- **ulSBHandle** : Specify the SBHandle to be acquisition target.
- **ulSBAttrID** : Specify the SBAttrID to be acquisition target.
- **ulSize** : Specify 0.

After getting value, the following members of NkMAIDSBAttrValue are set.

- ulSize : The buffer size of the current attribute value for the flash.

■Step.2. Get the current attribute value for the flash.

The client must set the following member of NkMAIDSBAttrValue.

- `ulSBHandle` : Specify the SBHandle to be acquisition target.
- `ulSBAttrID` : Specify the SBAttrID to be acquisition target.

- ulSize : Specify the size of pData (The buffer size for the acquisition to SBHandleAttributeDescList.)
- pData : Specify the pointer of SBHandleAttributeDescList.

After getting value, the following members of NkMAIDSBAttrValue are set.

- ulSize : The buffer size of SBAttributeList, which is set to pData.
- pData : SBAttributeList.

[In case of Set]

When the client sends kNkMAIDCapOperation_Set, the client must set the following member of NkMAIDSBAttrValue.

- ulSBHandle : Not used in set.
- ulSBAttrID : Not used in set.
- ulSize : The size of SBAttributeList be set to pData. Specify the size of actual data size specified in pData.
- pData : The pointer of SBAttributeList.

The following cases, this Capability cannot be set into.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

InvalidSBAttributeValue	The specified SB attribute value does not supported.
ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command. When valid SB information cannot be acquired.

3.211.1. SBHandleAttributeList

Field	DataSize	Data Type or Data Set	Description
NumberOfElements	4	UINT32	The number of SBHandle in the list.
Element1 SBHandle	4	SBHandle	SBHandle for element1.
Element1 SBAttributeList	DTS	Refer to SBAttributeList	SBAttributeList for element1.
Element2 SBHandle	4	SBHandle	SBHandle for element2.
Element2 SBAttributeList	DTS	Refer to SBAttributeList	SBAttributeList for element2.
...			
ElementN SBHandle	4	SBHandle	SBHandle for element N.
ElementN SBAttributeList	DTS	Refer to SBAttributeList	SBAttributeList for element N.

3.211.2. *SBAAttributeList*

Field	Size (Byte)	DataType or Data Set	Description
NumberOfElements	4	UINT32	The number of the attribute in the list.
Element1 Attribute	DTS	Refer to SBAttribute	SBAAttribute for element 1
Element2 Attribute	DTS	Refer to SBAttribute	SBAAttribute for element 2
...			
ElementN Attribute	DTS	Refer to SBAttribute	SBAAttribute for element N

3.211.3. *SBAAttribute*

Field	Size (Byte)	DataType	Description
AttributeID	2	UINT16	SBAAttrID or SBGroupAttrID
DataType	2	UINT16	Data type of the attribute The value varies depending on the attribute. Refer to DataTypeCode
CurrentValue	DTS	DTS	The current value in the camera The value varies depending on the attribute.

- ulSize : The buffer size of SBGroupAttributeDescList, which is set to pData.
 - pData : SBGroupAttributeDescList.
- * When it is specified eNkMAIDSBGroupAttribute which does not support by the mounted flash, the value may be indefinite.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.212.1. SBGroupAttributeDescList

Field	DataSize	Data Type or Data Set	Description
NumberOfElements	4	UINT32	The number of SBGroupID in the list.
Element1 SBGroupID	4	SBGroupID	SBGroupID for element1.
Element1 SBAttributeDescList	DTS	Refer to SBAttributeDescList	Attribute descriptor for the element1 flash.
Element2 SBGroupID	4	SBGroupID	SBGroupID for element2.
Element2 SBAttributeDescList	DTS	Refer to SBAttributeDescList	Attribute descriptor for the element2 flash.
...			
ElementN SBGroupID	4	SBGroupID	SBGroupID for element N.
ElementN SBAttributeDescList	DTS	Refer to SBAttributeDescList	Attribute descriptor for the element N flash.

3.213. SBGroupAttrValue

This capability obtains or updates the current attribute value for the specified flash group.

Capability	kNkMAIDCapability_SBGroupAttrValue
Object types	Source
ulType	kNkMAIDCapType_Generic
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	pointer to NkMAIDSBGroupAttrValue structure <pre>typedef struct tag NkMAIDSBGroupAttrValue { ULONG ulSBGroupID;-----0 : all flash group, N : one of SBGroupID (one of eNkMAIDSBGroupID) ※Use only get ULONG ulSBGroupAttrID; -----0 : all SBGroupAttrID, N : one of SBGroupAttrID (one of eNkMAIDSBGroupAttribute) ※Use only get ULONG ulSize;----- The size of pData. void* pData;----- The pointer of SBGroupAttributeList※ } NkMAIDSBGroupAttrValue, FAR* LPNkMAIDSBGroupAttrValue;</pre> ※The format of SBGroupAttributeList is shown at “SBGroupAttributeList”

[In case of Get]

The client gets the attribute for the flash group in the following procedure.

- Step.1. Get the size of the current attribute value for the flash group.
- Step.2. Get the current attribute value for the flash group.

If you want to run the Get after running the Set, it is necessary to always executed from Step 1.

- Step.1. Get the size of the current attribute value for the flash group.

The client must set the following member of `NkMAIDSBGroupAttrValue`.

- **ulSBGroupID** : Specify the SBGroupID to be acquisition target.
- **ulSBGroupAttrID** : Specify the SBGroupAttrID to be acquisition target.
- **ulSize** : Specify 0.

After getting value, the following members of `NkMAIDSBGroupAttrValue` are set.

- ulSize : The buffer size of the current attribute value for the flash group.

- Step.2. Get the current attribute value for the flash group.

The client must set the following member of `NkMAIDSBGroupAttrValue`.

- **ulSBGroupID** : Specify the SBGroupID to be acquisition target.
- **ulSBGroupAttrID** : Specify the SBGroupAttrID to be acquisition target.
- **ulSize** : Specify the size of pData (The buffer size for the acquisition to SBGroupIDAttributeDescList.)

- pData : Specify the pointer of SBGroupIDAttributeDescList.

After getting value, the following members of NkMAIDSBGroupAttrValue are set.

- ulSize : The buffer size of SBGroupAttributeList, which is set to pData.
- pData : SBGroupAttributeList.

* When it is specified eNkMAIDSBGroupAttribute which does not support by the mounted flash, the value may be indefinite.

[In case of Set]

When the client sends kNkMAIDCapOperation_Set, the client must set the following member of NkMAIDSBGroupAttrValue.

- ulSBGroupID : Not used in set.
- ulSBGroupAttrID : Not used in set.
- ulSize : The size of SBGroupAttributeList be set to pData. Specify the size of actual data size specified in pData.
- pData : The pointer of SBGroupAttributeList.

The following cases, this Capability cannot be set into.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

InvalidSBAttributeValue	The specified SB attribute value does not supported.
ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.213.1. SBGroupAttributeList

Field	DataSize	Data Type or Data Set	Description
NumberOfElements	4	UINT32	The number of SBGroupID in the list.
Element1 SBGroupID	4	SBGroupID	SBGroupID for element1.
Element1 SBAttributeList	DTS	Refer to SBAttributeList	SBAttributeList for element1.
Element2 SBGroupID	4	SBGroupID	SBGroupID for element2.
Element2 SBAttributeList	DTS	Refer to SBAttributeList	SBAttributeList for element2.
...			
ElementN SBGroupID	4	SBGroupID	SBGroupID for element N.
ElementN SBAttributeList	DTS	Refer to SBAttributeList	SBAttributeList for element N.

3.214. TestFlash

This capability makes the flash fire for testing.

Capability	kNkMAIDCapability_TestFlash
Object types	Source
ulType	kNkMAIDCapType_Generic
ulOperations	kNkMAIDCapOperation_Start
Data	pointer to NkMAIDTestFlash structure typedef struct tagNkMAIDTestFlash { ULONG ulType;----- Flash mode Specify kNkMAIDTestFlash_Test. ULONG ulParam;----- SBHandle for testing 0 : all flashes, N : one of SBHandle } NkMAIDTestFlash, FAR* LPNkMAIDTestFlash;

The following case, this Capability cannot be started.

- The value of Capability_SpotWBMode is 1(ON).

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.215. CaptureAsync

This will take a picture and save the image to specified media.

Capability	kNkMAIDCapability_CaptureAsync
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start

When the client runs this Capability, module returns control without waiting for the completion of the photographing operation. Therefore, for example, it is possible to check the state of the AE lock even during shooting. The result of this Capability is confirmed by Capability_DeviceReady.

When the release mode is set to continuous shooting, refer to [Capability_ShootingMode](#) for the maximum number of shots possible.

When the module prepared to get a main image, the module issues kNkMAIDEvent_AddChild to source object.

If the client executes this capability during live view, live view will be stopped by camera, and the camera take a picture with AF position set on live view without Auto focus.

The taken image will be saved to the destination specified by Capability_SaveMedia.

Bulb exposure shootng can be executed when Capability_ExposureMode is Manual and Capability_ShutterSpeed is bulb. The bulb shooting is started by this capability and terminated by Capability_TerminateCapture.

In the following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

[Z 7II, Z 6II_FU1]

- When the value of Capability_ShutterSpeed is “Time”.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_SpotWBMode is “1(ON)”.
- When the value of Capability_MovieLogOutput is 1(Outputting).
- When the value of Capability_MovieScreenSize is “22 : 3840x2160 60p”, “23 : 3840x2160 50p”, during movie file recording.

[Z 6II]

- When the value of Capability_ShutterSpeed is “Time”.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_SpotWBMode is “1(ON)”.
- When the value of Capability_MovieLogOutput is 1(Outputting).

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
--------------	---

DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media
NoMedia	• If the card is under being formatted • If no card is inserted and Capability_SaveMedia is "card", or "card and SDRAM". • When battery is low.
InvalidMedia	If the card error has occurred on camera and Capability_SaveMedia is "card", or "card and SDRAM".
UnformattedMedia	If the card has unformatted and Capability_SaveMedia is "card", or "card and SDRAM".
MediaReadOnly	If the card is protected.
ApertureFEE	Aperture ring is not minimum.
BatteryDontWork	When battery is low.
ShutterBulb	Under bulb warning
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.216. AFCaptureAsync

This will take a picture after auto focus and save an image to specified media.

Capability kNkMAIDCapability_AFCaptureAsync

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

When the client runs this Capability, module returns control without waiting for the completion of the photographing operation. Therefore, for example, it is possible to check the state of the AE lock even during shooting.

The result of this Capability is confirmed by Capability_DeviceReady.

This will take a picture after auto focus. If the Capability_FocusMode is MF (0) or lens is not attached, the camera does shooting immediately without auto focus.

When auto focus failed, whether taking a picture or returning out of focus error, that is depend on the setting of Capability_FocusMode, Capability_AFsPriority, Capability_AFcPriority., and Capability_LiveViewSelector.

When Capability_LiveViewSelector is "1: movie live view", it always shifts to shooting operation irrespective of the state after AF driving. Since the camera is fixed "priority release".

When the release mode is set to continuous shooting, refer to Capability_ShootingMode for the maximum number of shots possible.

When the module prepared to get a main image, the module issues

kNkMAIDEvent_AddChild to source object.

The taken image will be saved to the destination specified by Capability_SaveMedia.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

[Z 7II, Z 6II_FU1]

- The value of Capability_SpotWBMode is 1(ON).
- When the value of Capability_ShutterSpeed is "Time".
- When the value of Capability_RetractableLensWarningStatus is "True".
- When the value of Capability_MovieLogOutput is 1(Outputting).
- When the value of Capability_MovieScreenSize is "22 : 3840x2160 60p", "23 : 3840x2160 50p", during movie file recording.

[Z 6II]

- When the value of Capability_SpotWBMode is "1(ON)".
- When the value of Capability_ShutterSpeed is "Time".
- When the value of Capability_RetractableLensWarningStatus is "True".
- CapabilityWhen the value of Capability_MovieLogOutput is 1(Outputting).

Result Codes

OutOfFocus	When the following all conditions are met. • AF mode of operation is the focus priority. • The AF operation is out-of-focus, • The value of Capability_LiveViewSelector is “0: live view photography”.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media
NoMedia	• If the card is under being formatted • If no card is inserted and Capability_SaveMedia is "card", or "card and SDRAM". • When battery is low.
InvalidMedia	If the card error has occurred on camera and Capability_SaveMedia is "card", or "card and SDRAM".
UnformattedMedia	If the card has unformatted and Capability_SaveMedia is "card", or "card and SDRAM".
MediaReadOnly	If the card is protected.
ApertureFEE	Aperture ring is not minimum.
BatteryDontWork	When battery is low.
ShutterBulb	Under bulb warning.
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.217. DeviceReady

This will check the status of Capture_CaptureAsync, Capability_AFCaptureAsync, or Capability_PreCaptureAsync.

Capability kNkMAIDCapability_DeviceReady

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

The Capability is used to check the operating status of Capability_CaptureAsync, Capability_AFCaptureAsync, or Capability_PreCaptureAsync.

For example, if the Client wants to check the status of the AE locked during shooting, starts shooting by Capability_CaptureAsync, and gets the state of the AE lock in Capability_LockExposure,

the operating result of Capability_CaptureAsync is obtained by this Capability.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
OutOfFocus	If auto focus failed by Capability_CaptureAsync, Capability_AFCaptureAsync, or Capability_PreCaptureAsync
DeviceBusy	• If the camera is in the Busy state. • During the camera is operating by Capability_CaptureAsync, Capability_AFCaptureAsync, or Capability_PreCaptureAsync
MFDriveEnd	If the focus position reached the end of focus area.
BulbReleaseBusy	During bulb exposure shooting.
CaptureFailure	If the measuring white balance is failed.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.218. ElectronicVR

This will set “Electric VR”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_ElectronicVR
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDElectronicVR <u>0: OFF</u> 1: ON

This capability is used for movie recording only.

The following cases, this Capability is read only.

[Z 7II, Z 6II_FU1]

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”, “22: 3840x2160 60p”, “23: 3840x2160 50p”, “17 : 1920x1080 24p x5(slow-mo)”, “18 : 1920x1080 120p”or “19 : 1920x1080 100p”.

[Z 6II]

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)”, “17 : 1920x1080 24p x5(slow-mo)”, “18 : 1920x1080 120p”or

“19 : 1920x1080 100p”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.219. MovieActive_D_Lighting

This will set “Active D-Lighting”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieActive_D_Lighting
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDMovieActive_D_Lighting

<u>0 : Off</u>
1 : Low
2 : Normal
3 : High
4 : Extra high
5 : Same as photo settings

This capability operates in a setting of Capability_Active_D_Lighting, when it is set to "5 : Same as photo settings ". This capability operates in "2 : Normal ", when the setting of Capability_Active_D_Lighting is "6 : Auto ".

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.
- The value of Capability_MovieLogOutput is 1(Outputting)

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.220. **FlickerReductionSetting**

This will set “Flicker reduction shooting”(Photo Shooting Menu).

Capability	kNkMAIDCapability_FlickerReductionSetting
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault
Data	one of eNkMAIDFlickerReductionSetting

<u>0 : OFF</u>
1 : ON

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_ExposureDelayEx is except “3 : Off”
- The value of Capability_HDRMode is except 0(OFF).
- The value of Capability_LiveViewPhotoShootingMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.221. **SBIIntegrationFlashReady**

This will get the information about the ready state of the external flash displayed in the flash info of the INFO display.

Capability	kNkMAIDCapability_SBIIntegrationFlashReady
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray
Data	one of eNkMAIDSBIIntegrationFlashReady <u>0: Uncharged</u> 1: Ready

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.222. ExternalRecordingControl

This will set “HDMI -advanced setting- External recording control”. (SETUP)

Capability	kNkMAIDCapability_ExternalRecordingControl
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDExternalRecordingControl <u>0: OFF</u> 1: ON

In the following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.223. LiveViewPhotoShootingMode

This will set “Silent photography”. (Photo Shooting Menu)

Capability	kNkMAIDCapability_LiveViewPhotoShootingMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDLiveViewPhotoShootingMode <u>0: OFF</u>

1: ON

In the following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.224. DetectionPeaking

This will set the [focus peaking] – [peaking level] to be used in live view (Custom Setting Menu d10).

Capability	kNkMAIDCapability_DetectionPeaking
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDDetectionPeaking <u>0: OFF</u> 1: Low sensitivity 2: Standard 3: High sensitivity

In the following case, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.225. HighlightBrightness

This will set “Highlight brightness”. (Custom Setting Menu g6)

Capability	kNkMAIDCapability_HighlightBrightness
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDHighlightBrightness 0: 180 1: 191 2: 202 3: 213 4: 224 5: 235 <u>6: 248</u> 7: 255

In the following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.226. MovieAttenuator

This will set “Attenuator”. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieAttenuator
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieAttenuator <u>0: Disable</u> 1: Enable

In the following cases, this Capability is read only.

- During movie file recording and the value of Capability_MovieRecMicrophone is “4: Microphone off”.
- The value of Capability_SpotWBMode is 1(ON).
- The value of Capability_MovieScreenSize is “15 : 1920x1080 30p x4(slow-mo)”, “16 : 1920x1080 25p x4(slow-mo)” or “17 : 1920x1080 24p x5(slow-mo)”,

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.227. SaveCameraSetting

Save camera settings

Capability	kNkMAIDCapability_SaveCameraSetting
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start
Data	None

Operation by this Capability is the same function as the saving the camera setting when the power switch is turned off.

The setting values are stored in the nonvolatile area of the camera. However, if the operation of the camera cannot be completed, the operation of the camera will not be guaranteed thereafter.

After operating this Capability, it's necessary that SourceObject is closed. And the other Capability cannot be operated, and the camera operations other than camera power OFF should not be performed.

The following cases, the Visibility of this capability is invalid and the Operations are invalid.

- During movie file recording.
- When the value of Capability_LockCamera is False:Unlock(PC-camera mode)

Result Codes

NotSupported	• If you run the Operation that it is not supported against Capability. • If the setting of the camera cannot be saved.
DeviceBusy	If the camera is in the Busy state.(included during shooting)
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.228. MovieFileType

This will set movie file type. (Movie Shooting Menu)

Capability	kNkMAIDCapability_MovieFileType
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieFileType <u>0: MOV</u> 1: MP4

In the following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.229. InternalFlashComp

This will set “Frash compensation” in units of 1/6 EV (Photo Shooting Menu).

Capability	kNkMAIDCapability_InternalFlashComp
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	-3.0EV~+1.0EV (Default:0EV)

The unit of the value of this Capability depends on the value of Capability_EVInterval.

In the following cases, this Capability is read only.

- The value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When Capability_HDRMode is 1: On (single photo) or 2: On (series).

- When the exposure mode are setting except P, S, A, and M.
- During the warning of SB setting.
- During acquisition of image dust off data.

Result Codes

ValueOutOfBounds	• If an out-of-range value has been set. • If the value that cannot set is set to this capability according to the EVInterval setting.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.230. GetRecordingInfo

This will get the information of movie recording.

Capability	kNkMAIDCapability_GetRecordingInfo
Object types	Video
ulType	kNkMAIDCapType_Generic
ulOperations	kNkMAIDCapOperation_Get
Data	pointer to NkMAIDGetRecordingInfo structure typedef struct tagNkMAIDGetRecordingInfo { ULONG ulIndexOfMov; --- Index of the target movie file ULONG ulTotalMovCount; --- The number of divided movie files NK_UINT_64 ullTotalMovSize; ---- Total size of all divided movie files } NkMAIDGetRecordingInfo, FAR* LPNkMAIDGetRecordingInfo;

Movie recording information of the target VideoObject is acquired by this Capability.

In the case of the movie file size is over 4GBytes, it is able to get the following information by this capability.

- Index of the target movie file
- The number of divided movie files
- Total size of all divided movie files

Example:

1st file ... 4GB

2nd file ... 2GB

The results that is recorded 6GB movie in total,

* 1st VideoObject

ulIndexOfMov = 0

ulTotalMovCount = 2

ullTotalMovSize = 6442450944 (6GB)

* 2nd VideoObject

ulIndexOfMov = 1

ulTotalMovCount = 2

ullTotalMovSize = 6442450944 (6GB)

If movie file is not divided, ulIndexOfMov = 0, ulTotalMovCount = 1 is returned. Also, movie file size of target VideoObject is returned as ullTotalMovieSize.

If it wants to get the size of each divided movie file, Capability_GetVideoImageEx can be used.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.231. DiffractionCompensation

This will set “Diffraction compensation” of Photo Shooting Menu.

Capability	kNkMAIDCapability_DiffractionCompensation
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDDiffractionCompensation 0: OFF <u>1: ON</u>

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.232. VibrationReduction

This will set “Vibration reduction” of Photo Shooting Menu.

Capability	kNkMAIDCapability_VibrationReduction
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDVibrationReduction

If the lens has Normal and Sport anti-vibration modes.

0: OFF

1: Normal

2: Sport

If the lens only has Normal anti-vibration mode.

0: OFF

1: ON

The value supported by this Capability is different depending on type of lens attached to the camera.

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When the lens attached to the camera has VR switch.
- When Capability_LensType indicate 0(not CPU lens) and Capability_FmmManual is 0(No setting).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.233. MovieAutoDistortion

This will set “Auto distortion control” of Movie Shooting Menu.

Capability	kNkMAIDCapability_MovieAutoDistortion
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieAutoDistortion 0: OFF <u>1: ON</u>

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When Capability_LensType indicates 0x00000080(Always correct distortion)

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.234. MovieAfAreaMode

This will set “AF-area mode” of Movie Shooting Menu.

Capability kNkMAIDCapability_MovieAfAreaMode

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

Menu	String	Capability_MovieFocusMode
Single-point AF	“Single”	MF, AF-S, AF-C, AF-F
<u>Auto-area AF</u>	“Auto”	AF-S, AF-C, AF-F
Wide-area AF(S)	“Wide(S)”	AF-S, AF-C, AF-F
Wide-area AF(L)	“Wide(L)”	AF-S, AF-C, AF-F
Wide-area AF (L-people)	“Wide(L-people)”	AF-S, AF-C, AF-F
Wide-area AF (L-animals)	“Wide(L-animals)”	AF-S, AF-C, AF-F
Auto-area AF (people)	“Auto(people)”	AF-S, AF-C, AF-F
Auto-area AF (animals)	“Auto(animals)”	AF-S, AF-C, AF-F

If AF-area mode can't set or use depending on the value of Capability_MovieFocusMode, this Capability will be set Single-point AF. If AF-area mode can set and use depending on the value of Capability_MovieFocusMode, this Capability will be set the previous value.

The value of this Capability is not the setting value of the menu but the control value.

※Regarding Enum Value: A value included in the enum value can be set as one of menu

value. However it may not be controlled by the state of the camera in the set value. In this case, immediately after setting this capability value, CapChange is issued. The value gotten by kNkMAIDCapOperation_Get becomes the control value.(The value of this Capability may not match the value of the menu.)

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- When Capability_MovieFocusMode is set to MF by user operation.
 1. Change Capability_MovieFocusMode by using module
 2. Change FocusMode from Movie Shooting Menu of the camera
 3. Change FocusMode from i-menu of the camera
during movie live view.

*Except for the CPU lens is not attached or the focus mode controlling is manual focus by the setting of the operation parts of the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.235. MovieVibrationReduction

This will set “Vibration reduction” of Movie Shooting Menu

Capability	kNkMAIDCapability_MovieVibrationReduction
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieVibrationReduction

If the lens has Normal and Sport anti-vibration modes.

0: OFF

1: Normal

2: Sport

100: Same as photo setting

If the lens only has Normal anti-vibration mode.

0: OFF

1: ON

100: Same as photo setting

The value supported by this Capability is different depending on type of lens attached to the camera.

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- When a VR lens with an image stabilization switch is attached to the camera.
- When a VR lens is not attached to the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.236. MovieFocusMode

This will set the focus mode for movie (Movie Shooting Menu).

Capability	kNkMAIDCapability_MovieFocusMode
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieFocusMode 0: AF-S 1: AF-C <u>2: AF-F</u> 4: MF

The value of this Capability is not the setting value of the menu but the control value.

※Regarding Enum Value: A value included in the enum value can be set as one of menu value. However it may not be controlled by the state of the camera in the set value. In this case, immediately after setting this capability value, CapChange is issued. The value gotten

by kNkMAIDCapOperation_Get becomes the control value.(The value of this Capability may not match the value of the menu.)

The following case, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.237. MovieVignetteControl

This will set “Vignette control” of Movie Shooting Menu.

Capability	kNkMAIDCapability_MovieVignetteControl
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieVignetteControl 0: OFF 1: Low <u>2: Normal</u> 3: High 100: Same as photo setting

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.

UnexpectedError	By the unexpected error, if it fails to execute the command.
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3.238. MovieDiffractionCompensation

This will set “Diffraction compensation” of Movie Shooting Menu.

Capability	kNkMAIDCapability_MovieDiffractionCompensation
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieDiffractionCompensation 0: OFF <u>1: ON</u>

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.239. RecordTimeCodes

This will set “Timecode – Record timecodes” of Movie Shooting Menu.

Capability	kNkMAIDCapability_RecordTimeCodes
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDRecordTimeCodes <u>0: OFF</u> 1: ON 2: ON(With HDMI external output)

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When a warning is occurring about time is not set.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.240. CountUpMethod

This will set “Timecode - Count-up method” of Movie Shooting Menu.

Capability	kNkMAIDCapability_CountUpMethod
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDCountUpMethod <u>0: Record run</u> 1: Free run

The following cases, this Capability is read only.

- When the value of Capability_RecordTimeCodes is 0(OFF).
- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When a warning is occurring about time is not set.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.241. TimeCodeOrigin

This will set “Timecode - Timecode origin” of Movie Shooting Menu

Capability	kNkMAIDCapability_TimeCodeOrigin
Object types	Source
ulType	kNkMAIDCapType_Generic
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	pointer to NkMAIDTimeCodeOrigin structure <pre>typedef struct tagNkMAIDTimeCodeOrigin { UCHAR ucTimeCodeInfo; ----- 0 : Reset / 1 : Enter Manually / <u>2 : Current Time</u> UCHAR ucTimeCode[3]; ----- [0]Hour(00-23) (Default : 0) [1] Minute(00-59) (Default : 0) [2] Second(00-59) (Default : 0) ULONG ulFrame; ----- Frame (*) (Default : 0) In the case of 24fps(0-23) In the case of 25/50/100fps(0-24) In the case of 30/60/120fps(0-29) } NkMAIDTimeCodeOrigin, FAR* LPNkMAIDTimeCodeOrigin;</pre>

(*)The range of values that can be set changes depending on the Frame rate of Capability_MovieScreenSize.

ucTimeCode[3] and ulFrame is used only when ucTimeCodeInfo is “1 : Enter Manually”

The following cases, this Capability is read only.

- When the value of Capability_RecordTimeCodes is 0(OFF).
- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When a warning is occurring about time is not set.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.242. DropFrame

This will set “Timecode - Drop frame” of Movie Shooting Menu

Capability	kNkMAIDCapability_DropFrame
-------------------	-----------------------------

Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDDropFrame 0: OFF <u>1: ON</u>

The following cases, this Capability is read only.

- When the value of Capability_RecordTimeCodes is 0(OFF).
- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.
- When a warning is occurring about time is not set.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.243. MovieLogOutput

This will get the state of N-Log/HDR (HLG) control.

Capability	kNkMAIDCapability_MovieLogOutput
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray
Data	one of eNkMAIDMovieLogOutput <u>0: Not outputting</u> 1: Outputting

For the output destination, refer to Capability_MovieLogSetting.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.244. MovieLogSetting

This will set “HDMI - Advanced - N-Log/HDR (HLG) output options ” of Setup Menu

Capability	kNkMAIDCapability_MovieLogSetting
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieLogSetting <u>0: OFF</u> 2: N-Log 3: HDR(HLG)

The following cases, this Capability is read only.

- When the value of Capability_HDMIOutputDataDepth is 0(8bit).
- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.245. HDMIOutputDataDepth

This will set “HDMI - Advanced - Output data depth” of Setup Menu

Capability	kNkMAIDCapability_HDMIOutputDataDepth
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDHDMIOutputDataDepth <u>0: 8bit</u> 1: 10bit

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.246. LowLightAF

This will set the setting of “Low-light AF”. (Custom setting menu a11)

Capability	kNkMAIDCapability_LowLightAF
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDLowLightAF <u>0: OFF</u> 1: ON

The following cases, this Capability is read only.

- When the exposure mode are setting except P, S, A, and M.
- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.247. ApplyLiveViewSetting

This will set the Apply settings to live view (Custom setting menu d8).

Capability	kNkMAIDCapability_ApplyLiveViewSetting
Object types	Source

ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDApplyLiveViewSetting 0: OFF <u>1: ON</u>

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.248. LiveViewImageStatus

This will get the state of the live view image to be acquired with
Capability_GetLiveViewImage.

Capability	kNkMAIDCapability_LiveViewImageStatus
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray
Data	one of eNkMAIDLiveViewImageStatus <u>0: Live view image can't be acquired</u> 1: Live view image can be acquired

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.249. LiveViewZoomArea

This will set the zoom area of live view image.

Capability	kNkMAIDCapability_LiveViewZoomArea
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDLiveViewZoomArea

eNkMAIDLiveViewZoomArea	Display device		Contents
	Tilting monitor	Viewfinder	
0	o	o	full screen
512	o	—	MagnifyRange : 512 x 342 Magnification : High(200%)
640	—	o	MagnifyRange : 640 x 426 Magnification : High(200%)
1024	o	—	MagnifyRange : 1024 x 682 Magnification : actual (100%)
1280	—	o	MagnifyRange : 1280 x 854 Magnification : actual (100%)
2048	o	—	MagnifyRange : 2048 x 1364 Magnification : Low(50%)
2560	—	o	MagnifyRange : 2560 x 1706 Magnification : Low(50%)

※The magnification is the magnification relative to the size of the display device.

The following cases, the values that can be set are restricted, and the enumerated values vary.

- When the value of Capability_SpotWBMode is 1(ON) , values for which the magnification is greater than 100%.
- Values other than 100% during movie file recording.

The following cases, this Capability is read only.

- When the value of Capability_LiveViewStatus is 0(OFF).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	• If you run the Operation that it is not supported against Capability. • If you changed the value of this Capability during Photo Shooting.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.250. TrackingAFArea

During remote live view, this will start tracking operation at the auto area AF mode or change the selected face,

Capability kNkMAIDCapability_TrackingAFArea

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Set

Data pointer to NkMAIDTrackingAFArea structure

```
typedef struct tagNkMAIDTrackingAFArea
```

```
{
```

```
    ULONG    ulTrackingStatus;        -----1: Start tracking / 0: Stop tracking
```

```
    NkMAIDPoint stAfPoint;           ----- The coordinates of the AF area
```

```
                                     (※Used only when ulTrackingStatus = 1)
```

```
} NkMAIDTrackingAFArea, FAR* LPNkMAIDTrackingAFArea;
```

When starting the tracking operation, set ulTrackingStatus to 1 and set the AF area with a focus on the coordinates specified by X (X axis coordinates) and Y (Y choosing axis coordinates) of the NkMAIDPoint structure. (please refer Capability_ContrastAFArea)

The value range of X and Y is defined by “total size” of “Display information” in live view header get by Capability_GetLiveViewImage.

However the range that can be actually set becomes an area where "size of the AF frame" length and breadth size half was subtracted from the length and breadth size of "total size" respectively. When the value over the range that can be set to X and Y, the value changed to maximum or minimum will be set.

When stopping the tracking operation, set to 0 for ulTrackingStatus. And then

NkMAIDPoint structure is not used.

The change of the live view header information is as follows.

Condition		Value of live view header information which changes after this command		
Face detecting	Tracking state	Tracking state	AF frame center coordinates	Index of detected face
There is a detected face at the specified coordinates	Waiting Selecting Tracking	Waiting	-	Index of faces existing at specified coordinates
There is not a detected face at the specified coordinates	Waiting Selecting Tracking	Tracking	Specified coordinates	-

The following cases, this Capability is read only.

- When the value of Capability_LiveViewStatus is except “3(ON(remote live view))”
- When the value of Capability_FocusAreaMode is except "Auto" , "Auto(people)", "Auto(animals)", and live view photography is starting.
- When the value of Capability_MovieFocusAreaMode is except "Auto" , "Auto(people)", "Auto(animals)", and movie live view_ is starting.
- When the value of Capability_SpotWBMode is not “1(ON)”

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
NotLiveView	If live view does not be started.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.251. MovieAfSpeed

This will set “AF speed” (Custom setting menu g4).

Capability	kNkMAIDCapability_MovieAfSpeed
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get,kNkMAIDCapOperation_Set
Data	-5(low speed)~5(high speed)(Default:0)

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.252. MovieAfSpeedWhenToApply

This will set “AF speed - When to apply” (Custom setting menu g4).

Capability	kNkMAIDCapability_MovieAfSpeedWhenToApply
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovieAfSpeedWhenToApply <u>0 : Always active</u> 1 : Active only during recording

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.253. MovieAfTrackingSensitivity

This will set “AF tracking sensitivity” (Custom setting menu g5).

Capability	kNkMAIDCapability_MovieAfTrackingSensitivity
Object types	Source
ulType	kNkMAIDCapType_Range
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
Data	1(sensitive)~7(insensitive) (Default:4)

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.254. PreCapture

This will take a picture for the presetting white balance.

Capability	kNkMAIDCapability_PreCapture
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start
Data	None

This capability is used for white balance for still image.

The following cases, the Visibility of this capability is invalid and the Operations are invalid.

- When the value of Capability_ShutterSpeed is “Time” or greater than 30s.
- When the value of Capability_LiveViewSelector is “1(Movie live view)”.
- When the value of Capability_RetractableLensWarningStatus is "True".
- When the value of Capability_HDRMode is except “Off”.
- When the value of Capability_MovieLogOutput is “1(Outputting)”
- When the exposure mode is other than P, S, A, M.
- When the value of Capability_SpotWBMode is 1 (ON).

Result Codes

CaptureFailure	If the measuring white balance is failed.
NotSupported	• If you run the Operation that it is not supported against Capability. • If the value of Capability_LockCamera is “False”.
DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media.
NoMedia	• Recording card is initializing. • Capability_SaveMedia is "card" or "card and SDRAM"

	and then the recording card does not exist. • When battery is low.
InvalidMedia	Capability_SaveMedia is "card" or "card and SDRAM" and then the card access error has occurred in the camera.
UnformattedMedia	Capability_SaveMedia is "card" or "card and SDRAM" and then the card is not formatted.
MediaReadOnly	If the card is protected.
ApertureFEE	Aperture ring is not minimum.
BatteryDontWork	When battery is low.
ShutterBulb	Under bulb warning
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.255. GetManualSettingLensData

This will get “Non-CPU lens data” of Setup Menu

Capability kNkMAIDCapability_GetManualSettingLensData

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Get

Data pointer to NkMAIDGetManualSettingLensData structure

typedef struct tagGetManualSettingLensData

{

ULONG ulLensID;-----0xFFFFFFFF : All lens setting information,

N : Arbitrary LensID

ULONG ulSize;----- Data size of pData

void* pData;----- The Pointer of ManualSettingLens DataSet※

} NkMAIDGetManualSettingLensData, FAR*

LPNkMAIDGetManualSettingLensData;

※The format of ManualSettingLens DataSet is shown at “ManualSettingLens DataSet”

Get by the following procedure.

1. Get the size to get the current value.
2. Get the current value of lens setting information.

It is necessary to execute from step 1.

- 1. Get the size to get the current value.

Set the following items of the structure.

- ulLensID : Set the Lens ID to be getted

- ulSize : Set to 0

After getting value, the following items are set.

- ulSize : Size for getting current value of LensID

■2. Get the current value of lens setting information.

Set the following items of the structure.

- ulLensID : Set the Lens ID to be getted
- ulSize : Data size of pData
- pData : Data pointer for getting

After getting value, the following items are set.

- pData : ManualSettingLens DataSet

Result Codes

ValueOutOfBounds	If an invalid value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.255.1. ManualSettingLens DataSet

Field		Size	Description
Setting lens No.		4	Refer to ManualSettingLensNo
The number of lens data (N)		4	Get the specified lens data: 1/ Get all lens data: 20
Manual setting lens data [0]	Focal Length	1	Information of No.1 lens data ※ ※In the case of individual acquisition, information on the lens No. specified by ulLensID can be acquired Focal Length : Refer to FmmManual . Maximum aperture : Refer to F0Manual .
	Maximum aperture	1	
	Reserved	2	
Manual setting lens data [1]	Focal Length	1	Information of No.2 lens data
	Maximum aperture	1	
	Reserved	2	
Manual setting lens data [N - 1]	Focal Length	1	Information of No.N lens data
	Maximum aperture	1	
	Reserved	2	

3.256. AFModeAtLiveView

This will set the Focus mode of the Photo Shooting Menu.

Capability kNkMAIDCapability_AFModeAtLiveView

Object types Source

ulType kNkMAIDCapType_Enum
kNkMAIDArrayType_Unsigned

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data one of eNkMAIDAFModeAtLiveView

0: AF·S

1: AF·C

4: MF

The value of this Capability is not the setting value of the menu but the control value.

※Regarding Enum Value: A value included in the enum value can be set as one of menu value. However it may not be controlled by the state of the camera in the set value. In this case, immediately after setting this capability value, CapChange is issued. The value gotten by kNkMAIDCapOperation_Get becomes the control value.(The value of this Capability may not match the value of the menu.)

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.257. ElectronicFrontCurtainShutterEx

This will set “Shutter type”. (Custom Setting Menu d5)

Capability	kNkMAIDCapability_ElectronicFrontCurtainShutterEx
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDElectronicFrontCurtainShutter 0: Mechanical shutter 1: Electronic front-curtain shutter <u>2: Auto</u>

When a lens that can not set a mechanical shutter is attached, 0: Mechanical shutter is not included in the enumeration.

In the following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.
- When the Capability_LiveViewPhotoShootingMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.258. MovieRecDestination

This will set the Destination of the Movie Shooting Menu.

Capability	kNkMAIDCapability_MovieRecDestination
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDMovRecDestination2 <u>0: Slot 1</u> 1: Slot 2

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1 (ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.259. Slot2ImageSaveMode

This will set the RAW primary - JPEG secondary of the Photo Shooting Menu.

Capability	kNkMAIDCapability_Slot2ImageSaveMode
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDSlot2ImageSaveMode <u>0: Overflow</u> 1: Backup 2: RAW Slot 1 - JPEG Slot 2

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1 (ON).
- During movie file recording.
- When the value of Capability_HDRMode is other than “OFF”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.260. PrimarySlot

This will select primary slot selection of Photo shooting menu. (Photo shooting menu)

Capability	kNkMAIDCapability_PrimarySlot
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set
Data	one of eNkMAIDPrimarySlot2 <u>0 : slot1</u> 1 : slot2

In the following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the value of Capability_HDRMode is except “Off”.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.261. ActiveSlot

This will show the record destination slot when the the record destination is card.

Capability	kNkMAIDCapability_ActiveSlot
Object types	Source
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get
Data	one of eNkMAIDActiveSlot <u>0: a card is not inserted</u> 1: Slot 1 2: Slot 2 3: Slot 1 & Slot 2

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.262. ChangeMonitorOffStatus

Request switching the camera monitor forced off status, or show the current camera monitor status.

Capability	kNkMAIDCapability_ChangeMonitorOffStatus
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDChangeMonitorOffStatus <u>0: Off</u> 1: On

If you set the value of this Capability to “1: On” while the monitor is turned on, the monitor will be switched off.

If you set the value of this Capability to “0: Off” while the monitor is turned off, it will either be switched on or stay off depending on the state of the camera.

If you execute kNkMAIDCapOperation_Set on this Capability and kNkMAIDResult_NoError is returned, it indicates that the monitor off state change request has been accepted, but the completion of the operation is not guaranteed.

The completion of the operation can be confirmed by receiving a CapChangeValueOnly event from this Capability.

If you execute kNkMAIDCapOperation_Get on this Capability, the current status of the monitor will be returned. Thus, this Capability does not hold the value specified at the moment of the request.

This Capability applies to the monitors below.

- Rear monitor.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.263. FocusAreaMode

This will set the AF area mode(Photo Shooting Menu).

Capability kNkMAIDCapability_FocusAreaMode

Object types Source

ulType kNkMAIDCapType_Enum

kNkMAIDArrayType_PackedString

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

Data

Menu	String	Capability_AFModeAtLiveView
Dynamic AF	“Dynamic”	AF-C
Single <u>AF</u>	“Single”	MF, AF-S, AF-C
<u>Auto-areaAF</u>	“Auto”	AF-S, AF-C
Pinpoint AF	“Pinpoint”	AF-S
Wide-area AF(S)	“Wide(S)”	AF-S, AF-C
Wide-area AF(L)	“Wide(L)”	AF-S, AF-C
Wide-area AF (L-people)	“Wide(L-people)”	AF-S, AF-C
Wide-area AF (L-animals)	“Wide(L-animals)”	AF-S, AF-C
Auto-area AF (people)	“Auto(people)”	AF-S, AF-C
Auto-area AF (animals)	“Auto(animals)”	AF-S, AF-C

In the case of the AF area mode which can not be set / used depending on the value of Capability_AFModeAtLiveView, this capability will become Single AF.

The value of this Capability is not the setting value of the menu but the control value.

※Regarding Enum Value: A value included in the enum value can be set as one of menu value. However it may not be controlled by the state of the camera in the set value. In this case, immediately after setting this capability value, CapChange is issued. The value gotten by kNkMAIDCapOperation_Get becomes the control value.(The value of this Capability may not match the value of the menu.)

The following cases, this Capability is read only

- When the value of Capability_SpotWBMode is 1 (ON).
- During movie file recording.
- When Capability_AFModeAtLiveView is set to MF by user operation.
 1. When changing Capability_AFModeAtLiveView using Module.
 2. When changing from Photo Shooting Menu to FocusMode.
 3. When changing from imenu to Focus Mode during live view photography.

*Except for the CPU lens is not attached or the focus mode controlling is manual focus by the setting of the operation parts of the camera.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability. If the CPU lens does not attached.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.264. RemainingExposureTime

This will get the remaining exposure time during still image shooting.

Capability kNkMAIDCapability_RemainingExposureTime

Object types Source

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

Data 0 ~ 900 (sec) (Default : 0)

If you perform a capture when the exposure mode is M and the value of Custom setting menu [d. Shootig/display] – [d6. Extended shutter speeds (M)] is “On”, a CapChangeValueOnly event is issued only once. Otherwise, even if the value of this Capability is modified, no event is issued.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.265. ExtendedShutterSpeedsManual

This will set the value of Custom setting menu [d. Shooting/display] – [d6. Extended shutter speeds (M)].

Capability	kNkMAIDCapability_ExtendedShutterSpeedsManual
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDExtendedShutterSpeedsManual <u>0: Off</u> 1: On

The following cases, this Capability is read only.

- During movie file recording.
- When the value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.266. AutoFlashIsoSensitivityControl

This will set “Auto flash ISO sensitivity control”. (Custom setting menu e4)

Capability	kNkMAIDCapability_AutoFlashIsoSensitivityControl
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDAutoFlashIsoSensitivityControl <u>0: Subject and background</u> 1: Subject only

The following cases, this Capability is read only.

- During movie file recording.
- The value of Capability_SpotWBMode is 1(ON).
- When the exposure mode are setting except P, S, A, and M.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.267. StillCaptureModeSaveFrame

This will set the value of “Release mode (save frame)” from the Movie shooting menu.

Capability	kNkMAIDCapability_StillCaptureModeSaveFrame
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDStillCaptureModeSaveFrame <u>1: Single frame</u> 2: Continuous

When the release mode is set to continuous shooting, refer to [Capability_ShootingMode](#) for the maximum number of shots possible.

The following cases, this Capability is read only.

- When the value of Capability_MovieReleaseButton is “1: Record movies”, during movie live view.
- During movie recording.
- When the value of Capability_SpotWBMode is 1(ON).

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.268. InitiatePixelMapping

This will start pixel mapping.

Capability kNkMAIDCapability_InitiatePixelMapping

Object types Source

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

If you execute this Capability and kNkMAIDResult_NoError is returned, it indicates that the pixel mapping has successfully started.

The completion of the pixel mapping is notified by a PixelMappingComplete event.

The following cases, the ulVisibility and ulOperations of this capability become invalid.

- During movie file recording.
- When sequence error has occurred.
- When battery is low.

Result Codes

NotSupported	• If you run the Operation that it is not supported against Capability. • If the camera temperature exceeds a certain (specified) value. • If a wireless remote controller is connected.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.269. PreCaptureAsync

This will take a picture for white balance presetting.

Capability	kNkMAIDCapability_PreCaptureAsync
Object types	Source
ulType	kNkMAIDCapType_Process
ulOperations	kNkMAIDCapOperation_Start
Data	None

When executing this Capability, the module returns control to the client without waiting for the shooting to complete. Therefore, it is for example possible to get the value of another Capability even during shooting. The result of this Capability is confirmed by Capability_DeviceReady.

This Capability is used only for white balance for still image shooting.

The following cases, the Visibility of this Capability is invalid and the Operations are invalid.

- When the value of Capability_ShutterSpeed is "Time".
- When the value of Capability_LiveViewSelector is "1: Movie live view".
- When the value of Capability_RetractableLensWarningStatus is "True".
- When the value of Capability_HDRMode is other than "0: Off".
- When the exposure mode are setting except P, S, A, and M.
- The value of Capability_SpotWBMode is 1 (ON).

Result Codes

CaptureFailure	If the measuring white balance is failed.
NotSupported	• If you run the Operation that it is not supported against Capability. • If the value of Capability_LockCamera is "False".
DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media.
NoMedia	• Recording card is initializing. • Capability_SaveMedia is "card" or "card and SDRAM" and then the recording card does not exist. • When battery is low.
InvalidMedia	Capability_SaveMedia is "card" or "card and SDRAM" and then the card access error has occurred in the camera.
UnformattedMedia	Capability_SaveMedia is "card" or "card and SDRAM" and then the card is not formatted.
MediaReadOnly	If the card is protected.

MirrorUpCatpure_A lready_Start	If the mirror is up due to camera operation.
ApertureFEE	Aperture ring is not minimum.
BatteryDontWork	When battery is low.
ShutterBulb	Under bulb warning
UnexpectedError	By the unexpected error, if it fails to execute the command.

3.270. LiveViewImageCompression

This will set the image quality of the live view image acquired in Capability_GetLiveViewImage.

Capability	kNkMAIDCapability_LiveViewImageCompression
Object types	Source
ulType	kNkMAIDArrayType_Unsigned kNkMAIDCapType_Enum
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set
Data	one of eNkMAIDLIVEViewImageCompression <u>0: BASIC (Size priority)</u> 1: BASIC (Image quality priority) 2: NORMAL (Size priority) 3: NORMAL (Image quality priority) 4: FINE (Size priority) 5: FINE (Image quality priority)

When the camera is connected and it's possible to set this Capability, the Module sets “5: FINE (Image quality priority)” to this Capability.

The following cases, this Capability is read only.

- When the value of Capability_SpotWBMode is 1(ON).
- During movie file recording.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

4. SB definition

4.1. SBHandle

The SBHandle is used for identifying the flashes (the internal flash, external flash or other flash units).

The SBHandle takes an unsigned 32-bit value, and the unique number is assigned to each flash.

4.1.1. eNkMAIDSBHandle

eNkMAIDSBHandle	ID	Description
kNkMAIDSBHandle_ALL	0	All flashes

4.2. eNkMAIDSBGroupID

eNkMAIDSBGroupID	ID	Description
kNkMAIDSBGroupID_ALL	0	All flash group
kNkMAIDSBGroupID_Master	0x00000001	Master flash
kNkMAIDSBGroupID_A	0x00000002	Group A
kNkMAIDSBGroupID_B	0x00000004	Group B
kNkMAIDSBGroupID_C	0x00000008	Group C
kNkMAIDSBGroupID_D	0x00000010	Group D
kNkMAIDSBGroupID_E	0x00000020	Group E
kNkMAIDSBGroupID_F	0x00000040	Group F

4.3. eNkMAIDSBAttribute

eNkMAIDSBAttribute	ID	Description
kNkMAIDSBAttribute_ALL	0	All SBAttrID
kNkMAIDSBAttribute_Name	0x1001	This indicates the string to express the name or the nickname of the remote flash.
kNkMAIDSBAttribute_GroupID	0x1002	This indicates the SBGroupID of the flash.
kNkMAIDSBAttribute_Status	0x1003	This indicates the ready state of the remote flash.
kNkMAIDSBAttribute_TestFlashDisable	0x1005	This indicates the disabled state for the test firing of the flash.

4.4. eNkMAIDSBGroupAttribute

eNkMAIDSBGroupAttribute	ID	Description
kNkMAIDSBGroupAttribute_ALL	0	All SBGroupAttrID
kNkMAIDSBGroupAttribute_FlashMode	0x4001	This indicates the flash mode of the flash group.
kNkMAIDSBGroupAttribute_FlashCompensation	0x4002	This indicates the flash group compensation value in units of 1/6 EV.
kNkMAIDSBGroupAttribute_FlashRatio	0x4003	This indicates the ratio of the flash group.
kNkMAIDSBGroupAttribute_FlashLevel	0x4004	This indicates the flash level of the flash group.
kNkMAIDSBGroupAttribute_FlashRange	0x4005	This indicates the distance of the flash group.
kNkMAIDSBGroupAttribute_Repeat	0x4006	This indicates the repeat setting of the flash group.
kNkMAIDSBGroupAttribute_RepeatCount	0x4007	This indicates the firing count of the flash group.
kNkMAIDSBGroupAttribute_RepeatInterval	0x4008	This indicates the repeat interval of the flash group.
kNkMAIDSBGroupAttribute_Invalid	0x4009	This indicates the valid state of the flash group.

5. Standard Capabilities

5.1. AsyncRate

Capability	kNkMAIDCapability_AsyncRate
Object types	Module
ulType	kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get

5.2. ProgressProc

Capability	kNkMAIDCapability_ProgressProc
Object types	Source, Image, Thumbnail
ulType	kNkMAIDCapType_Callback
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

5.3. EventProc

Capability	kNkMAIDCapability_EventProc
Object types	Module, Source, Item, Image, Thumbnail
ulType	kNkMAIDCapType_Callback
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

5.4. DataProc

Capability	kNkMAIDCapability_DataProc
Object types	Image, Thumbnail
ulType	kNkMAIDCapType_Callback
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

5.5. UIRequestProc

Capability	kNkMAIDCapability_UIRequestProc
Object types	Module
ulType	kNkMAIDCapType_Callback
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

5.6. IsAlive

Capability	kNkMAIDCapability_IsAlive
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Object types	Module, Source, Item, Image, Thumbnail,
ulType	kNkMAIDCapType_Boolean
ulOperations	kNkMAIDCapOperation_Get

5.7. Children

Capability	kNkMAIDCapability_Children
Object types	Module, Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray

5.8. State

Capability	kNkMAIDCapability_State
	Not supported

5.9. Name

Capability	kNkMAIDCapability_Name
Object types	Module, Source, Item, Image, Thumbnail, Video
ulType	kNkMAIDCapType_String
ulOperations	kNkMAIDCapOperation_Get

The image saved on SDRAM is taken a picture on the mode Capability_SaveMedia is "1:SDRAM", the value of this capability about Item, Image, Thumbnail is "DSC_0000.xxx".

The image saved on SDRAM is taken a picture on the mode Capability_SaveMedia is "2:Card + SDRAM", the value of this capability about Item, Image, Thumbnail is "folder name¥filename.xxx". However, when the image doesn't exist the on the card (For the reasons card was not inserted), "DSC_0000.xxx" is used.

5.10. Description

Capability	kNkMAIDCapability_Description
	Not supported

5.11. Interface

Capability	kNkMAIDCapability_Interface
Object types	Source
ulType	kNkMAIDCapType_String
ulOperations	kNkMAIDCapOperation_Get

5.12. DataTypes

Capability	kNkMAIDCapability_DataTypes
Object types	Source, Item
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get

5.13. DateTime

Capability	kNkMAIDCapability_DateTime
Object types	Item
ulType	kNkMAIDCapType_DateTime
ulOperations	kNkMAIDCapOperation_Get

5.14. StoredBytes

Capability	kNkMAIDCapability_StoredBytes
Object types	Item, Image, Thumbnail
ulType	kNkMAIDCapType_Unsigned
ulOperations	kNkMAIDCapOperation_Get

5.15. Eject

Capability	kNkMAIDCapability_Eject
	Not supported

5.16. Feed

Capability	kNkMAIDCapability_Feed
	Not supported

5.17. Capture

This will take a picture and save the image to specified media.

Capability kNkMAIDCapability_Capture
Object types Source
ulType kNkMAIDCapType_Process
ulOperations kNkMAIDCapOperation_Start

When the release mode is set to continuous shooting, refer to [Capability_ShootingMode](#) for the maximum number of shots possible.

When the module prepared to get a main image, the module issues kNkMAIDEvent_AddChild to source object.

If the client executes this capability during live view, live view will be stopped by the camera, and the camera will take a picture with AF position set on live view without Auto focus.

The taken image will be saved to the destination specified by Capability_SaveMedia. When there is no free space in the specified media, kMAIDResult_MediaFull will return.

Bulb exposure shooting can be executed when Capability_ExposureMode is Manual and Capability_ShutterSpeed is bulb.

The bulb shooting is started by this capability and terminated by Capability_TerminateCapture.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.
[Z 7II, Z 6II_FU1]

- When the value of Capability_ShutterSpeed is “Time” or greater than 30s.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_SpotWBMode is “1(ON)”.
- When the value of Capability_MovieLogOutput is “1(Outputting)”.
- When the value of Capability_MovieScreenSize is “22 : 3840x2160 60p”, “23 : 3840x2160 50p”, during movie file recording.

[Z 6II]

- When the value of Capability_ShutterSpeed is “Time” or greater than 30s.
- When the value of Capability_RetractableLensWarningStatus is “True”.
- When the value of Capability_SpotWBMode is “1(ON)”.
- When the value of Capability_MovieLogOutput is “1(Outputting)”.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
HardwareError	The error occurs on camera body.
MediaFull	If there is not free space in specified media
NoMedia	• If the card is under being formatted

	• If no card is inserted and Capability_SaveMedia is "card", or "card and SDRAM". • When battery is low.
InvalidMedia	If the card error has occurred on camera and Capability_SaveMedia is "card", or "card and SDRAM".
UnformattedMedia	If the card has unformatted and Capability_SaveMedia is "card", or "card and SDRAM".
MediaReadOnly	If the card is protected.
ApertureFEE	Aperture ring is not minimum.
BatteryDontWork	When battery is low.
ShutterBulb	Under bulb warning
MountAdapterFWVer	If the attached mount adapter needs a version upgrade.
UnexpectedError	By the unexpected error, if it fails to execute the command.

5.18. Mode

Capability kNkMAIDCapability_Mode

Not supported

5.19. Acquire

Capability kNkMAIDCapability_Acquire

Object types Image, Thumbnail

ulType kNkMAIDCapType_Process

ulOperations kNkMAIDCapOperation_Start

5.20. Start

Capability kNkMAIDCapability_Start

Not supported

5.21. Length

Capability kNkMAIDCapability_Length

Not supported

5.22. SampleRate

Capability kNkMAIDCapability_SampleRate

Not supported

5.23. Stereo

Capability kNkMAIDCapability_Stereo
Not supported

5.24. Samples

Capability kNkMAIDCapability_Samples
Not supported

5.25. Filter

Capability kNkMAIDCapability_Filter
Not supported

5.26. Prescan

Capability kNkMAIDCapability_Prescan
Not supported

5.27. AutoFocus

Capability kNkMAIDCapability_AutoFocus
Not supported

5.28. AutoFocusPt

Capability kNkMAIDCapability_AutoFocusPt
Not supported

5.29. Focus

Capability kNkMAIDCapability_Focus
Not supported

5.30. Coords

Capability kNkMAIDCapability_Coords
Not supported

5.31. Resolution

Capability kNkMAIDCapability_Resolution
Not supported

5.32. Preview

Capability kNkMAIDCapability_Preview
Not supported

5.33. Negative

Capability kNkMAIDCapability_Negative
Not supported

5.34. Bits

Capability kNkMAIDCapability_Bits
Not supported

5.35. Planar

Capability kNkMAIDCapability_Planar
Not supported

5.36. Lut

Capability kNkMAIDCapability_Lut

Not supported

5.37. Transparency

Capability kNkMAIDCapability_Transparency

Not supported

5.38. Threshold

Capability kNkMAIDCapability_Threshold

Not supported

5.39. Pixels

Capability kNkMAIDCapability_Pixels

Object types Image, Thumbnail

ulType kNkMAIDCapType_Size

ulOperations kNkMAIDCapOperation_Get

5.40. ForceScan

Capability kNkMAIDCapability_ForceScan

Not supported

5.41. ForcePrescan

Capability kNkMAIDCapability_ForcePrescan

Not supported

5.42. ForceAutoFocus

Capability kNkMAIDCapability_ForceAutoFocus

Not supported

5.43. NegativeDefault

Capability kNkMAIDCapability_NegativeDefault

Not supported

5.44. Firmware

Capability kNkMAIDCapability_Firmware

Not supported

5.45. CommunicationLevel1

Capability kNkMAIDCapability_CommunicationLevel1

Not supported

5.46. CommunicationLevel2

Capability kNkMAIDCapability_CommunicationLevel2

Not supported

5.47. BatteryLevel

Capability kNkMAIDCapability_BatteryLevel

Object types Source

ulType kNkMAIDCapType_Integer

ulOperations kNkMAIDCapOperation_Get

Data 1, 20, 40, 60, 80, 100

This will show the remain of battery by percent.

The camera returns the 6 kind of value, 1, 20, 40, 60, 80 and 100.

When the value of this capability is 1, the current camera status will be set to the prohibition of taking a picture, and the value of Capability_LiveViewProhibit is set to 0x00000100 ("battery shortage").

When the external power supply is used, this capability returns -1.

Result Codes

NotSupported	If you run the Operation that it is not supported against Capability.
UnexpectedError	By the unexpected error, if it fails to execute the command.

5.48. FreeBytes

Not supported

5.49. FreeItems

Not supported

5.50. Remove

Not supported

5.51. FlashMode

Capability	kNkMAIDCapability_FlashMode
Object types	Source
ulType	kNkMAIDCapType_Enum kNkMAIDArrayType_Unsigned
ulOperations	kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set,
Data	one of eNkMAIDFlashMode, eNkMAIDFlashModeDX2 6: Fill flash 7: Rear-curtain sync 8: Slow sync 9: Red-eye reduction 10: Slow sync + red-eye reduction 262: Flash Off

Flash mode ExposureMode SceneMode	Fill flash	Slow sync	Rear-curtain sync()	Red-eye reduction	Slow sync + red-eye reduction	Flash Off
P/A	Fill flash	Slow sync	Rear-curtain sync	Red-eye reduction	Slow sync + red-eye reduction	Flash Off
S,M	Fill flash	—	Rear-curtain sync	Red-eye reduction	—	Flash Off
AUTO	Fill flash	—	—	Red-eye reduction	—	Flash Off
U1/U2/U3	Follow the shooting mode set for the user mode.					

①The display state of the camera differs depending on the value of Capability_ExposureMode. It also differs depending on external flash light.

When Capability_ExternalNewTypeFlashMode is 7 (“Repeating flash”) and Capability_ExposureMode is P,S,A,M, “Rear-curtain sync” cannot be set and 0 “Fill flash” will be set.

In the following cases, the value of this Capability is fixed to "262: flash off" and other values are not included in enumerated values.

- When the value of Capability_SpotWBMode is 1(ON).
- When the value of Capability_HDRMode is except 0(OFF).
- When the value of Capability_LiveViewPhotoShootingMode is 1(ON).
- During movie live view.
- During the warning of SB setting.
- During acquisition of image dust off data.

Result Codes

ValueOutOfBounds	If an out-of-range value has been set.
NotSupported	If you run the Operation that it is not supported against Capability.
DeviceBusy	If the camera is in the Busy state.
UnexpectedError	By the unexpected error, if it fails to execute the command.

5.52. ModuleType

Capability kNkMAIDCapability_ModuleType

Object types Module

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

5.53. AcquireStreamStart

Capability kNkMAIDCapability_AcquireStreamStart

Not supported

5.54. AcquireStreamStop

Capability kNkMAIDCapability_AcquireStreamStop

Not supported

5.55. AcceptDiskAcquisition

Capability kNkMAIDCapability_AcceptDiskAcquisition

Object types Source

ulType kNkMAIDCapType_Generic

ulOperations kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

5.56. Version

Capability kNkMAIDCapability_Version

Object types Module

ulType kNkMAIDCapType_Unsigned

ulOperations kNkMAIDCapOperation_Get

5.57. FilmFormat

Capability kNkMAIDCapability_FilmFormat

Not supported

5.58. TotalBytes

Capability kNkMAIDCapability_TotalBytes

Not supported

6. Event

The client can't receive the event as follows while opening item object.

6.1. AddChild

This event will be issued when the child is added under an object.

Event kNkMAIDEvent_AddChild

Object types Module, Source, Item

data parameter Added child ID

When the added child is Item Object, Item ID will be set to the data parameter of call back function.

This event is issued only about the image preserved in SDRAM. The image preserved on the card is not issued.

6.2. RemoveChild

This event will be issued when the child is removed under an object.

Event kNkMAIDEvent_RemoveChild

Object types Module, Source, Item

data parameter Removed child ID

6.3. WarmingUp

Event kNkMAIDEvent_WarmingUp

Not supported

6.4. WarmedUp

Event kNkMAIDEvent_WarmedUp

Not supported

6.5. CapChange

This event will be issued when the information of Capability is changed.

Event kNkMAIDEvent_CapChange

Object types Module, Source, Item

data parameter Capability ID

In the following cases, this event will be issued.

- When the information of structure "NkMAIDCapInfo" of capability was changed.
- When the enum configuration of Capability with type of kNkMAIDCapType_Array has been changed

※When there is a change for Capability_FmmManual, Capability_F0Manual, Capability_ManualSetLensNo, this event is not issued and Event_ManualSettingLensDataChanged will be issue

6.6. OrphanedChildren

Event kNkMAIDEvent_OrphanedChildren
Not supported

6.7. CapChangeValueOnly

This event will be issued when the current value of capability is changed.

Event kNkMAIDEvent_CapChangeValueOnly

Object types Module, Source, Item, Data

data parameter Capability ID

This event will be issued when only the current value of capability is changed (the array data, ulVisibility, ulOperations is not changed).

※When there is a change for Capability_FmmManual, Capability_F0Manual, Capability_ManualSetLensNo, this event is not issued and Event_ManualSettingLensDataChanged will be issued.

6.8. CaptureComplete

This will be issued when the acquisition or deletion of all images of which it takes a picture is completed.

Event kNkMAIDEvent_CaptureComplete

Object types Source

data parameter 0 : The all SDRAM images by Capture, AFCapture, CaptureAsync, AFCaptureAsync, CaptureDustImage are finished to read or deleted.
1: The all images by Capture, AFCapture, CaptureAsync, AFCaptureAsync, CaptureDustImage are finished to record in card.

It shows that the all images are finished to record in card or the all SDRAM images were finished to read or deleted.

This event is not issued in case of shooting by shutter-release button.

This event will issue per issuing Capture, or AFCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync.

6.9. AddChildInCard

This event will be issued when the child is added in card.

Event kNkMAIDEvent_AddChildInCard

Object types Item

data parameter Added child ID

When the added child is Item Object, Item ID will be set to the data parameter of call back function.

This event is issued only about the movie data preserved on the card. The still image data preserved on the card is not issued.

Depending on the format of the recording device to be saved, this event is issued multiple times by one movie recording. It is issued at least once every 4GB.

6.10. RecordingInterrupted

This event will be issued the factor of discontinue when the recording of movie was stopped with error occurring.

Event kNkMAIDEvent_RecordingInterrupted

Object types Source

data parameter pointer to NkMAIDEventParam

The contents of the NkMAIDEventParam structure are as follows.

NkMAIDEventParam.ulEvent = kNkMAIDEvent_RecordingInterrupted;

NkMAIDEventParam.ulElements = 2;

NkMAIDEventParam.ulParam[0] : Error type

1: Some error.

0: Low access speed card error.

NkMAIDEventParam.ulParam[1] : Recording device

0: Recording to card

1: Recording to external device

2: Recording to both card and external device

When the movie recording is stopped by interruptions of camera factor, this event is issued after kNkMAIDEvent_AddChildInCard event for all video files.

When movie files are divided by one movie recording, the end of the movie can be known at this event. It can be determined that all kNkMAIDEvent_AddChildInCard events issued between the stop of recording movie and issue of this event are movie data generated for one movie recording.

6.11. CapChangeOperationOnly

Operation value or visibility of Capability has changed.

Event kNkMAIDEvent_CapChangeOperationOnly

Object types Module, Source, Item, Data

dataparamater Capability ID

Events to notify (the number of elements in the array, enumeration value configuration, value, etc. has not changed) if only Operation or visibility · Invalid attribute of Capability has changed.

6.12. SBAdded

It is used to inform the flash unit is newly attached to the camera.

Event kNkMAIDEvent_SBAdded

Object types Source

dataparamater pointer to NkMAIDEventParam

The contents of NkMAIDEventParam structure, becomes as follows.

NkMAIDEventParam.ulEvent = kNkMAIDEvent_SBAdded

NkMAIDEventParam.ulElements= 1

NkMAIDEventParam.ulParam[0] = SBHandle

6.13. SBRemoved

It is used to inform the flash unit is detached from the camera.

Event kNkMAIDEvent_SBRemoved

Object types Source

dataparamater pointer to NkMAIDEventParam

The contents of NkMAIDEventParam structure, becomes as follows.

NkMAIDEventParam.ulEvent = kNkMAIDEvent_SBRemoved;

NkMAIDEventParam.ulElements= 1;

NkMAIDEventParam.ulParam[0] = SBHandle;

NkMAIDEventParam.ulParam[0] = SBHandle;

6.14. SBAttrChanged

It is used to inform the attributes of the flash unit have changed.

This event is not issued when the attributes of the flash have changed by receiving Capability_SBAttrValue command from the host.

Event kNkMAIDEvent_SBAttrChanged

Object types Source

dataparamater pointer to NkMAIDEventParam

The contents of NkMAIDEventParam structure, becomes as follows.

NkMAIDEventParam.ulEvent = kNkMAIDEvent_SBAttrChanged;

NkMAIDEventParam.ulElements= 2;

NkMAIDEventParam.ulParam[0] = SBHandle;

NkMAIDEventParam.ulParam[1] = SBAttrID;

6.15. SBGroupAttrChanged

It is used to inform the attribute descriptor for the flash group or the current value has changed.

This event is not issued when the attributes of the flash have changed by receiving Capability_SBGroupAttrValue command from the host.

Event kNkMAIDEvent_SBGroupAttrChanged

Object types Source

dataparamater pointer to NkMAIDEventParam

The contents of NkMAIDEventParam structure, becomes as follows.

NkMAIDEventParam.ulEvent = kNkMAIDEvent_SBGroupAttrChanged;

NkMAIDEventParam.ulElements= 2;

NkMAIDEventParam.ulParam[0] = SBGroupID;

NkMAIDEventParam.ulParam[1] = SBGroupAttrID;

If the Capability_SBWirelessMode is set to other than “Single flash” and the Capability_SBWirelessMultipleFlashMode is set to “Repeating”, SBGroupID of the master flash is used to inform the changes for the following flash group regardless of the value of Capability_SBUisableGroup.

- kNkMAIDSBGroupAttribute_FlashLevel
- kNkMAIDSBGroupAttribute_RepeatCount
- kNkMAIDSBGroupAttribute_RepeatInterval

6.16. MovieRecordComplete

This event will be issued the host that the movie recording is successfully completed.

Event kNkMAIDEvent_MovieRecordComplete

Object types Source

dataparamater Recording device

0: Recording to card

1: Recording to external device

2: Recording to both card and external device

This event will be issued when recording movie is completed by Capability_MovRecInCardStatus(OFF), or the maximum recording time reaching. This event is issued after kNkMAIDEvent_AddChildInCard event for all video files.

When movie files are divided by one movie recording, the end of the movie can be known at this event. It can be determined that all kNkMAIDEvent_AddChildInCard events issued between the stop of recording movie and issue of this event are movie data generated for one movie recording.

6.17. PictureControlAdjustChanged

This event will be issued the host that the setting related to Picture Control has changed.

Event kNkMAIDEvent_PictureControlAdjustChanged

Object types Source

data parameter pointer to NkMAIDEventParam

The contents of the NkMAIDEventParam structure are as follows.

NkMAIDEventParam.ulEvent = kNkMAIDEvent_PictureControlAdjustChanged

NkMAIDEventParam.ulElements = 2

NkMAIDEventParam.ulParam[0] : PicCtrlItem

NkMAIDEventParam.ulParam[1] : ShootingMode

0: Photo Shooting Menu

1: Movie Shooting Menu

This event will be issued when the following operations are made.

- When changing the setting of each item of picture control.
- When each item of the custom picture control is edited, registered, renamed or deleted.

The picture control item specified for PicCtrlItem will be changed depending on the setting value of ShootingMode.

[In case of Photo Shooting Menu]

The range of values enumerated is Capability_PictureControl

[In case of Movie Shooting Menu]

The range of values enumerated is Capability_MoviePictureControl (excluding "100: same as photo setting ") .

6.18. StartMovieRecord

This event will be issued the host that the camera has started movie recording.

Event kNkMAIDEvent_StartMovieRecord

Object types Source

dataparameter. Recording device

0: Recording to card

1: Recording to external device

2: Recording to both card and external device

This event will be issued the host that the camera has started movie recording by pressing the recording button of the camera. Recording device is specified for parameter. This event is not issued in case the starting of movie recording by Capability_MovRecInCardStatus.

6.19. ManualSettingLensDataChanged

This event will be issued the host that “Non-CPU lens data” has changed.

Event kNkMAIDEvent_ManualSettingLensDataChanged

Object types Source

data parameter LensNo. to be changed

0~19 : No1~No20

LensNo. is specified for parameter.

6.20. PixelMappingComplete

This event will be issued when the pixel mapping operation has completed.

Event	kNkMAIDEvent_PixelMappingComplete
Object types	Source
Data paramater	1: Success 0: Failure

This event notifies the completion of the pixel mapping by the execution of Capability_InitiatePixelMapping.

7. Vendor Unique Results

7.1. ApertureFEE

The aperture is not set maximum F number.

Result kNkMAIDResult_ApertureFEE

Command Start

Capability Capture, AFCapture, PreCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync

Explanation If the ExposureMode is set to Program or SpeedPriority and the aperture ring of the lens is not set to maximum F number, the camera cannot execute capture command.

Expected Action The client displays the message to set the aperture to maximum F number and is waiting for next command.

7.2. BufferNotReady

This is not used in the current module.

7.3. NormalTTL

The speedlight is set TTL mode.

Result kNkMAIDResult_NormalTTL

Command Start

Capability Capture, CaptureAsync

Explanation The camera cannot take a picture when an external speedlight is attached and it is set TTL(measuring through the lens) mode.

Expected Action The client displays the message that the camera cannot take a picture and is waiting for next command.

7.4. MediaFull

There are neither free spaces that can be recorded on the card nor a free space that can be recorded with built-in the camera SDRAM.

Result kNkMAIDResult_MediaFull

Command Start

Capability Capture, AFCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync, PreCapture

Explanation There is no free space at the specified media, so the client can not take a picture.

Expected Action The client displays the message that the camera cannot take a picture and is

waiting for next command.

7.5. InvalidMedia

It shows that the client can not take a picture because recording media is broken.

Result kNkMAIDResult_InvalidMedia

Command Start

Capability Capture, AFCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync

Explanation When Capability_SaveMedia is "Card" or "Card + SDRAM" it shows the client can not take a picture because the card is broken.

Expected Action The client displays the message that the camera cannot take a picture and is waiting for next command.

7.6. EraseFailure

This is not used in the current module.

7.7. CameraNotFound

The module did not find a camera on the bus.

Result kNkMAIDResult_CameraNotFound

Command The commands need access to the camera. (most of the commands for the Source, the Item or the Data object.)

Explanation The camera was disconnected. If the client sends Async command to the Module object at intervals, it can tell that the camera is reconnected by AddChild event.

Expected Action The client displays the message that the camera was disconnected and is waiting for next command.

7.8. BatteryDontWork

The main battery in the camera is used up.

Result kNkMAIDResult_BatteryDontWork

Command Start

Capability Capture, AFCapture, CaptureDustImage, PreCapture, CaptureAsync, AFCaptureAsync

Explanation The camera cannot take a picture because of the battery.

Expected Action The client displays the message that the camera cannot take a picture and suggests changing battery.

7.9. ShutterBulb

The exposure time is set to Bulb.

Result	kNkMAIDResult_ShutterBulb
Command	Start
Capability	Capture, AFCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync
Explanation	The camera cannot execute capture command if the Capability_ShutterSpeed is set to bulb.
Expected Action	The client displays the message that the camera cannot take a picture and is waiting for next command.

7.10. OutOfFocus

Auto focus operation is failed.

Result	kNkMAIDResult_OutOfFocus
Command	Start
Capability	Capture, AFCapture, CaptureAsync, AFCaptureAsync
Explanation	When the Capability_FocusMode is AF-S(1) and auto focus operation is failed, the camera cannot take a picture. Then this error is returned for the start of Capability_Capture, Capability_AFCapture, Capability_CaptureAsync or Capability_AFCaptureAsync.
Expected Action	The client displays the message that the camera is out of focus and is waiting for next command.

7.11. Protected

This is not used in the current module.

7.12. FileExists

This is not used in the current module.

7.13. SharingViolation

This is not used in the current module.

7.14. DataTransFailure

An error occurred while data transference.

Result	kNkMAIDResult_DataTransFailure
Command	Start, Async
Capability	Acquire
Explanation	If this error occurs while the client read an image from DRAM, it will lose the image.
Expected Action	The client aborts the data transference.

7.15. SessionFailure

The module cannot open source object because the camera cannot open more session.

Result	kNkMAIDResult_SessionFailure
Command	Open
Capability	-
Explanation	The camera can open 1 session. If the client tries to open more source object, the module returns this error.
Expected Action	The client displays an error message and is waiting for next command.

7.16. FileRemoved

This is not used in the current module.

7.17. BusReset

This command was aborted because bus-reset occurred.

Result	kNkMAIDResult_BusReset
Command	any command
Capability	any capability
Explanation	If bus-reset occurred, the command, which the module is executing, is aborted. Then the module returns this result for the command.
Expected Action	The client sends the command again.

7.18. NonCPULens

This is not used in the current module.

7.19. ReleaseButtonPressed

This is not used in the current module.

7.20. BatteryExhausted

This is not used in the current module.

7.21. CaptureFailure

The camera failed in setting the value for white balance preset data.

Result kNkMAIDResult_CaptureFailure

Command Start

Capability PreCapture

Explanation When it fails in white balance measurement(Capability_PreCapture), this error is returned.

Expected Action The client displays the message to confirm that the specified index of the preset is not protected and set again and is waiting for next command.

7.22. InvalidString

This is not used in the current module.

7.23. NotInitialized

This is not used in the current module.

7.24. CaptureDisable

This is not used in the current module.

7.25. DeviceBusy

A camera did not receive a command.

Result kNkMAIDResult_DeviceBusy

Command any command

Capability any capability

Explanation Since a camera is in the state where the command is not receivable, when it is not able to perform, this error returns.

Expected Action This command is sent again or a display of a user interface is returned to the state before command execution.

7.26. CaptureDustFailure

The camera failed in taking a dust off ref photo.

Result	kNkMAIDResult_CaptureDustFailure
Command	Start
Capability	CaptureDustImage
Explanation	When it fails in taking a dust off ref photo(Capability_CaptureDustImage), this error is returned.
Expected Action	Do nothing.

7.27. ICADown

This is not used in the current module.

7.28. NotLiveView

Live view was automatically stopped by the factor of the camera. (Included the case of that the live view time limit passed.)

Result	kNkMAIDResult_NotLiveView
Command	Start, Set
Capability	GetLiveViewImage
Explanation	When live view was automatically stopped by the factor of the camera (Include the case of that the live view time limit passed.), this error is returned.
Expected Action	The client displays an error message and is waiting for next command.

7.29. MFDriveEnd

The focus position reached the end of focus area in manual focus.

Result	kNkMAIDResult_MFDriveEnd
Command	Set
Capability	DeviceReady
Explanation	When the focus position reached the end of focus area by Capability_MFDrive, this error is returned.
Expected Action	After kNkMAIDCapability_MFDrive run, If the focus position reached the end of focus area, the module returns this error code about kNkMAIDCapability_DeviceReady .

7.30. UnformattedMedia

It shows that the client can not take a picture because the card is unformatted.

Result	kNkMAIDResult_UnformattedMedia
Command	Start
Capability	Capture, AFCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync
Explanation	When Capability_SaveMedia is "Card" or "Card + SDRAM" it shows the client can not take a picture because the card is unformatted.
Expected Action	The client displays the message that the camera cannot take a picture and is waiting for next command.

7.31. MediaReadOnly

It shows that the client can not take a picture because the card is protected.

Result	kNkMAIDResult_MediaReadOnly
Command	Start
Capability	Capture, AFCapture, CaptureDustImage, CaptureAsync, AFCaptureAsync
Explanation	When Capability_SaveMedia is "Card" or "Card + SDRAM" it shows the client can not take a picture because the card is protected.
Expected Action	The client displays the message that the camera cannot take a picture and is waiting for next command.

7.32. DuringUpdate

This is not used in the current module.

7.33. BulbReleaseBusy

It shows during bulb exposure shooting.

Result	kNkMAIDResult_BulbReleaseBusy
Command	Start, Set
Capability	Capture, CaptureAsync
Explanation	It shows during bulb exposure shooting
Expected Action	If bulb exposure shooting started by Capability_Capture, Capability_CaptureAsync, kNkMAIDResult_BulbReleaseBusy will be returned until bulb exposure shooting is terminated.

7.34. SilentReleaseBusy

It shows during silent shooting.

Result	kNkMAIDResult_SilentReleaseBusy
Command	Start
Capability	DeviceReady

Explanation	It shows during silent shooting.
Expected Action	This result will be returned when the following all conditions will be met. • Capability_LiveViewPhotoShootingMode is "1: ON" • After performed Capability_AFCapture, Capability_AFCaptureAsync, Capability_Capture or Capability_CaptureAsync • Perform Capability_DeviceReady before the silent shooting is completed

7.35. MovieFrameReleaseBusy

This is not used in the current module.

7.36. InvalidSBAttributeValue

It shows that it does not support the specified SB attribute value.

Result	kNkMAIDResult_InvalidSBAttributeValue
Command	Set
Capability	SBAAttrValue, SBGroupAttrValue
Explanation	It shows that it does not support the specified SB attribute value.
Expected Action	The client displays the error message and is waiting for next command.

7.37. Waiting_2ndRelease

This is not used in the current module.

7.38. MirrorUpCapture_Already_Start

This is not used in the current module.

7.39. RecInCard

It shows that the start of recording by card was successful.

Result	kNkMAIDResult_RecInCard
Command	Set
Capability	MovRecInCardStatus
Explanation	It shows that the start of recording by card was successful.
Expected Action	When Capability_MovRecInCardStatus is 1(ON) and the start of recording by card was successful. It is returned to tell whether it records into the card.

7.40. RecInExternalDevice

This is not used in the current module.

7.41. RecInCardAndExternalDevice

This is not used in the current module.

7.42. MountAdapterFWVer

It shows that the attached mount adapter needs a version upgrade.

Result	kNkMAIDResult_MountAdapterFWVer
Command	Start, Set, Get, GetArray
Capability	Capture, AFCapture, CaptureAsync, AFCaptureAsync, PreCapture, PreCaptureAsync, LiveViewStatus, GetLiveViewImage
Explanation	It shows that the attached mount adapter needs a version upgrade.
Expected Action	Do nothing.

8. kNkMAIDDataObjType_Video

Capability for which data object type kNkMAIDDataObjType_Video can be used by this module applies to the content described in not the MAID3.1 rule but this document.

9. ASCII code table

Capability of camera comments only the following ASCII code 90 characters can be input.

SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
:	;	<	=	>	?	@	[	]	_	{	}				
0	1	2	3	4	5	6	7	8	9						
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Q	R	S	T	U	V	W	X	Y	Z						
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
q	r	s	t	u	v	w	x	y	z						

For each Capability of UserComment, ArtistName, CopyrightInfo, you can also enter the following five characters.

¥	^	`		~
---	---	---	--	---

10. DataTypeCode

The basic data types used by the camera are as follows.

DataTypeCode	Type	Description
0x0001	INT8	Signed 8-bit integer
0x0002	UINT8	Unsigned 8-bit integer
0x0003	INT16	Signed 16-bit integer
0x0004	UINT16	Unsigned 16-bit integer
0x0005	INT32	Signed 32-bit integer
0x0006	UINT32	Unsigned 32-bit integer
0x0007	INT64	Signed 64-bit integer
0x0008	UINT64	Unsigned 64-bit integer
0x0009	INT128	Signed 128-bit integer
0x000A	UINT128	Unsigned 128-bit integer
0x4001	AIN8	Signed 8-bit integer array
0x4002	AUINT8	Unsigned 8-bit integer array
0x4003	AIN16	Signed 16-bit integer array
0x4004	AUINT16	Unsigned 16-bit integer array
0x4005	AIN32	Signed 32-bit integer array
0x4006	AUINT32	Unsigned 32-bit integer array
0x4007	AIN64	Signed 64-bit integer array
0x4008	AUINT64	Unsigned 64-bit integer array
0x4009	AIN128	Signed 128-bit integer array
0x400A	AUINT128	Unsigned 128-bit integer array
0xFFFF	STR	Variable length Unicode character string

11. Capability table that can be set during live view photography, movie live view, movie recording, SpotWB.

The following table shows the capabilities that can be set during live view.

Capability	Live view photograph	Movie live view	During movie file recording	SpotWB
3. Vendor Unique Capabilities				
ImageSize	o	x	x	x
RawImageSize	o	x	x	x

CompressionLevel	0	X	X	X
WBMode	0	0	0	X
Sensitivity	0	0	0	X
ResetMenuBank	0	0	X	X
CompressRAWEx	0	0	X	X
WBTuneAuto	0	0	0	X
WBAutoType	0	0	0	X
WBTuneIncandescent	0	0	0	X
WBFluorescentType	0	0	0	X
WBTuneFluorescent	0	0	0	X
WBTuneSunny	0	0	0	X
WBTuneFlash	0	0	0	X
WBTuneShade	0	0	0	X
WBTuneCloudy	0	0	0	X
WBTuneColorTempEx	0	0	0	X
WBTuneColorAdjust	0	0	0	X
WBTuneNatural	0	0	0	X
WBTunePreset1	0	0	0	X
WBTunePreset2	0	0	0	X
WBTunePreset3	0	0	0	X
WBTunePreset4	0	0	0	X
WBTunePreset5	0	0	0	X
WBTunePreset6	0	0	0	X
WBPresetProtect1	0	0	X	X
WBPresetProtect2	0	0	X	X
WBPresetProtect3	0	0	X	X
WBPresetProtect4	0	0	X	X
WBPresetProtect5	0	0	X	X
WBPresetProtect6	0	0	X	X
WBPresetNumber	0	0	0	X
WBPresetName	0	0	X	X
CCDDDataMode	0	0	X	X
ImageColorSpace	0	0	X	X
IsoControl	0	0	X	X
NoiseReduction	0	0	X	X
NoiseReductionHighISO	0	0	X	X

CompressRAWBitMode	0	0	X	X
PictureControl	0	0	X	X
PictureControlDataEx2	0	0	X	X
DeleteCustomPictureControl	0	0	X	0
Active D Lighting	0	0	X	X
ISOAutoShutterTime	0	0	X	X
ISOAutoShutterTimeAutoValue	0	0	X	X
ISOAutoHiLimit	0	0	0	X
VignetteControl	0	0	X	X
MovieScreenSize	0	0	X	X
MovieImageQuality	0	0	X	X
MovieRecMicrophone	0	0	0	X
MovieRecMicrophoneValue	0	0	0	X
AutoDistortion	0	0	X	X
UserMode1	--	--	--	--
UserMode2	--	--	--	--
UserMode3	--	--	--	--
HDRMode	0	X	X	X
HDRExposure	0	X	X	X
HDRSmoothing	0	X	X	X
HDRSaveIndividualImages	0	X	X	X
FlashISOAutoHighLimit	0	0	X	X
ResetCustomSetting	0	0	X	X
AFcPriority	0	0	X	X
AFsPriority	0	0	X	X
AFLockOnAcross	0	0	X	X
AFAreaPoint	0	0	X	X
EVInterval	0	0	X	X
CWMeteringDiameter	0	0	X	X
ExpBaseMatrix	0	0	X	X
ExpBaseCenter	0	0	X	X
ExpBaseSpot	0	0	X	X
ExpBaseHighlight	0	0	X	X
ShootingSpeed	0	0	X	X
ShootingLimit	0	0	X	X
ExposureDelayEx	0	0	X	X

NumberingMode	0	0	X	X
ResetFileNumber	0	0	X	X
FlashSyncTime	0	0	X	X
FlashSlowLimit	0	0	X	X
ExposureCompFlashUsed	0	0	X	X
BracketingVary	0	0	X	X
BracketingFactor	0	0	X	X
BracketingOrder	0	0	X	X
ShutterSpeedLockSetting	0	0	X	X
ApertureLockSetting	0	0	X	X
MovieReleaseButton	0	0	X	X
SBWirelessMode	0	X	X	X
SBWirelessMultipleFlashMode	0	X	X	X
SBUsableGroup	--	--	--	--
WirelessCLSEntryMode	0	0	X	X
SBPINCode	0	0	X	X
RadioMultipleFlashChannel	--	--	--	--
OpticalMultipleFlashChannel	0	X	X	X
FlashRangeDisplay	0	X	X	X
AllTestFiringDisable	--	--	--	--
SBSettingMemberLock	0	X	X	X
ShootNoCard	0	0	X	X
UserComment	0	0	X	X
EnableComment	0	0	X	X
ClockDateTime	0	0	X	X
ManualSetLensNo	0	0	X	X
FmmManual	0	0	X	X
F0Manual	0	0	X	X
EnableCopyright	0	0	X	X
ArtistName	0	0	X	X
CopyrightInfo	0	0	X	X
ShutterSpeed	0	X	X	X
FlexibleProgram	0	0	X	X
Aperture	0	0	0	X
MeteringMode	0	0	X	X
ExposureMode	0	0	X	0

ExposureComp	0	0	0	X
ShootingMode	0	0	X	0
ContinuousShootingNum	0	0	0	X
FocusAreaMode	0	0	X	X
EnableBracketing	0	X	X	X
AEBracketingStep	0	0	X	X
WBBracketingStep	0	0	X	X
BracketingType	0	0	X	X
ADLBracketingType	0	0	X	X
ADLBracketingStep	0	0	X	X
LiveViewStatus	0	0	0	0
LiveViewProhibit	--	--	--	--
CameraInclination	--	--	--	--
RemainContinuousShooting	--	--	--	--
RemainCountInMedia	--	--	--	--
LockExposure	--	--	--	--
LockFocus	--	--	--	--
LockFV	--	--	--	--
ExposureStatus	--	--	--	--
InfoDisplayErrStatus	--	--	--	--
FocalLength	--	--	--	--
FocusMode	--	--	--	--
BracketingCount	--	--	--	--
ExternalFlashStatus	--	--	--	--
ExternalFlashComp	--	--	--	--
ExternalFlashSort	--	--	--	--
ExternalNewTypeFlashMode	--	--	--	--
LensInfo	--	--	--	--
AFCapture	0	X	X	X
ContrastAF	0	0	0	0
MFDriveStep	0	0	0	0
MFDrive	0	0	0	0
ContrastAFArea	0	0	0	X
CaptureDustImage	0	0	X	X
DeleteDramImage	0	0	X	0
RawJpegImageStatus	--	--	--	--

<u>CurrentItemID</u>	0	0	0	0
<u>GetLiveViewImage</u>	0	0	0	0
<u>GetVideoImageEx</u>	0	0	X	0
<u>LockCamera</u>	0	0	X	X
<u>CameraType</u>	--	--	--	--
<u>LensType</u>	--	--	--	--
<u>MovRecInCardStatus</u>	X	0	0	X
<u>MovRecInCardProhibit</u>	--	--	--	--
<u>AngleLevel</u>	--	--	--	--
<u>SaveMedia</u>	0	0	X	X
<u>TerminateCapture</u>	0	0	X	0
<u>BlinkingStatus</u>	--	--	--	--
<u>LiveViewSelector</u>	0	0	X	X
<u>ResetWBMode</u>	0	0	X	X
<u>MovieShutterSpeed</u>	X	0	0	X
<u>MovieAperture</u>	X	0	0	X
<u>MovieSensitivity</u>	0	0	0	X
<u>MovieExposureComp</u>	X	0	0	X
<u>MovieRecFrameCount</u>	--	--	--	--
<u>RetractableLensWarningStatus</u>	--	--	--	--
<u>MovieWindNoiseReduction</u>	0	0	0	X
<u>MovieRecordingZone</u>	0	0	0	X
<u>MovieISOControl</u>	0	0	0	X
<u>MovieISOAutoHiLimit</u>	0	0	0	X
<u>MoviePictureControlDataEx2</u>	0	0	X	X
<u>DeleteMovieCustomPictureControl</u>	0	0	X	0
<u>MovieCCDDataMode</u>	0	0	X	X
<u>MovieWBMode</u>	0	0	0	X
<u>MovieWBTuneAuto</u>	0	0	0	X
<u>MovieWBAutoType</u>	0	0	0	X
<u>MovieWBTuneIncandescent</u>	0	0	0	X
<u>MovieWBFluorescentType</u>	0	0	0	X
<u>MovieWBTuneFluorescent</u>	0	0	0	X
<u>MovieWBTuneSunny</u>	0	0	0	X
<u>MovieWBTuneShade</u>	0	0	0	X
<u>MovieWBTuneCloudy</u>	0	0	0	X

MovieWB TuneColorTempEx	0	0	0	X
MovieWB TuneColorAdjust	0	0	0	X
MovieWB TuneNatural	0	0	0	X
MovieWB TunePreset1	0	0	0	X
MovieWB TunePreset2	0	0	0	X
MovieWB TunePreset3	0	0	0	X
MovieWB TunePreset4	0	0	0	X
MovieWB TunePreset5	0	0	0	X
MovieWB TunePreset6	0	0	0	X
MovieWB PresetProtect1	0	0	X	X
MovieWB PresetProtect2	0	0	X	X
MovieWB PresetProtect3	0	0	X	X
MovieWB PresetProtect4	0	0	X	X
MovieWB PresetProtect5	0	0	X	X
MovieWB PresetProtect6	0	0	X	X
MovieWB PresetName	0	0	X	X
MovieResetWB Mode	0	0	X	X
MovieNoiseReductionHighISO	0	0	X	X
MoviePictureControl	0	0	X	X
ISOControlSensitivity	--	--	--	--
MovieMeteringMode	0	0	X	X
LiveViewImageSize	0	0	X	X
SpotWB Mode	X	0	X	0
SpotWB Measure	0	0	X	0
SpotWB ChangeArea	0	0	X	0
SpotWB ResultDispEnd	0	0	X	0
RawJpegTrasferStatus	0	0	0	0
GetSBHandles	--	--	--	--
GetSBAttrDesc	--	--	--	--
SBAttrValue	0	0	X	X
GetSBGroupAttrDesc	--	--	--	--
SBGroupAttrValue	0	0	X	X
TestFlash	0	0	0	X
CaptureAsync	0	0	0	X
AFCaptureAsync	0	0	0	X
DeviceReady	0	0	0	0

ElectronicVR	0	0	X	X
MovieActive D Lighting	0	0	X	X
FlickerReductionSetting	0	0	X	X
SBIntegrationFlashReady	--	--	--	--
ExternalRecordingControl	0	0	X	X
LiveViewPhotoShootingMode	0	0	X	X
DetectionPeaking	0	0	0	X
HighlightBrightness	0	0	0	X
MovieAttenuator	0	0	0	X
SaveCameraSetting	0	0	X	0
MovieFileType	0	0	X	X
InternalFlashComp	0	X	X	X
GetRecordingInfo	--	--	--	--
DiffractionCompensation	0	0	X	X
VibrationReduction	0	0	X	X
MovieAutoDistortion	0	0	X	X
MovieAfAreaMode	0	0	0	X
MovieVibrationReduction	0	0	X	X
MovieFocusMode	0	0	0	X
MovieVignetteControl	0	0	X	X
MovieDiffractionCompensation	0	0	X	X
RecordTimeCodes	0	0	X	X
CountUpMethod	0	0	X	X
TimeCodeOrigin	0	0	X	X
DropFrame	0	0	X	X
MovieLogOutput	--	--	--	--
MovieLogSetting	0	0	X	X
HDMIOutputDataDepth	0	0	X	X
LowLightAF	0	0	X	X
ApplyLiveViewSetting	0	0	X	X
LiveViewImageStatus	--	--	--	--
LiveViewZoomArea	0	0	X	0
TrackingAFArea	0	0	0	X
MovieAfSpeed	0	0	X	X
MovieAfSpeedWhenToApply	0	0	X	X
MovieAfTrackingSensitivity	0	0	X	X

PreCapture	0	0	X	X
GetManualSettingLensData	0	0	0	0
AFModeAtLiveView	0	0	X	X
ElectronicFrontCurtainShutterEx	0	0	X	X
MovieRecDestination	0	0	X	X
Slot2ImageSaveMode	0	0	X	X
ChangeMonitorOffStatus	0	0	0	0
PrimarySlot	0	0	X	X
ActiveSlot	--	--	--	--
RemainingExposureTime	--	--	--	--
ExtendedShutterSpeedsManual	0	0	X	X
AutoFlashIsoSensitivityControl	0	0	X	X
StillCaptureModeSaveFrame	0	0	X	X
InitiatePixelMapping	0	0	X	X
PreCaptureAsync	0	0	X	X
LiveViewImageCompression	0	0	X	X
5. Standard Capabilities				
AsyncRate	--	--	--	--
ProgressProc	0	0	0	0
EventProc	0	0	0	0
DataProc	0	0	0	0
UIRequestProc	0	0	0	0
IsAlive	--	--	--	--
Children	0	0	0	0
State	--	--	--	--
Name	--	--	--	--
Description	--	--	--	--
Interface	--	--	--	--
DataTypes	--	--	--	--
DateTime	--	--	--	--
StoredBytes	--	--	--	--
Eject	--	--	--	--
Feed	--	--	--	--
Capture	0	0	0	X
Mode	--	--	--	--
Acquire	0	0	0	0

Start	--	--	--	--
Length	--	--	--	--
SampleRate	--	--	--	--
Stereo	--	--	--	--
Samples	--	--	--	--
Filter	--	--	--	--
Prescan	--	--	--	--
AutoFocus	--	--	--	--
AutoFocusPt	--	--	--	--
Focus	--	--	--	--
Coords	--	--	--	--
Resolution	--	--	--	--
Preview	--	--	--	--
Negative	--	--	--	--
Bits	--	--	--	--
Planar	--	--	--	--
Lut	--	--	--	--
Transparency	--	--	--	--
Threshold	--	--	--	--
Pixels	--	--	--	--
ForceScan	--	--	--	--
ForcePrescan	--	--	--	--
ForceAutoFocus	--	--	--	--
NegativeDefault	--	--	--	--
Firmware	--	--	--	--
CommunicationLevel1	--	--	--	--
CommunicationLevel2	--	--	--	--
BatteryLevel	--	--	--	--
FreeBytes	--	--	--	--
FreeItems	--	--	--	--
Remove	--	--	--	--
FlashMode	0	X	X	X
ModuleType	--	--	--	--
AcquireStreamStart	--	--	--	--
AcquireStreamStop	--	--	--	--
AcceptDiskAcquisition	--	--	--	--

Version	--	--	--	--
FilmFormat	--	--	--	--
TotalBytes	--	--	--	--

12. History

12.1. Ver.1.1 Rev.1.1 October 29, 2021.

- Added of the following capabilities.
 - 3.270. [LiveViewImageCompression](#)
 - 11. [Capability table that can be set during live view photography, movie live view, movie recording, SpotWB](#)
- Changed Description in the following.
 - 3.147 [GetLiveViewImage](#)
 - 3.147.1 [LiveView data format](#)
- Deleted of the following capabilities.
 - MovieWBPresetsNumber
 - 10. [Capability table that can be set during live view photography, movie live view, movie recording, SpotWB.](#)

12.2. Ver.1.1 Rev.1.0 January 29, 2021.

- Added informations about Z 6II_FU1(Z 6II firmup1).
 - 2. [Supported camera](#)
 - 3.1. [ImageSize](#)
 - 3.5. [Sensitivity](#)
 - 3.34. [CCDDataMode](#)
 - 3.46. [ISOAutoHiLimit](#)
 - 3.48. [MovieScreenSize](#)
 - 3.49. [MovieImageQuality](#)
 - 3.60. [FlashISOAutoHighLimit](#)
 - 3.138. [AFCapture](#)
 - 3.150. [CameraType](#)
 - 3.153. [MovRecInCardProhibit](#)
 - 3.162. [MovieSensitivity](#)
 - 3.164. [MovieRecFrameCount](#)
 - 3.169. [MovieISOAutoHiLimit](#)
 - 3.172. [MovieCCDDataMode](#)
 - 3.202. [ISOControlSensitivity](#)
 - 3.216. [CaptureAsync](#)
 - 3.217. [AFCaptureAsync](#)
 - 3.219. [ElectronicVR](#)
 - 5.17. [Capture](#)

12.3. Ver.1.0 Rev.1.0 November 12, 2020.